HYDRAULICS

INS3211 LECTURE NOTES

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Pipe Flow

7.1 Introduction

The transport of a fluid (liquid or gas) in a closed conduit (commonly called a pipe if it is of round cross section or a duct if it is not round) under pressure is extremely important in our daily operations (Fig.7.1a). The water pipes in our homes and distribution system that delivers the water from the city well to house. Numerous hoses and pipes carry hydraulic fluid or other fluids to various components of vehicles and machines. The air quality within our buildings is maintained at comfortable levels by the distribution of conditioned (heated, cooled, humidified/dehumidified) air through a maze of pipes and ducts. Another interesting example to pipes is that carry blood throughout our body and air into and out of our lungs. Although all of these systems are different, the fluid-mechanics principles governing the fluid motions are common. In this chapter we will apply the basic principles to a specific, important topic- the flow of viscous, incompressible fluids in pipes and ducts.

Some of the basic components of a pipe system may include more than one diameter, the various fittings used to connect the individual pipes to form the desired system, the flow rate control devices (valves), and the pumps or turbines that add energy to or remove energy from the system.

Even a closed conduit is not completely filled or open conduit involves the flow of a liquid that is called open channel. In open channel flow, fluid does not fill conduit even in closed (Fig.7.1b). A free surface exists between the flowing fluid (usually water) and fluid above it (usually the atmosphere). For example, the storm severe which is a closed conduit is designed as open channel flow because the flow in the sewer is expected to maintain a free surface most of the time. The main driving force for such flows is the fluid weight-gravity forces the fluid to flow downhill.

The difference between open channel flow and the pipe flow is in the fundamental mechanism that drives the flow. For open channel flow, gravity alone is the driving force- the water flows down a hill. For pipe flow, gravity may be important (the pipe need not be horizontal), but the main driving force is likely to be a pressure gradient along the pipe. If the pipe is not full, it is not possible to maintain this pressure difference, p_1-p_2.

Although not all conduits used to transport fluid from one location to another are round in cross sections, most of the common ones are. These include typical water pipes, hydraulic hoses, and other conduits that are designed to withstand a considerable pressure difference across their walls without undue distortion of their shape.

The flow of a fluid in a pipe may be laminar flow or it may be turbulent flow. Any fluid flowing in a pipe had to enter the pipe at some location. Fluid typically enters the pipe with a nearly uniform velocity profile. As the fluid moves through the pipe, viscous effects cause it to stick to the pipe wall (the no-slip boundary condition). Thus, a boundary layer in which viscous effects are important is produced along the pipe wall such that the initial velocity profile changes with distance along the pipe until the fluid reaches the end of the entrance length, beyond which the velocity profile does not vary with distance. The boundary layer has grown in thickness to completely fill the pipe. Viscous effects are of considerable importance within the boundary layer.

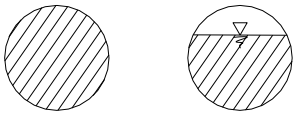


Figure 7.1 (a) Pressurized flow (Pipe) (b) Free surface flow (Channel)

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The fluid in contact with the wall has zero velocity; “no slip” and a velocity gradient exist between the fluid in the free stream and the plate surface if the conduit has enough section.

The major difference is that, in the pipe flow case, there is a limit to boundary layer thickness growth, namely the pipe radius.

If, therefore, this limit is reach before transition occurs, i.e. if laminar boundary layers meet at the pipe centre, the flow in the remainder of the pipe will be laminar. On the other hand, if transition within the boundary layer occurs before fill the pipe, the flow in the rest of the pipe will be turbulent.

Once boundary layer, whether laminar or turbulent in nature, has grown to fill the whole pipe cross-section, the flow may be said to be fully developed and no further changes in velocity profile are to be expected downstream, provided that the pipeline characteristics remain constant.

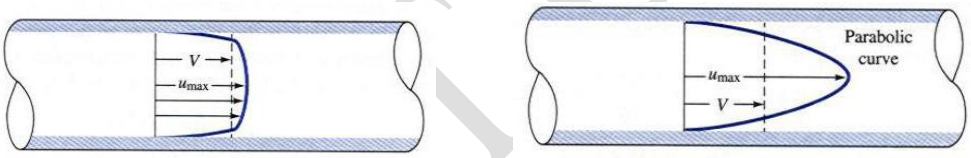
Typical entry lengths for establishment of fully developed laminar or turbulent flow may be taken as 120 and 60 pipe diameters, respectively.

For laminar flow, the velocity profile is parabolic and maximum velocity is at axis due to viscous effect (Fig 7.2a). For turbulent flow, the velocity profile is more uniform than the velocity profile of laminar flow due to momentum exchange and the maximum velocity is also at axis of pipe (Fig.7.2b), it changes as logarithmic.

Average velocity (V) in fully developed turbulent flow is 0.8 times of the maximum velocity in a pipe. However, average velocity in laminar flow is 0.5 times of the maximum velocity in a pipe, while 0.66 times of the maximum velocity between parallel plates.

For laminar flow $V=0.5\ u_{max}$

For turbulent flow $V=0.8u_{max}$



a) Laminar Flow (b) Turbulent Flow

Figure 7.2 Velocity gradients

7.2 Distribution of Velocity in Pipe Flow

Aim; to find a relation between mean velocity on the one hand and slope of the total head (energy line) on the other hand. The formula should be valid for all conditions and for any cross section. They should be as simple as possible; the accuracy should be compatible with the unavoidable uncertainty in conditions of walls (roughness) and of the fluid (viscosity).

From energy –slope to mean velocity there are two steps to make; (i) from **slope to shear stress** between fluid and the wall and (ii) from **shear to velocity**.

The most important parameter is average velocity (V) to define the fluid flow. It is very difficult to determine the velocity gradient even in a cross-section through a conduit, so researchers want to use the average velocity.

For the conditions shown in Fig. 7.3, the velocity distribution in a pipe is directly linked to the shear stress distribution; hence it is important to understand the latter. To determine the shear stress distribution, we start with the equation of equilibrium applied to a cylindrical element of fluid that is oriented coaxially with the pipe as shown in Fig.7.3.

Pressure against fore front and back front are different. The equilibrium of the forces against fore front and back front (pressures), shear along perimeter and gravity now leads to (no acceleration):

$$(p+dp)\pi r^2 - p\pi r^2 - (\rho g\pi r^2 ds)\sin\alpha - \tau_y(2\pi r ds) = 0$$

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$$dp\pi r^2 - \rho g\pi r^2 ds\sin\alpha - \tau_y 2\pi r ds = 0$$

$$\left(\frac{dp}{ds} - \rho g\sin\alpha\right)\frac{r}{2} = \tau_y$$

$$\tau_y = \left(\frac{1}{\rho g}\frac{dp}{ds} - \sin\alpha\right)\rho g\frac{r}{2}$$

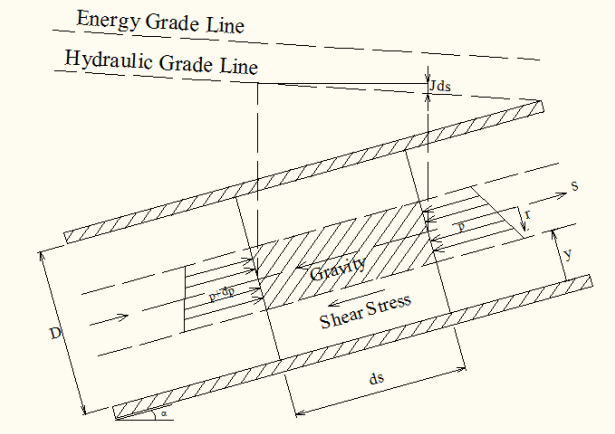


Figure 7.3 Forces on the differential circular volume

$$\Delta h_L = \left(\frac{V^2}{2g} + \frac{(p+\Delta p)}{\gamma} + z_1\right) - \left(\frac{V^2}{2g} + \frac{p}{\gamma} + z_2\right) = \frac{\Delta p}{\gamma} + (z_1 - z_2);$$

$$z_2 - z_1 = ds.\sin\alpha$$

$$\Delta h_L = \frac{dp}{\rho g} - ds\sin\alpha \rightarrow \frac{\Delta h_L}{ds} = J = \frac{1}{\rho g}\frac{dp}{ds} - \sin\alpha$$

if the slope of energy substitutes into the shear stress equation we get

$$\tau_y = \rho g\frac{r}{2}J$$

$$r = \frac{D}{2} - y$$

$$\tau_y = \rho g\left(\frac{D}{4} - \frac{y}{2}\right)J \qquad (7.1)$$

or on the wall;

$$\tau_0 = \rho g\left(\frac{D}{4}\right)J$$

where J is the loss of head per unit length of the conduit. Evidently hydraulic radius is $R=(\pi/4)D^2/\pi D=D/4$, so equation above is valid in this case too.

$$\tau_0 = \rho gRJ \qquad (7.2)$$

it gives the total shear stress on the wall.

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Since velocity distribution is different for laminar and turbulent flows, it is better to examine the second step at two stages. Contrary to the first step, in which it is indifferent if the flow is laminar or turbulent.

The criterion is Reynolds number, which is a velocity times a length over the kinematic viscosity. It is obvious that the velocity may be the mean velocity. In the case of a flat cross-section the depth h may be used as length in Reynolds number; in the case of a circular pipeline flowing full, it is customary to take the diameter D. Instead, we now take the hydraulic radius R, because we have learnt, when taking our first step, to consider that as a length, defining any cross section. This means that our values of Re are the same as the ones which are generally used for open conduits, but that pipe lines they are only fourth of the value which is generally given in literature (R=D/4).

The value for the transition of laminar to turbulent flow and vice versa in pipes, 2320, is based on the use of D in the Reynolds number. In the term of R of expressing Re the critical value is 2320/4=580. The transition is not always very sudden; better to say that it occurs at Reynolds number (defined with R) somewhere between 500 and 600. However, in practice it is taken ≅500.

This agrees with experiments with open conduits, where transition, not very sudden, occurs at Re≅500 or lower. It is evident that in this case the length used in Re is either the mean depth h_m, or the hydraulic radius, R, which is practically equal to h_m (slightly lower).

7.2.1 Laminar Flow through Pipes

The relation between velocity distribution, discharge and pressure can be found in analytical form for laminar flow.

A relation between shear stress inside a fluid and the gradient of velocity is given by the definition of kinematic viscosity,

$$\nu = \left(\tau_y/\rho\right)/\left(du/dy\right) \tag{7.3}$$

This is valid for a Newtonian (normal) fluid and for laminar flow

Shear stress is taken from Eq.(7.3) and put into Eq. (7.1);

$$du = \left(g\,J/\nu\right)\left(\frac{D}{4}-\frac{y}{2}\right)dy$$

Integrated;

$$u = \left(g\ J/\nu\right)\left(\ y\frac{D}{4}-\frac{y^2}{4}\right)+\text{constant} \tag{7.4}$$

Since no laminar flow along the wall, velocity will be zero and the constant is zero. Velocity at $y=\frac{D}{2}$ (in the axis);

$$u_{axis} = \left(gJ/\nu\right)\frac{D^2}{16} = \left(gJ/\nu\right)R^2 \tag{7.5}$$

It is seen from Eq. (7.5), velocity has a parabolic distribution.

The average velocity is one half of this

$$V = \left(gJ/\nu\right)\frac{D^2}{32} \tag{7.6}$$

or

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$$V = (gJ/\nu)\frac{R^2}{2} \tag{7.7}$$

This equation is known as **Hagen-Poiseuille**. The discharge can be obtained using Eq.(7.7)

$$Q = \frac{\gamma \pi D^4 J}{128\mu} \quad (7.8)$$

From Eq. (7.8), for two points having a pressure drop of Δp $\left(\text{remember } J = \frac{\Delta p/\gamma}{L}\right)$

$$\Delta p = \frac{128\mu LQ}{\pi D^4} \tag{7.9}$$

We may obtain the conclusions that head loss changes linearly with viscosity, length and discharge but forth power of the diameter. Head loss relates with one power of the velocity in laminar flow. Fig.7.4 shows the variation of the velocity and shear stress in the laminar flow.

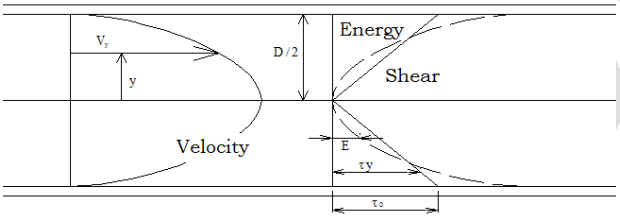


Figure 7.4 Distribution of velocity, shear stress and energy loss

7.2.2 Turbulent Flow through Pipes

In turbulent flow Eq. (7.3) is not valid. By the mixing of adjacent layers of fluid, momentum is transported from one layer to another. This leads to the conclusion that the shear will not be proportional to the gradient of velocity, but to the square of that gradient [$(du/dy)^2$]. The dimension of the ratio shear stress and gradient $(\tau_y/\rho)/(du/dy)^2$ gives square of length (L^2).

$$L^2 = (\tau_y/\rho)/(du/dy)^2 \tag{7.10}$$

or

$$\sqrt{\tau_y/\rho} = L(du/dy) \tag{7.11}$$

The length L has physical meaning. The greater L gives the smaller velocity gradient for a given value of $\sqrt{\tau_y/\rho}$. In other words, the greater L causes the stronger the mixing. Therefore, we call L *mixing length*. We may consider it as an average of sidewise excursion, which the fluid particles make. Near a (smooth) wall the mixing length must be very small. Farther away it will increase. KARMAN assumed that χ will be proportional to the distance from the wall.

$$L = \chi y \tag{7.12a}$$

Karman's parameter (χ) kappa is considered as a constant, at least for Newtonian fluids.

$$\chi = 0.4$$

In suspension, where turbulence is suppressed by the suspended particles, kappa may perhaps be less than 0.4.

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In circular pipes, there is no surface, however, to bring L back to zero in axis. It will grow until the center of the pipe is reached. Further it decreases again slowly at first, faster then, until it is zero again at opposite wall. The solid line in sketch may represent this.

There are many ways of describing such a line by a mathematical equation. We do not know if any of these formulae is the correct one; our knowledge of turbulence is not sufficient for that. So, theoretically we cannot reach an undisputable result. That gives us the freedom to assume a relation between L and y. Afterwards we can compare the results that have been obtained with various assumptions. An equation may be given as follows.

$$L = \chi y(1 - 2y/D)^{1/2} \tag{7.12b}$$

which is represented by the dash-dot-line in Fig. 7.5. The difference between Eq. (7.12a) and (7.12b) is about one percent. We use Eq.(7.12a).

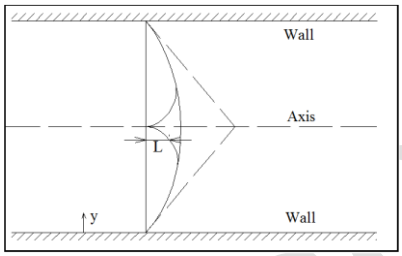


Figure 7.5 Mixing-length in pipes

The velocity near the wall of pipes is quite small and it is zero just on the wall. Along the wall of pipes turbulence is completely suppressed where the laminar sub-layer is defined by Prandtl. The thickness of laminar sub-layer in the turbulent boundary layer is small compared to the local boundary layer thickness. The flow is basically laminar in laminar sub-layer but it is not exactly true. However, in the laminar sub-layer viscosity is predominant. Hence instead of saying laminar sub-layer, this layer is called viscous sub-layer. The velocity profile across this sub-layer is assumed linear and tangential to the velocity profile up through the turbulent layer. Above the upper limit of the viscous sub-layer, i.e. outer sub-layer, is called outer zone (Fig. 7.6).

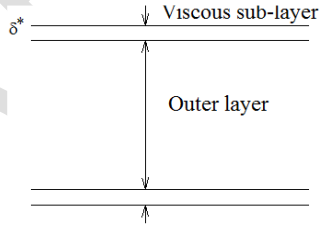


Figure 7.6 Flow regions in turbulent pipe flow

Viscous Sub-layer

In the sub-layer, which is very thin, we may take τ_0 for the shear (error quite negligible) because viscous forces are dominant in the viscous sub-layer, so there

$$\tau_0 = \mu \frac{du}{dy} \tag{7.13}$$

Integrated and u=0 may be written for y=0, so velocity distribution may be found in the viscous sub-layer as follows;

$$u = \frac{\tau_0}{\mu} y$$

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This equation may be written in non-dimensional form by dividing the term of $\sqrt{\frac{\tau_0}{\rho}}$,

$$\frac{u}{\sqrt{\frac{\tau_0}{\rho}}} = \frac{\tau_0}{\mu\sqrt{\frac{\tau_0}{\rho}}} y \quad \rightarrow \quad \frac{u}{\sqrt{\frac{\tau_0}{\rho}}} = \frac{\tau_0\sqrt{\frac{\tau_0}{\rho}}}{\mu\frac{\tau_0}{\rho}} y \quad \rightarrow \quad \frac{u}{\sqrt{\frac{\tau_0}{\rho}}} = \frac{\sqrt{\frac{\tau_0}{\rho}}}{\nu} y$$

$\sqrt{\frac{\tau_0}{\rho}}$ has the dimension of a velocity? This velocity (a real velocity) is called *shear velocity* (friction velocity), u_* which characterizes the shear at the boundary or turbulence strength. Hence the velocity in the sublayer is

$$\frac{u}{u_*} = \frac{u_*}{\nu} y$$

or

$$u = \frac{u_*^2}{\nu} y \tag{7.14}$$

As it is shown from Eq. (7.14), the velocity distribution is linear.

$$u_* = \sqrt{\frac{\tau_0}{\rho}} = \sqrt{gRJ} \tag{7.15}$$

Reynolds number is

$$Re = \frac{u_*\delta_*}{\nu}$$

in the sub-layer, where δ_* is the thickness of the viscous sub-layer, the experimental studies show that Re number equals a constant in the sub-layer.

$$11.6 = \frac{u_*\delta_*}{\nu}$$

from this equation the thickness of the sub-layer may be

$$\delta_* = 11.6\frac{\nu}{u_*} \tag{7.16}$$

However, outside the viscous sub-layer the viscous shear stress is negligible compared with the resistance resulting from turbulence. Viscous shear stress decreases and finally it may be negligible compared with turbulent shear stress (Fig.7.7). There is a transition region where both viscous and turbulent shear stresses are equally dominant, so the total shear stress is

$$\tau = \mu\frac{du}{dy} + \left(-\rho\overline{u'v'}\right) \tag{7.17}$$

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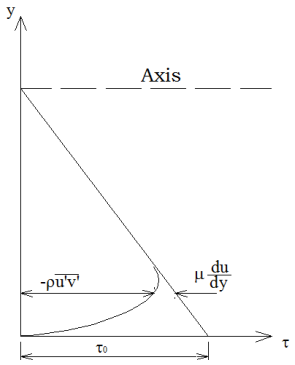


Figure 7.7 Distribution of shear stress

In outer zone, shear stress is $\tau = -\rho\overline{u'v'}$, since $\mu\dfrac{du}{dy} << -\rho\overline{u'v'}$. It is zero at the pipe center and increases to a maximum near the wall, as shown in Fig 7.7. Near the wall in the viscous sub-layer, turbulent shear stress reduces to zero because all of the shear stress is in the form of viscous shear stress. Turbulent shear stress may be written according to the Eq. (7.11),

$$\tau = \rho L^2\left(\frac{du}{dy}\right)^2 \tag{7.18}$$

or

$$\tau = \varepsilon\left(\frac{du}{dy}\right) \tag{7.19}$$

where,

$$\varepsilon = \rho L^2\left(\frac{du}{dy}\right) \tag{7.20}$$

is called “Eddy viscosity or Turbulence viscosity”

$$\tau = -\rho\overline{u'v'} = \varepsilon\left(\frac{du}{dy}\right)$$

This equation is known as Prandtl’s mixing length theory.

Mixing length $L = \chi y$ may be assumed from y=0 (wall) to y=D/2 (axis) and velocity distribution is symmetric relative to the pipe axis.

We assume $\tau = \tau_0 = \text{constant}$ because it is assumed that the shear stress is uniform and approximately equal to the shear stress at the wall for the zone of flow near the boundary. Thus the foregoing equation becomes

$$\tau_0 = \rho\chi^2\left(y\frac{du}{dy}\right)^2$$

Taking the square root of each side of the above equation

$$u_* = \sqrt{\frac{\tau_0}{\rho}} = \chi\left(y\frac{du}{dy}\right)$$

or rearranging yield

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$$du = \frac{u_*}{\chi}\frac{dy}{y}$$

Integrated;

$$u = \frac{u_*}{\chi}(\ln y + \text{constant}) \tag{7.21}$$

Eq. (7.21) shows that the velocity gradient has *logarithmic profile* in outer zone next to viscous sub-layer in which turbulence is dominant. Maximum velocity occurs at axis of pipe ($y = \frac{D}{2}$), so

$$u_{max} = \frac{u_*}{\chi}\left(\ln\frac{D}{2} + \text{constant}\right) \tag{7.22}$$

hence Eq. (7.21) is subtracted from Eq. (7.22)

$$\frac{u_{max} - u}{u_*} = \frac{1}{\chi}\ln\frac{D}{2y} \tag{7.23}$$

or if we consider logarithm based on 10.

$$\frac{u_{max} - u}{u_*} = 5.75\log\frac{D}{2y} \tag{7.24}$$

it is known as "velocity-defect law". For flow near the center of the pipe, as far flow near the outer limit of the boundary layer, the velocity defect law is applicable. Another important point about this law is that it applies to rough as well as smooth surfaces. Fig. 7.8 shows velocity gradient in the sub-layer and outer layer. There is a smooth transition of velocity between the viscous sublayer and the zone of logarithmic velocity distribution. This region is called the buffer zone in which neither law applies.

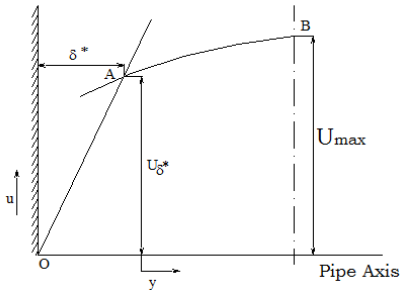


Figure 7.8 Velocity distributions in pipes

Since along the wall turbulence is completely suppressed, the viscous sub-layer, which is very thin, always exists over the pipe wall. All surfaces are rough to some extent. When viewed under a microscope the surface of almost every material seems to have nail-like projections on it. The height, shape, size and spacing of these projections depend on the type of the material. In practice usually, the height of the roughness (k) projections is taken to be the measure of the pipe roughness. Again, length of surface projections in absolute terms may not be a useful indicator since identical lengths of roughness projections may have significantly different effects in a small diameter pipe and in a large diameter pipe. Therefore, effects of pipe roughness are usually expressed in terms of *relative roughness*, which is defined as the ratio of the roughness projections to the diameter of the pipe.

When the roughness is considerably less than the viscous sub-layer thickness, such pipes are called hydraulic *smooth*. If the roughness height is more than the viscous sub-layer thickness, pipes are called

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hydraulic rough. There are three cases when the roughness of the pipe wall is compared with the viscous sub-layer thickness.

What happens if the wall is not smooth enough to allow the laminar sub-layer to develop which we express by saying that the wall is hydraulically *rough* instead of *hydraulically smooth*?

The well-known experiments of Nikuradse give an answer to this question. Nikuradse covered the walls of his experimental conduit by layer of grains of sand of approximately uniform diameter, k.

In this case, too, the logarithmic distribution of the velocity is obeyed which is confirmed by many experiments. So, Eq. (7.21) may again be applied, but the integration constant will depend on the roughness of the wall instead of the sub-layer, which absent now.

i) For $k<\delta_*$ the height of roughness is submerged in the viscous sub-layer, the roughness does not affect to turbulent boundary layer and this is called "hydraulic smooth pipe" (Fig. 7.9).

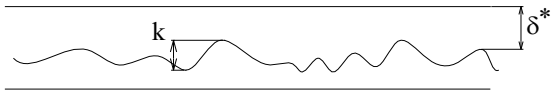


Figure 7.9 Hydraulic smooth regime

ii) For $k>6\delta_*$; the roughness affects and eddies will be developed from the wall. It is called "hydraulic rough pipe" (Fig. 7.10).

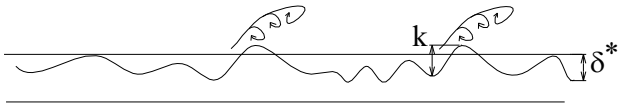


Figure 7.10 Hydraulic rough regime

iii) For $k\approx\delta_*$; the "transition pipe flow" exists.

Velocity Distribution for Hydraulic Smooth Pipe

From Eq. (7.14), the velocity is at the top of the sub-layer ($y=\delta_*$)

$$u_{\delta_*} = \frac{u_*^2}{\nu}\delta_* \tag{7.25}$$

or, if the thickness of viscous sub-layer (7.16) substitutes into Eq. (7.25)

$$\frac{u_{\delta_*}}{u_*} = \frac{u_*}{\nu}11.6\frac{\nu}{u_*}$$

$$\frac{u_{\delta_*}}{u^*} = 11.6 = \text{constant} \tag{7.26}$$

Both Eq. (7.26) and Eq. (7.24) must give the same values at point A ($y=\delta_*$) in Fig 7.8. If Eq. (7.26) and Eq. (7.16) substitute into Eq. (7.24) for the boundary condition, $y=\delta_*$.

$$\frac{u_{\delta_*}}{u_*} = \frac{u_{maks}}{u_*} - 5.75\log\frac{D}{2\delta_*}$$

$$11.6 = \frac{u_{max}}{u_*} - 5.75\log\frac{Du_*}{2\nu}\frac{1}{11.6}$$

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$$\frac{u_{max}}{u_*} = (11.6 - 5.75\log 11.6) + 5.75\log\frac{Du_*}{2\nu}$$

$$\frac{u_{max}}{u_*} = 5.5 + 5.75\log\frac{Du_*}{2\nu} \tag{7.27}$$

when Eq. (7.27) which is obtained from the boundary condition, $y=\delta_*$ substitutes into Eq. (7.24),

$$\frac{u}{u_*} = 5.5 + 5.75\log\frac{Du_*}{2\nu} - 5.75\log\frac{D}{2\nu}$$

$$\frac{u}{u_*} = 5.5 + 5.75\log\frac{yu_*}{\nu} \tag{7.28}$$

As indicated in Fig.7.11, for regions not to close the smooth wall, but not all the way out to the pipe center, Eq. (7.28) gives a reasonable correlation with the experimental data except the region between $u_* y/\nu = 8$ and 70 (transition region or buffer zone). Eq. (7.28) is valid for values of yu_*/ν ranging from approximately 20 to 10^5.

If we integrate the Eq. (7.28) for discharge passing through the cross-sectional area (dA) of the pipe, the average velocity can be found.

q=udA

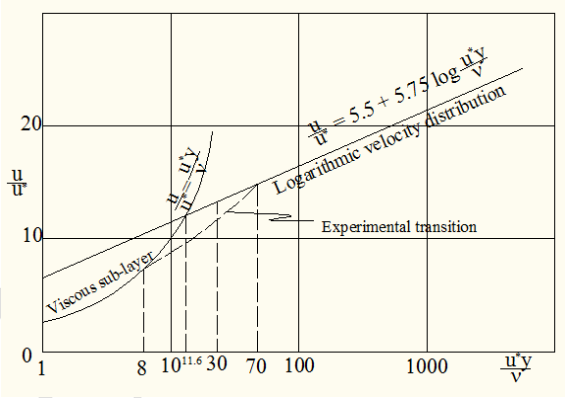


Figure 7.11 u/u_* versus Re number

Total discharge is

$$Q = \int_0^{D/2} udA = AV \tag{7.29}$$

where A = cross-sectional area of pipe.

If the velocity is constant in the cross-section and it equals to maximum velocity (u_{max}), discharge;

$$\int_0^{D/2} u_{max}\,dA = u_{max}A \tag{7.30}$$

if Eq. (7.29) is subtracted from Eq. (7.30)

$$\int_0^{D/2} (u_{max} - u)dA = (u_{max} - V)A \tag{7.31}$$

if Eq. (7.23) substitutes into Eq. (7.31)

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$$\int_0^{D/2} \frac{u_*}{\chi} \ln \frac{D}{2y} dA = (u_{max} - V)A$$

$$dA = 2\pi\left(\frac{D}{2} - y\right)dy$$

$$\int_0^{D/2} \frac{u_*}{\chi} \ln \frac{D}{2y} 2\pi\left(\frac{D}{2} - y\right)dy = (u_{max} - V)A$$

$$\frac{2\pi}{\chi} u_* \left\{ \left[\frac{D}{2} y - \frac{y^2}{4} \right]_0^{D/2} - \ln \frac{2y}{D} \left[\frac{D}{2} y - \frac{y^2}{2} \right]_0^{D/2} \right\} = (u_{max} - V)A$$

$$\frac{u_{max} - V}{u_*} = \frac{3}{2\chi} = \text{constant}$$

$$\frac{u_{max} - V}{u_*} = 3.75 \qquad \text{for } \chi=0.4$$

$$\frac{V}{u_*} = \frac{u_{max}}{u_*} - 3.75 \tag{7.32}$$

According to (7.28), the velocity at any point in the cross-section of pipe,

$$\frac{u}{u_*} = 5.5 + 5.75 \log \frac{u_* y}{\nu}$$

$u=u_{max}$ for $y = \frac{D}{2}$

$$\frac{u_{max}}{u_*} = 5.5 + 5.75 \log \frac{u_* D}{2\nu} \tag{7.33}$$

If Eq.(7.33) substitutes into Eq.(7.32), average velocity may be found (V) for smooth pipe flow;

$$\frac{V}{u_*} = 1.75 + 5.75 \log \frac{u_* D}{2\nu} \tag{7.34}$$

Velocity Distribution for Hydraulic Rough Pipe

Eq.(7.28) was found using top sub-layer condition. When roughness is dominant relative to the thickness of viscous sub-layer, the boundary condition will change in Eq. (7.21). According to experimental data;

$u=0$ for $y = \frac{k}{30}$ (for $\chi=0.4$)

$$\text{constant} = -2.5 u_* \ln \frac{k}{30}$$

The constant substitutes into Eq. (7.21)

$$\frac{u}{u_*} = 2.5 \ln y - 2.5 \ln \frac{k}{30}$$

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$$\frac{u}{u_*} = 2.5 \ln \frac{30y}{k} \tag{7.35}$$

$$\frac{u}{u_*} = 5.75 \log\left(\frac{y}{k}\right) + 8.5 \tag{7.35a}$$

Average velocity can be found by following the similar way of hydraulic smooth pipe flow case,

$$\frac{V}{u_*} = 4.75 + 5.75 \log \frac{D}{2k} \tag{7.36}$$

Numerous tests on flow in rough pipes all showed that a semi-logarithmic velocity distribution is valid over most of the pipe section.

Transition Region

Eq.(7.36) ; $\frac{V}{u_*} = 5.75 \log \frac{6.7D}{2k} = \frac{13.4R}{k}$

Eq.(7.34); $\frac{V}{U^*} = 5.75 \log \frac{u\ D}{\gamma} = 5.75 \log \frac{11.6D}{\delta_*} = 5.75 \log \frac{46.4R}{\delta_*}$

If both equations are summed,

$$\frac{V}{u_*} = 5.75 \log\left(\frac{12R}{k + 0.3\delta_*}\right) \tag{7.37}$$

Both viscous sub-layer and roughness are dominant in the transition region, Eq.(7.37) valid in that region using the summation of Eq.(7.34) and (7.36).

Eq. (7.37) may be written as follows; $\left(u_* = \sqrt{gRJ} = 3.13\sqrt{RJ}\right)$

$$V = 18\sqrt{RJ} \log\left(\frac{12R}{k + 0.3\delta_*}\right) \tag{7.38}$$

When measuring y, we have to determine a smooth plane, which represents the wall of pipe. In the layer of sand, which presents the roughness approximately, one half of the volume is in sand and one half in water. So, we take the wall halfway the grains. It is assumed that the half way of the grains is mathematical wall because there is as much fluid outside of as there is solid matter inside. The tops of the bodies which constitute the roughness have a height of a=k/2 in respect to the mathematical wall. This is one of the reasons why we take the radius of sand grains, a for the roughness length instead of Nikuradse's sand roughness, which is diameter of the grains. In some literature, a is used instead of k. If we rewrite Eq.(7.38) with a;

$$V = 18\sqrt{RJ} \log\left(\frac{6R}{a + \frac{\delta_*}{7}}\right) \tag{7.39}$$

or

$$V = C\sqrt{RJ} \tag{7.40}$$

This is called Chezy equation. It is applicable only for turbulent flow, where,

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$$C = 18\log\frac{6R}{a + \frac{\delta_*}{7}} \qquad \left(m^{1/2}/s\right) \tag{7.41}$$

or

$$C = 18\log\frac{12R}{k + 0.3\delta_*} \qquad \left(m^{1/2}/s\right) \tag{7.42}$$

It is Chezy coefficient. If we summarize the formulas in turbulent pipe flow they are valid in the following conditions;

1) Hydraulically Smooth Regime;

$$k<\delta_* \qquad \left(= 11.6\frac{\nu}{u_*}\right) \tag{7.43}$$

2) Hydraulically Rough Regime;

$$k>6\delta_* \quad \left(= 70\frac{\nu}{u_*}\right) \tag{7.44}$$

3) Transitional Regime

$$\delta_*<k<6\delta_* \tag{7.45}$$

Note: For a turbulent flow the velocity profile can be represented by the empirical equation for $10^5<Re<10^7$. Power law formula is applicable everywhere except close the wall.

$$\frac{u}{u_{max}} = \left(\frac{y}{\delta_*}\right)^{1/7} \tag{7.46}$$

This empirical equation is called the *power*-law equation (Schlichting, 1970).

$$\frac{V}{u_{max}} = \left(y/r_0\right)^m \left\{\begin{array}{ccccccc} Re & 4\times10^3 & 2.3\times10^4 & 1.1\times10^5 & 1.1\times10^6 & 3.2\times10^6 \\ m & 1/6 & 1/6.6 & 1/7 & 1/8.8 & 1/10 \\ \frac{V}{u_{max}} & 0.791 & 0.807 & 0.817 & 0.850 & 0.865 \end{array}\right\} \tag{7.47}$$

where m varies with the Re number.

7.3 Energy Losses in Pipes

One of the problems of interest to engineers is the estimation of energy required to force a fluid through a conduit with a certain velocity or pressure. Since some of the energy is lost in overcoming frictional resistance, estimation of such losses is important before a reliable estimate of total required energy made. Loss by pipe friction means that the kinetic energy transforms to heat energy.

1- Head Loss Due to Friction

2- Local Loss (Minor Loss)

Head Loss Due to Friction

Mechanical energy transforms to heat energy by friction due to pipe wall.

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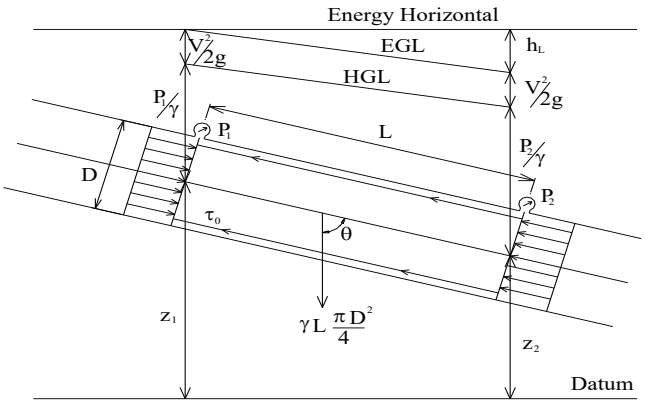


Figure 7.12

As shown in Fig. 7.12, energy loss between the sections (1) and (2) per unit time and per unit weight in pipe

$$\Delta h_L = \left(\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1\right) - \left(\frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2\right)$$

It is also called head loss due to friction. This equation may be written with $V_1=V_2$ because the pipe has constant diameter,

$$\Delta h_L = \left(\frac{P_1}{\gamma} + z_1\right) - \left(\frac{P_2}{\gamma} + z_2\right)$$

the equilibrium applies to a control volume for steady flow (acceleration equals to zero),

$$p_1 \frac{\pi D^2}{4} - p_2 \frac{\pi D^2}{4} + \gamma L \frac{\pi D^2}{4} \cos\theta - \tau_0 \pi D L = 0$$

hence

$$\tau_0 = \frac{D\gamma}{4L}\left(\frac{p_1 - p_2}{\gamma} + L\cos\theta\right)$$

or

$$\tau_0 = \frac{D\gamma}{4L}\Delta h_L$$

If we divide both sides of the above equation with $\rho V^2/8$,

$$\frac{\tau_0}{\rho V^2/8} = \frac{D\gamma \Delta h_L/4L}{\rho V^2/8} = \lambda \qquad (7.48)$$

non-dimensional frictional factor λ may be obtained, similar coefficient was found by dimensional analysis in the previous chapter. V is the average velocity in Eq.(7.48), and we rearrange Eq. (7.48), we may get,

$$h_L = \lambda \frac{L}{D}\frac{V^2}{2g} \qquad (7.49)$$

It is well known Darcy–Weisbach equation.

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Friction coefficient may be obtained for different type of pipes by numerous tests. The expressions of the coefficient in the form of non-dimensional form were found by dimensional analysis (see Chapter 6);

For Laminar flow;

$$h_L = \lambda \frac{L}{D}\frac{V^2}{2g} \quad ; \qquad \lambda = f(Re)$$

For Turbulent Flow;

i) In Hydraulic smooth pipe

$$h_L = \lambda \frac{L}{D}\frac{V^2}{2g} \quad ; \qquad \lambda = f(Re)$$

ii) In Hydraulic rough ripe

$$h_L = \lambda \frac{L}{D}\frac{V^2}{2g} \quad ; \qquad \lambda = f\left(\frac{k}{D}\right)$$

iii) In Transition

$$h_L = \lambda \frac{L}{D}\frac{V^2}{2g} \quad ; \qquad \lambda = f\left(Re, \frac{k}{D}\right)$$

In general, the coefficient λ is the function of both Re number and relative roughness parameter. Much of this information is a result of experiments conducted by J. Nikuradse in 1933 and amplified by many others since then. In commercially available pipes the roughness is defined by Moody. He gave a chart in Fig. 7.13 which shows functional dependence. It is called *Moody chart*. His data correlated the original data of Nikuradse in terms of the relative roughness of commercially available pipe materials.

Friction Factor

I-For turbulent Flow;

1- Hydraulic Smooth Pipes Using Average Velocity

$$\frac{V}{u_*} = 5.75\log\left(\frac{Du_*}{2\nu}\right) + 1.73$$

and $\lambda = 8\tau_0/\rho V^2$; $\lambda/8 = \frac{\tau_0}{\rho}\Big/V^2$; $8/\lambda = V^2/\tau_0/\rho$

$$\sqrt{\frac{8}{\lambda}} = \frac{V}{\sqrt{\frac{\tau_0}{\rho}}}$$

$$\sqrt{\frac{8}{\lambda}} = \frac{V}{u_*}$$

if the equations are rearranged

$$\sqrt{\frac{8}{\lambda}} = 5.75\log\left(\frac{VD}{\nu}\frac{u_*}{2V}\right) + 1.73$$

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$$\sqrt{\frac{8}{\lambda}} = 5.75\log\left(\mathrm{Re}\sqrt{\frac{\lambda}{32}}\right) + 1.73 \tag{7.50}$$

For $Re<10^5$ following equation is also proposed;

$$\lambda = 0.316/\mathrm{Re}^{1/4} \tag{7.51}$$

2- Hydraulic Rough Pipes Using Average Velocity

$$\frac{V}{u_*} = 5.75\log\left(\frac{D}{2k}\right) + 4.75$$

$$\frac{V}{u_*} = \sqrt{\frac{8}{\lambda}}$$

using above equations, the friction factor is

$$\sqrt{\frac{8}{\lambda}} = 5.75\log\frac{D}{2k} + 4.75 \tag{7.52}$$

3- Transition Regime (semi-empirical formula; Colebrook – White equation),

$$\frac{1}{\sqrt{\lambda}} = -2\log\left(\frac{k}{3.7D} + \frac{2.51}{\mathrm{Re}\sqrt{\lambda}}\right) \tag{7.53}$$

In fact, the Moody chart is a graphical representation of this equation, which is empirical fit of the pipe flow pressure drop data. Eq.(7.53) is called the *Colebrook formula*. A difficulty with its use is that it is implicit in the dependence of λ. It needs iterative solutions.

The computer-aided solution may be proposed as follows;

$$Q = -2A\sqrt{2gD\frac{\Delta h_L}{L}}\log\left(\frac{k}{3.7D} + \frac{2.51\nu}{D\sqrt{2gD\frac{\Delta h_L}{L}}}\right) \tag{7.54}$$

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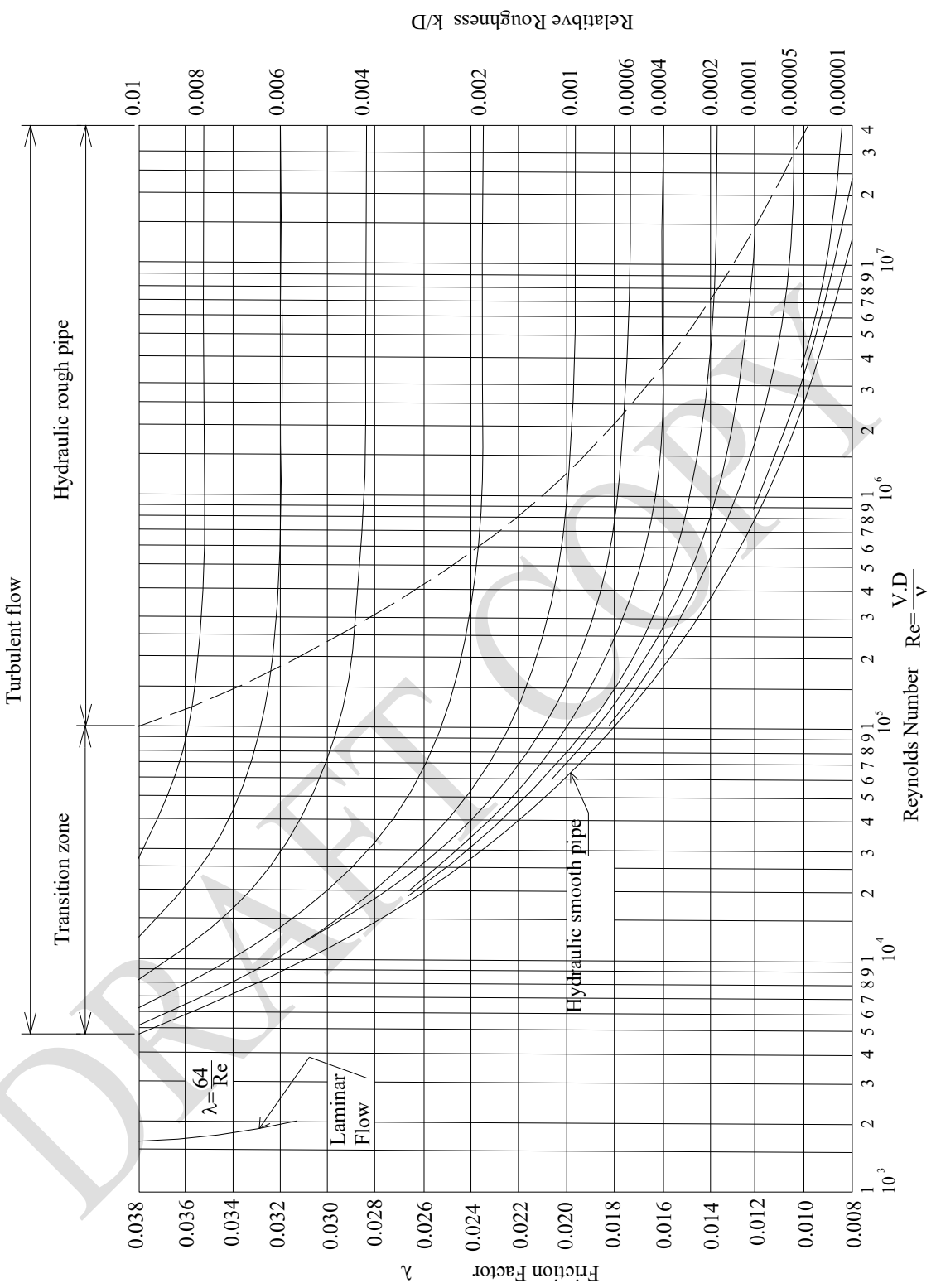


Figure 7.13 Moody chart

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II-For Laminar Flow

Average velocity,

$$V = (gJ/\nu)\cdot D^2/32$$

so

$J=\Delta h_L/L$

$$V = \frac{g\,\Delta h_L}{\nu\,L}\,\frac{D^2}{32}$$

$$\Delta h_L = \frac{32}{\frac{D}{\nu}}\frac{L}{D}\frac{V}{g}$$

if the above equation is multiplied with 2V

$$\Delta h_L = \frac{64}{\frac{VD}{\nu}}\frac{L}{D}\frac{V^2}{2g}$$

or

$$\Delta h_L = \frac{64}{Re}\frac{L}{D}\frac{V^2}{2g}$$

where

$$\lambda = \frac{64}{Re} \qquad (7.55)$$

hence, Darcy-Weisbach equation may be written,

$$\Delta h_L = \lambda\frac{L}{D}\frac{V^2}{2g}$$

so, Darcy-Weisbach equation is valid for both laminar and turbulent pipe flow.

Empirical Equations for Energy Loss Due to Friction

In addition to the Darcy-Weisbach equation, the other empirical equations are proposed. The general form of these equations is;

$$\Delta h_L = K\cdot\frac{L\cdot V^n}{D^x} \qquad (7.56)$$

The rate $\Delta h_L/L$ is energy (or head) loss due to friction per unit length, and it is called Hydraulic slope or gradient.

n value equals to 1 for laminar flow and 1.7~2.0 for turbulent flow, but x value equals to 2 for laminar flow and 1.0~1.3 for turbulent flow.

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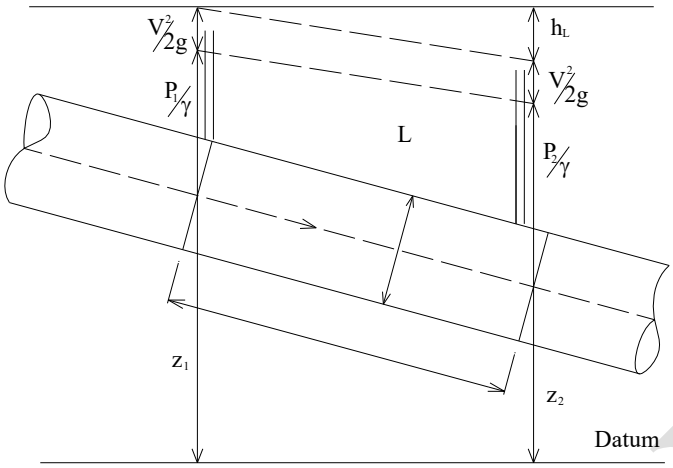


Figure 7.14 Variation of energy along the pipe

If the equation is rearranged,

$$V = \left(\frac{1}{K}\right)^{1/n} D^{x/n} J^{1/n}$$

and $(1/K)^{1/n} = C'$, x/n=y , 1/n=z are substituted into the above equation,

$$V = C' D^{y} J^{z}$$

using D=4R

$$V = \left(4^{y} \times C'\right) R^{y} J^{z}$$

or,

$$V = C^{x} R^{y} J^{z} \tag{7.57}$$

If the coefficient and the powers are written in terms of numerical values,

1- CHEZY Equation

$$V = C\sqrt{RJ} \tag{7.58}$$

C: Chezy coefficient in terms of Darcy friction factor;

$$C = \sqrt{\frac{8g}{\lambda}} \qquad m^{1/2}/s \tag{7.59}$$

For the Chezy coefficient, some equations have been proposed for pipe and channel flow;

Kutter formula $$C = \frac{100\sqrt{D}}{2m + \sqrt{D}} \tag{7.60}$$

or $$C = \frac{100\sqrt{R}}{m + \sqrt{R}} \tag{7.61}$$

where m; friction factor and it is known as Kutter coefficient.

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Table 7.1 Kutter Coefficient, m (Dönmezer, 1973)

Pipe types	Kutter coefficient (m)
New cast iron	0.20
Old cast iron	0.25
Steel	0.20
New ceramic pipe	0.25
Old ceramic pipe	0.30~0.35
Wastewater pipe	0.30~0.35
Cement-lined duct	0.20~0.25

2- MANNING – STRICKLER Formula

Manning

$$V = \frac{1}{n} R^{2/3} J^{1/2} \tag{7.62}$$

Strickler

$$V = k\ R^{2/3} J^{1/2} \tag{7.63}$$

where n is Manning friction factor, k is Strickler factor,

$$k = \frac{1}{n} \tag{7.64}$$

According to Strickler k is written in terms of roughness height;

$$k = \frac{21.1}{\sqrt[6]{k_{rough}}} \tag{7.65}$$

All the coefficients have relations as follows;

$$C = \frac{R^{1/6}}{n} = \sqrt{\frac{8g}{\lambda}} = 18 \log \frac{6R}{a + \delta_* / 7} \tag{7.66}$$

Table 7.2 Manning friction factor, n (Dönmezer, 1973)

Pipe type	Manning friction factor, n
Brass and glass	0.009-0.011
Asbestos cement	0.010-0.012
Steel	0.011-0.013
Wood	0.011-0.013
Cast iron, un-lined and clean	0.013-0.015
Cast iron, lined and clean	0.012-0.014
Galvanized iron	0.015-0.017
Concert pipe, well smoothed	0.011-0.012
Concert pipe, molded	0.012-0.014
Concert drainage pipe	0.012-0.014
Wastewater pipe, lined	0.013-0.015
Steel, riveted	0.015-0.017
Cast iron, not clean	0.015-0.035
Bend iron pipe	0.020-0.022

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3- WILLIAMS – HAZEN Equation

This equation is generally used in pipe networks

$$V = 0.85CR^{0.63}J^{0.54}$$

where C: Williams – Hazen coefficient.

Table 7.3 C Coefficients (Dönmezer, 1973)

Pipe types	Williams-Hazen coefficient, C
Cast iron	100-120
Steel	130-140
Concrete pipe	135-150
Brass, copper	120-140

From these equations, firstly, hydraulic gradient must be calculated and the loss due to friction is,

$$h_L = JL \tag{7.67}$$

7.3.2 Local (Minor) Head Loss

In general, the local losses are caused by the change of direction and magnitude of velocity and turbulence is generated in the changed cross-sections. Eddies are mostly placed at (20-50)D at the downstream of the source and gradually decreases. These losses that occur in pipelines because of bends, elbows, joints, valves, etc., are called minor losses. This is a misnomer, because in many situations they are more important especially for small pipe systems than the losses due to pipe friction considered so far in this section, but the name is conventional. In almost all cases minor loss is determined by experiment. However, one important exception is the head loss due to a sudden expansion in a pipeline.

7.3.2.1 Head Loss due to Sudden Expansion (Δh_a)

A sudden expansion (diverging flow) in a pipe is shown in Fig. 7.15. The streamlines emerging from the smaller diameter pipe cannot conform to the abrupt change in diameter at Section 1 and separate from the wall. The stream later expands and reattaches itself to the wall at 2. In the corner, *pockets of turbulent eddies are formed and energy is dissipated in the form of heat.*

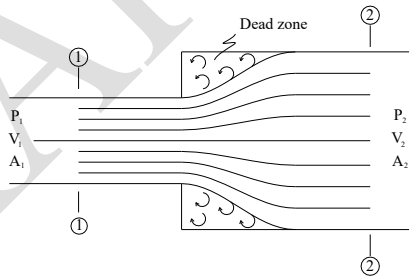


Figure 7.15 Streamlines in a sudden expansion.

If impulse-momentum equation is written between 1 and 2

$$p_1A_1 + p_1(A_2 - A_1) + \rho QV_1 - p_2A_2 - \rho QV_2 = 0$$

Continuity equation;

$Q=V_1A_1=V_2A_2$

Energy equation;

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + \Delta h_a$$

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from impulse – momentum and continuity equations

$$\frac{p_1 - p_2}{\gamma} = \frac{V_2(V_1 - V_2)}{g}$$

is obtained and if this equation is substituted into the energy equation

$$\Delta h_a = \frac{(V_1 - V_2)^2}{2g} \tag{7.68}$$

is obtained. By using the continuity equation, Eq.(7.68) can be written as;

$$\Delta h_a = \left(1 - \frac{A_1}{A_2}\right)^2 \frac{V_1^2}{2g} \tag{7.69}$$

$$K_a = \left(1 - \frac{A_1}{A_2}\right)^2 \tag{7.70}$$

$$\Delta h_a = K_a \frac{V_1^2}{2g} \tag{7.71}$$

where K_a sudden expansion head loss coefficient. The head loss due to sudden expansion can be greatly reduced by providing a gradual transition in the form of a diffuser. K depends on the expansion angle θ and the expansion ratio (D_2/D_1). The values of K for various angles and expansion ratios are given in Table 7.4

Table 7.4 Minor head loss coefficients for different gradual transitions

D_2/D_1	Angle (θ)											
	2°	6°	10°	15°	20°	25°	30°	35°	40°	45°	50°	60°
1.10	0.01	0.01	0.03	0.05	0.10	0.13	0.16	0.18	0.19	0.20	0.21	0.23
1.20	0.02	0.02	0.04	0.09	0.16	0.21	0.25	0.29	0.31	0.33	0.35	0.37
1.40	0.02	0.03	0.06	0.12	0.23	0.30	0.36	0.41	0.44	0.47	0.50	0.53
1.60	0.03	0.04	0.07	0.15	0.26	0.35	0.42	0.47	0.51	0.54	0.57	0.61
1.80	0.03	0.04	0.07	0.16	0.28	0.37	0.44	0.50	0.54	0.58	0.61	0.65
2.00	0.03	0.04	0.07	0.16	0.29	0.38	0.46	0.52	0.56	0.60	0.63	0.68
2.50	0.03	0.04	0.08	0.16	0.30	0.39	0.48	0.54	0.58	0.62	0.65	0.70
3.00	0.03	0.04	0.08	0.16	0.31	0.40	0.48	0.55	0.59	0.63	0.66	0.71
∞	0.03	0.05	0.08	0.16	0.31	0.40	0.49	0.56	0.60	0.64	0.67	0.72

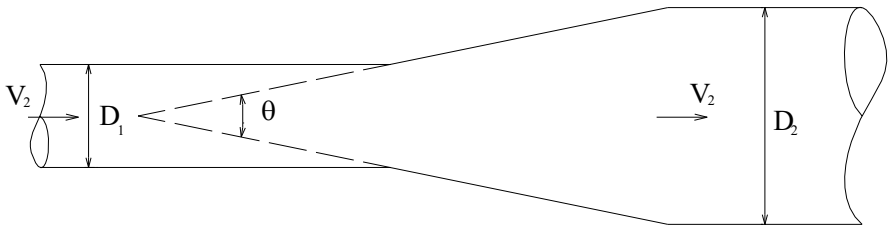


Figure 7.15 A diffuser (gradual transition)

$$\Delta h = K \frac{V_1^2}{2g}$$

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7.3.2.2 Head Loss due to Entrance into Reservoir

This is a special form of a sudden expansion. If the sudden expansion is from a pipe to a reservoir $D_1/D_2=0$ and the loss becomes $\frac{V_1^2}{2g}$; that is, the complete kinetic energy in the flow is converted into thermal energy

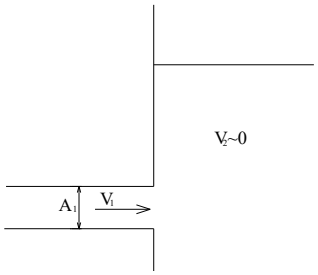


Figure 7.16 Entrance into a reservoir

The velocity at the reservoir will be $V_2 \approx 0$ (because $A_2 >> A_1$)

$$\Delta h_g = \frac{(V_1^2 - V_2^2)}{2g} = \frac{V_1^2}{2g} \tag{7.72}$$

For gradual varying connections

$$\Delta h_g = K_g \frac{V_1^2}{2g}$$

7.3.2.3 Head Loss due to Sudden Contraction (Δh_d)

A sudden contraction (converging flow) in a pipe is shown in Fig. 7.17. As the fluid approaches the contraction, the streamlines start to bend at section 1. This produces a zone of stagnant with some eddies from A to B. At B, the fluid separates from the pipe and a *vena contracta* is formed (the cross-section where the contraction is greatest is called vena contracta). The streamlines are parallel throughout the jet at vena contracta. The stream then expands and reattaches itself to the wall at the point C. Eddies are formed between the *vena contracta* and the wall and *it is here that most of the energy is dissipated as heat*. The value of K can be found experimentally by using various diameters.

The energy (head) loss between (3) and (2) can be thought as similar to minor head loss at expansion and head loss from section 1 to the *vena contracta* is small compared with the loss from section (3) to section (2) where velocity head is being reconverted into pressure head.

$$\Delta h_d = \frac{(V_c - V_2)^2}{2g}$$

is written.

With the continuity equation,

$A_2V_2=A_cV_c$

and

$A_c=C_2A_2$

(where C_c contraction coefficient)

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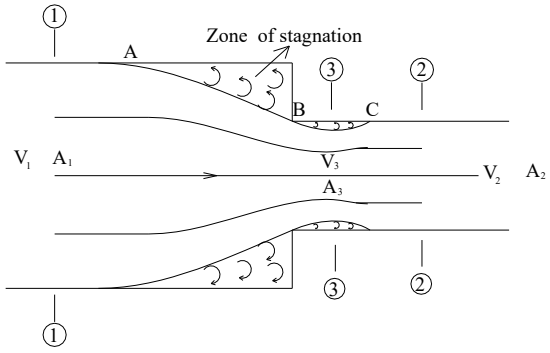


Figure 7.17 Sudden contraction

$$V_c = \frac{1}{C_c} \cdot V_2$$

So

$$\Delta h_d = \left(\frac{1}{C_c} - 1 \right)^2 \frac{V_2^2}{2g} \tag{7.73}$$

where

$$K_d = \left(\frac{1}{C_c} - 1 \right)^2 \tag{7.74}$$

$$\Delta h_d = K_d \frac{V_2^2}{2g} \tag{7.75}$$

where K_d: sudden contraction head loss coefficient.

Table 7.5 Sudden contraction head loss coefficient for different cross section for water

Contraction ratio, $\frac{A_2}{A_1}$	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
C_c	0.617	0.624	0.632	0.643	0.659	0.681	0.712	0.755	0.813	0.892	1.0
K_d	0.5	0.46	0.41	0.36	0.3	0.24	0.18	0.12	0.06	0.02	0

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7.3.2.4 Head Loss at the Entrance to a Pipeline from a Reservoir (Exit Loss) ($\Delta h_ç$)

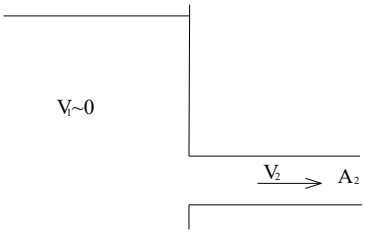


Figure 7.18 Entrance to a pipeline from a reservoir

A special case of a sudden contraction is encountered when a fluid enters into a pipe from a reservoir. In this case, a contraction ratio of zero may be assumed. The value of K depends on the conditions at the entrance to the pipe. The entrance may be squared, rounded, re-entrant or skewed.

It is possible to write $\Delta h_{ç} = 0.5\frac{V_2^2}{2g}$ for $A_1 >> A_2$, $A_2/A_1 \approx 0$, but by rounding the entrance to the pipeline the head loss could be varied. The head loss at the entrance to pipeline from a reservoir could be minimized by increasing the radius of curvature. The head loss could be obtained as minimum when radius of curvature is bigger than 0.14D.

Table 7.6 gives the head loss coefficients for different pipe entrances.

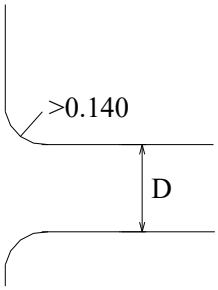
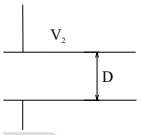
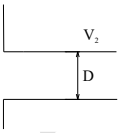
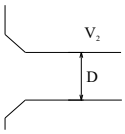
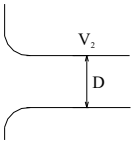


Figure 7.19 Transition to minimize the head loss

$$\Delta h_{ç} = K_{ç}\frac{V_2^2}{2g} \tag{7.76}$$

Table 7.6 Head loss coefficients K, for different pipe entrances

Figures	Head loss coefficient $K_ç$
 re-entrant	0.8-1.0
 squared	0.5
 skewed	0.25
 rounded	0.01-0.05

7.3.2.5 Head Loss in Pipe Fittings (Δh_v)

Additional head is lost in other pipe fittings such as valves, couplings, elbows, orifices, etc.

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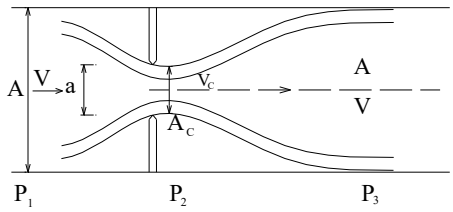


Figure 7.20 Orifice in a pipe

Orifice is an opening through which the fluid flows. It may be used for measuring the rate of flow out of a reservoir or through a pipe.

The head loss in an orifice may be expressed as

$$\Delta h_0 = \frac{(V_c - V)^2}{2g}$$

From the continuity equation

$A_c V_c = A.V$

since $A_c = C_c \cdot a$

a is the opening of the orifice, and C_c is contraction coefficient.

$$V_c = \frac{A}{A_c} V = \frac{A}{C_c \cdot a} V$$

The orifice head loss can be written as

$$\Delta h_0 = \frac{\left(\frac{A}{C_c\, a} V - V\right)^2}{2g} \qquad \text{or} \qquad \Delta h_0 = \left(\frac{A}{C_c\, a} - 1\right)^2 \frac{V^2}{2g}$$

where

$$K_0 = \left(\frac{A}{C_c a} - 1\right)^2$$

and is defined as orifice head loss coefficient.

$$\Delta h_0 = K_0 \frac{V^2}{2g} \tag{7.77}$$

Similarly, for a valve,

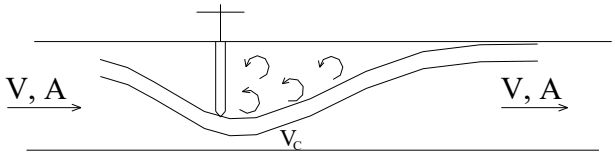


Figure 7.21 Valve

$$\Delta h_v = K_v \frac{V^2}{2g} \tag{7.78}$$

where K_v is the valve head loss coefficient.

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7.3.2.6 Head Loss in Bends (Δh_{dir})

When a fluid flows through a bend it suffers a head loss due to change in direction hence the increase of turbulence of flow. Consider a fluid flowing in a bend as shown in Fig. 7.22. The centrifugal forces cause the pressure to increase at the outer edge and decrease at the inner edge. The increase in pressure starts at point A, reaches a maximum at point B, and then decreases to normal. The decrease in pressure starts at C, reaches a minimum at D and then increases to normal. Thus, the fluid has to work against adverse pressure in the range from A to B and from D to E. This causes turbulent eddies and separation resulting in dissipation of energy as heat.

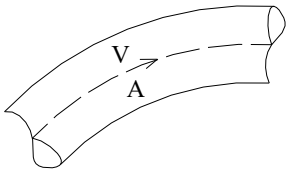


Figure 7.22 Bend

The head loss in a bend may be expressed as

$$\Delta h_d = K_d \frac{V^2}{2g} \tag{7.79}$$

in which K_d is called bend loss coefficient and depends on the radius of curvature and the angle of the bend.

Comments;
Although head loss due to changes of section, direction and the presence of obstructions, are termed minor head losses, they may be significant depending upon the nature of the flow system

All such losses are generally expressed as function of velocity head, or

$$\Delta h_g = K \frac{V^2}{2g} \tag{7.80}$$

where K= an experimentally determined coefficient. For all the equations given in the form of Eq. (7.80) the velocity is used where maximum $V=V_{max}$.

The Energy Variation in a reservoir-pipe system

As it is seen in Fig. 7.23 the below steps are followed to draw the variation of the energy grade line and hydraulic grade line because of the local losses and the friction losses.

- At the starting point $V^2/2g$, p/γ and z are examined and by using these values initial total energy is obtained.
- After the starting point, Δh head loss is subtracted from the previously obtained energy at each node and by connecting each node, energy grade line is obtained.
- If the head losses are minor head losses energy grade line decreases suddenly, if not decreases linearly.
- The hydraulic grade line at each point is obtained by subtracting the velocity head $V^2/2g$ from the energy grade line.
- Hydraulic grade line coincides with water surface in channels and reservoirs.
- If the reservoirs are big enough to assume the velocity is zero, the energy grade line coincides with the water surface.

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- The hydraulic grade line passes through axis of the pipeline, if one end of the pipe is open to atmosphere.
- At sudden expansion regions of the pipes, the hydraulic grade line doesn't decrease but it increases.

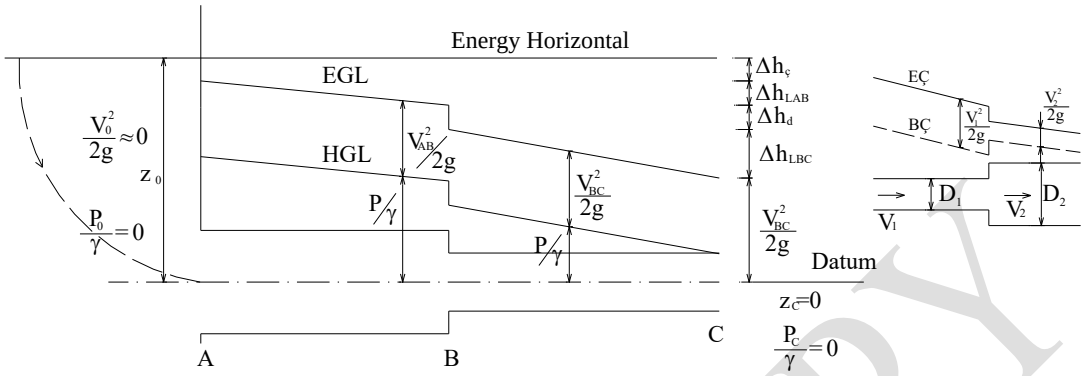


Figure 7.23 Energy variations in a reservoir-pipe system

The energy equation between O-C

$$\frac{V_0^2}{2g}^{0}+\frac{P_0}{\gamma}^{0}+z_0=\frac{V_{BC}^2}{2g}+\frac{P_C}{\gamma}^{0}+z_c^{0}+\Delta h_ç+\Delta h_{LAB}+\Delta h_d+\Delta h_{LBC}$$

$$z_0=\frac{V_{BC}^2}{2g}+K_ç\frac{V_{AB}^2}{2g}+\lambda_{AB}\frac{L_{AB}}{D_{AB}}\frac{V_{AB}^2}{2g}+K_d\frac{V_{BC}^2}{2g}+\lambda_{BC}\frac{L_{BC}}{D_{BC}}\frac{V_{BC}^2}{2g} \qquad (7.81)$$

7.4 Various Pipe Problems

The variation of energy in different pipe systems will be examined

1- Uniform Flow in a Constant Diameter Pipe Exiting from a Reservoir

a) Single Straight Pipeline

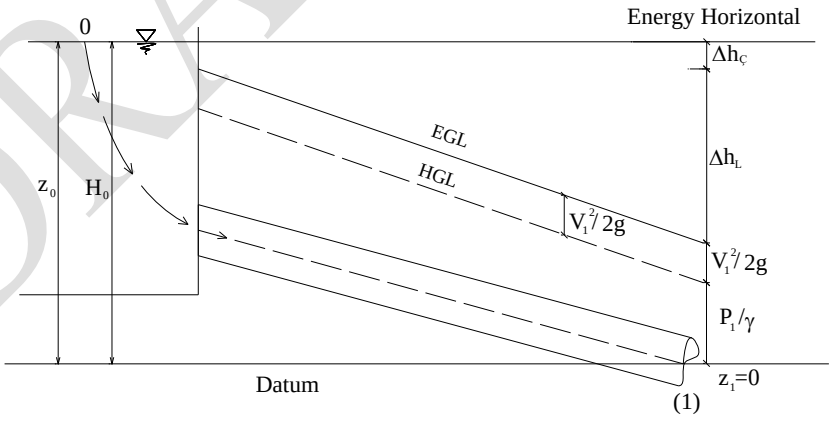


Figure 7.24 Straight pipeline exiting from a reservoir

As it is seen from Fig. 7.24, the energy equation is written between points (0) which is the reservoir free surface and node (1) in anywhere in a pipeline

$$z_0=\frac{V_1^2}{2g}+\frac{p_1}{\gamma}+\Delta h_L+\Delta h_ç \qquad (7.82)$$

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where V is the average velocity.

b) Pipeline Curved in Vertical Axis

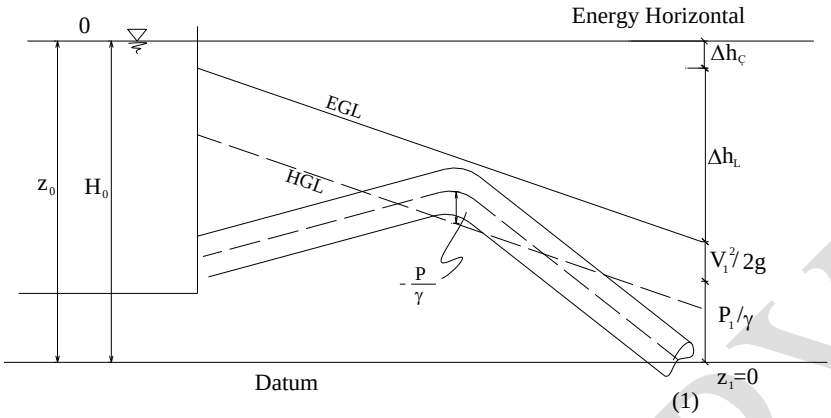


Figure 7.25 Reservoir curved pipeline

The energy equation is the same as Eq. (7.82), but since the hydraulic grade line intersects with the pipeline at points A and B, in the region in between these points pressure decreases to negative values and causes cavitation (Fig. 7.25).

When a liquids flow into a region where its absolute pressure falls to the vapor pressure of the liquid at the operating temperature, it boils and vapor bubbles form and then they suddenly collapse as the fluid moves into a region of higher pressure. This process is called cavitation. Cavitation causes physical damage. If the vapor bubbles are near to a solid boundary when they collapse, the forces exerted by the liquid into the cavities create very high-localized pressures that cause pitting of solid surface. The phenomenon is accompanied by noise and vibrations that have been described as similar to gravel going through a centrifugal pump.

Cavitation also causes reduction of discharge. Since the vapor pockets will cause to decrease the cross-section of the pipe the design discharge cannot be obtained. The minimum positive pressure should be obtained in such regions of the pipelines. Air vents are used to reach the pressure to atmospheric pressure (Fig. 7.26).

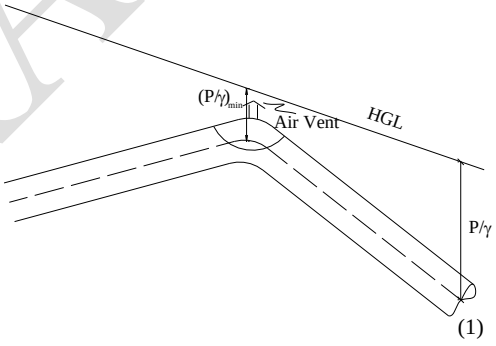


Figure 7.26 Hydraulic grade line in the curved region

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2- Flow between Reservoirs

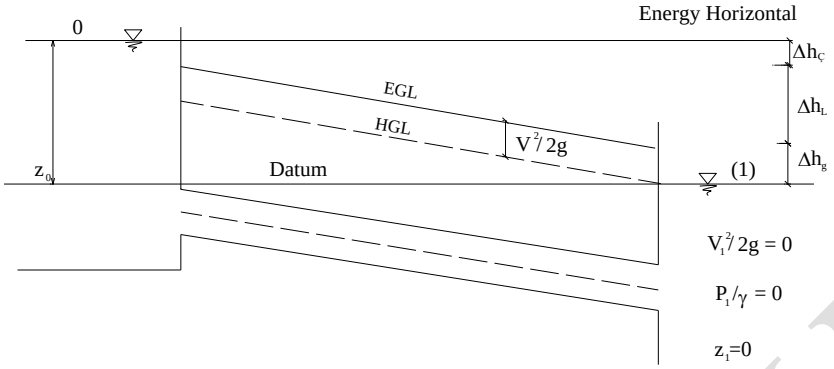


Figure 7.27 Flow between reservoirs

From Fig. 7.27 the energy equation between water surfaces (0) and (1) is written as;,

$$z_0 = \Delta h_ç + \Delta h_L + \Delta h_g \tag{7.83}$$

or

$$z_0 = \Sigma \Delta h \tag{7.84}$$

In this type of problem, the flow between two reservoirs exists only when the total energy loss is equal to the elevation difference of reservoirs.

3- Long Pipeline

When the length of the pipeline is very big than the pipe diameter, the velocity head $V^2/2g$ could be ignored. In this case the local head losses could be ignored as it is seen from Fig. 7.28 the energy grade line and hydraulic grade line are superimposed. The single grade line is commonly referred as the hydraulic grade line.

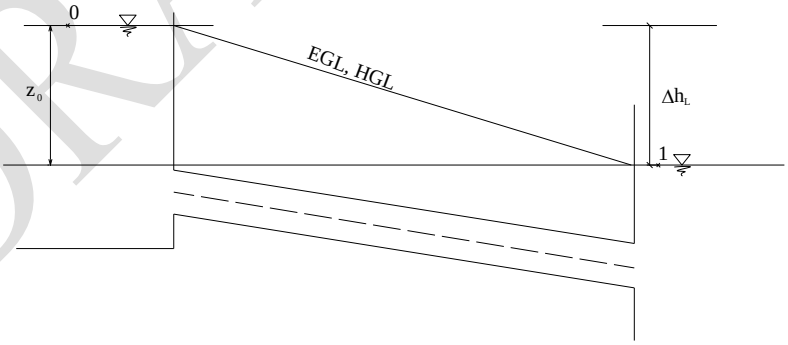


Figure 7.28 Hydraulic and energy grade line for long pipeline

$$z_0=\Delta h_L \tag{7.85}$$

In this case, the elevation difference of reservoirs is equal to the friction loss.

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4- Pipes in Series

Two or more pipes are said to be connected in series if the head of one is connected to the tail of the other and so on. Two important features of pipes connected in series are:

1) The head loss is cumulative- i.e., the total head loss is equal to the sum of the head losses in individual pipes.

The energy equation for the pipes in series in Fig. 7.29 is

$$z_0 = \Delta h_{ç} + \Delta h_{LAB} + \Delta h_a + \Delta h_{LBC} + \Delta h_d + \Delta h_{LCD} + \Delta h_g \quad (7.86)$$

$$z_0 = \Sigma \Delta h \quad (7.87)$$

The energy equation is the same as the other reservoir-pipe systems.

2) The discharge through each pipe is the same.

If the discharge is sought there are three unknowns in this equation. The unknowns are the velocities in each pipe. So, the continuity equation is written;

$$Q = V_{AB}A_{AB} = V_{BC}A_{BC} = V_{CD}A_{CD} \quad (7.88)$$

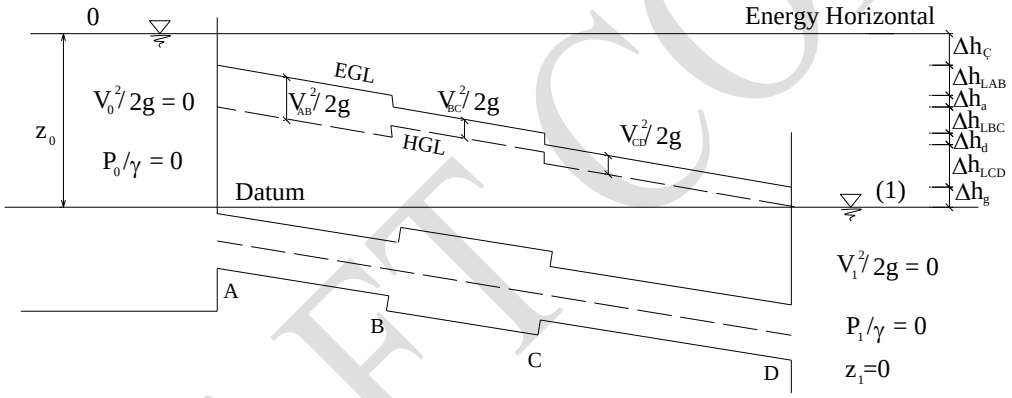


Figure 7.29 Pipes in series

All series pipe problems may fall into the following two categories:

1. Given the flow rate and pipe combinations, determine the total head loss.
2. Given the allowable head loss and the combination of the pipes, determine the flow rate.

5- Pipes in Parallel

A parallel pipe system consists of a combination of two or more pipes connected in such a way that the flow is divided among the pipes and *then is joined again. The junction at which the pipes branch out is called the* diverging node and the junction at which the pipes meet again is called *the converging node*. A parallel pipe system is shown in Fig. 7.30. Two important features of pipes connected in parallel are:

1. The head loss (pressure drop) through each pipe is the same, and is equal to the difference in the total energy between the diverging node and the converging node.
2. The discharge Q is cumulative-i.e., the discharge Q at point A will be distributed among the n parallel pipes in such a way that $Q = \sum_n Q_n$.

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In a parallel pipe problem, it is always advisable to first express all branches in terms of equivalent pipes so that only head losses due to friction are considered. Otherwise, the problems become complicated and difficult to solve.

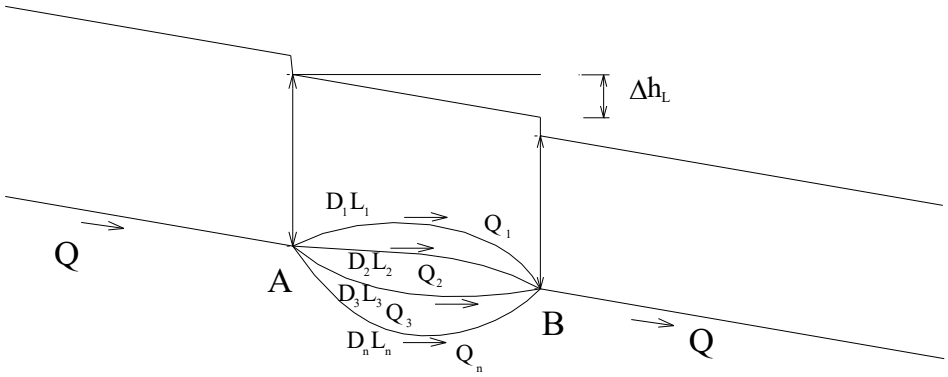


Figure 7.30 Pipes connected in parallel

In the parallel connected pipe system seen in Fig. 7.30:

a) Since the total energy between A and B is constant, the head loss through each pipe is the same.

$$\Delta h_{L1} = \Delta h_{L2} = \Delta h_{L3} = ... = \Delta h_{Ln} \tag{7.89}$$

b) Continuity equation

$$Q = Q_1 + Q_2 + Q_3 + ... + Q_n \tag{7.90}$$

c) Energy equation (the local head losses are ignored)

$$z_A + \frac{p_A}{\gamma} + \frac{V_A^2}{2g} = z_B + \frac{p_B}{\gamma} + \frac{V_B^2}{2g} + \Delta h_L \tag{7.91}$$

or for long pipe system;

$$z_A + \frac{p_A}{\gamma} = z_B + \frac{p_B}{\gamma} + \Delta h_L \tag{7.92}$$

(Note: In (a) there is only one continuous head loss.

These types of problems are solved by the help of above three equations.

6- Branching Pipes

A pipe system in which fluid is brought to a junction where three or more pipes meet is termed as a branching pipe system (Fig. 7.31). There are two important features of a branching pipe system:

1. The total amount of fluid entering into the junction must be equal to the total amount of fluid leaving the junction- i.e., the algebraic sum of all the flows at the junction must be zero, or $\sum Q = 0$;

2. All the pipes that meet at the junction (D) have the same pressure at the junction.

The direction of flow in a branch is determined by the relative positions of the total energy line at the two ends of the branch. For example, for the branching pipe system shown in Fig. 7.31, the total energy at junction is given by the elevation of the fluid in the manometer connected at junction D, and the total energy at the surface of each reservoir is given by the elevation of the fluid level in that

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reservoir. Therefore, fluid will flow from A to D; D to B and D to C. A typical branching pipe problem would be to determine the flow in each branch.

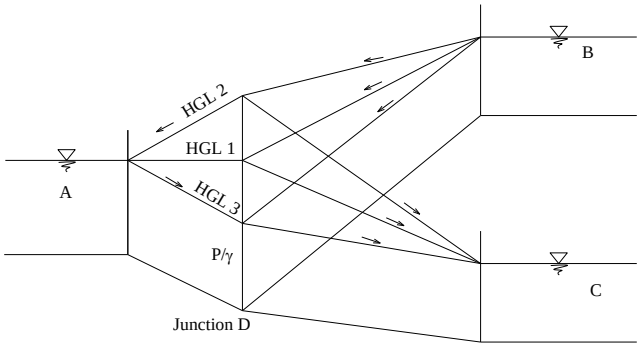


Figure 7.31 A branching pipe system

In three reservoir junction case the most important problem is to know the flow directions. If the pressure head is known at the junction, the flow directions could be easily obtained but on the contrary below steps must be followed.

a) Assume height of the total energy line at D (generally assumed as equal to the energy of reservoir A)

$$z_A = z_D + \frac{p_D}{\gamma} \tag{7.93}$$

in this case

$$Q_{BD} = Q_{DC} \tag{7.94}$$

if not, two conditions might be

i) If $Q_{BD} > Q_{DC}$

The flow is to C from A and B, and the continuity equation is written as

$$Q_{BD} = Q_{DA} + Q_{DC} \tag{7.95}$$

which means the head of junction D is higher than the elevation of reservoir A

$$z_D + \frac{p_D}{\gamma} = z_A + \Delta h_{LDA} \quad \text{or} \tag{7.96}$$

$$z_B = z_A + \Delta h_{LBD} + \Delta h_{LDA} \quad \text{and} \quad z_B = z_C + \Delta h_{LBD} + \Delta h_{LDC} \tag{7.97}$$

ii) If $Q_{BD}<Q_{DC}$

The flow is from C to A and B, and the continuity equation is written as;

$$Q_{DC}=Q_{BD}+Q_{AD} \tag{7.98}$$

which means the total head of D is lower than the elevation of reservoir A

$$z_A = \frac{p_D}{\gamma} + z_D + \Delta h_{LAD} \quad \text{or} \tag{7.99}$$

$$z_B = z_C + \Delta h_{LBD} + \Delta h_{LDC} \quad \text{and} \quad z_A = z_C + \Delta h_{LAD} + \Delta h_{LDC} \tag{7.100}$$

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7- Simple Pump in a Pipeline

Now we shall consider flow in which the head is developed by a pump. However, the head produced by a centrifugal pump is function of the discharge. Hence a direct solution is usually not immediately available.

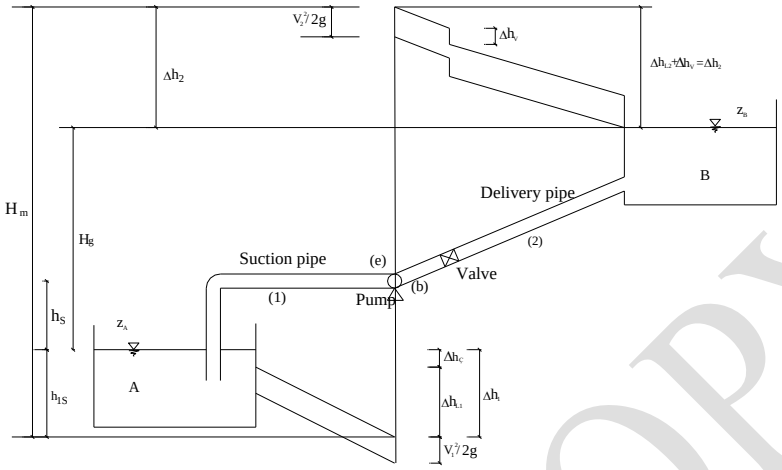


Figure 7.32 Pump and pipeline combination

The parameters used in Fig. 7.32 are expressed as below;

V_1=velocity in suction pipeline (V_s)

V_2=velocity in delivery pipe (V_d)

Δh_2=head loss in delivery pipe (friction, valves, etc.)

Δh_1=total head loss in suction pipeline

H_m=manometric head

H_g=static lift (elevation head)

If energy equation is written between the just right and left hand of the pump

$$H_m = \frac{p_d}{\gamma} + \frac{V_d^2}{2g} - \left(\frac{p_s}{\gamma} + \frac{V_s^2}{2g} \right) \tag{7.101}$$

Equation between reservoir A and left hand of the pump

$$\frac{p_s}{\gamma} = z_A - \frac{V_s^2}{2g} - \Delta h_1 \tag{7.102}$$

Equation between reservoir B and right hand of the pump

$$\frac{p_d}{\gamma} = z_B + \Delta h_2 - \frac{V_d^2}{2g} \tag{7.103}$$

Equations (7.102) and (7.103) are substituted into (7.101)

$$H_m = z_B - z_A + \Delta h_1 + \Delta h_2 \tag{7.104}$$

or

$$H_m = H_g + \Delta h_1 + \Delta h_2 \tag{7.105}$$

Which means manometric height is the summation of elevation head and the total losses.

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For long pipelines Fig 7.33 can be given as an example

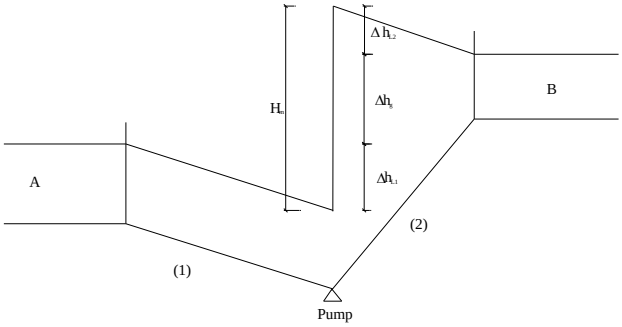


Figure 7.33 Pump in a long pipeline

The head losses within the pump itself are not included because such losses will affect efficiency of the pump. The power of pump which gives discharge Q;

$$N = \frac{\gamma(kN/m^3) * H_m(m) * Q(m^3/s)}{\eta} \quad (kW) \tag{7.106}$$

Here η is called as efficiency. The variation of efficiency is shown in Fig. 7.34

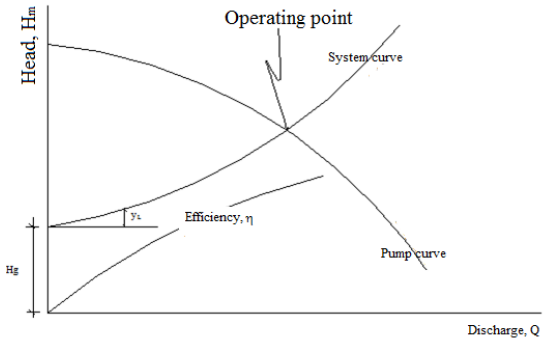


Figure 7.34 Pump and system curve

Hence for a given discharge, a certain head H_m must be supplied to maintain that flow. Thus, we can construct a head-versus-discharge curve is called as system curve. Any given centrifugal pump has a head-versus-discharge curve that is characteristic of that pump at a given pump speed. Such curves are supplied by the pump manufacturer; a typical one for a centrifugal pump is shown in Fig. 7.34.

Fig. 7.34 reveals that, as the discharge increases in a pipe, the required energy for flow also increases. However, the head that is produced by the pump decreases as the discharge increases. Consequently- the two curves intersect, and the operating point is at the point of intersection- that point where the head produced by the pump is just the amount needed to overcome the head loss in the pipeline.

The total head-discharge curves for a centrifugal pump can generally be expressed in the functional form

$$H_m = AQ^n + BQ + C$$

The coefficients A, B and C can be evaluated by taking three pairs of H_m and Q from a particular curve and solving equation above.

The civil engineer is mostly concerned with pump selection, in the design of pumping stations, and therefore the design of the shape of pump impellers will not be dealt with here.

1) Pump (conducts energy to fluid)

2) Motor

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3) Junctions

Pumps are classified into two groups according to the conduction of energy either continuous or alternating.

1) Volumetric Type

2) Turbine Type

a) Radial pumps (centrifugal pumps), H_m>60 m, N_s<2600

b) Semi-axial pumps 20 m<H_m<60 m, 2600<N_s<5000

c) Axial pumps 20 m>H_m, 5000<N_s<10000

For the same power input end efficiency the centrifugal type would generate a relatively large pressure head with a low discharge, the propeller type a relatively large discharge at a low head with the mixed flow having characteristics somewhere between the other two.

Pump types may be more explicitly defined by parameter called Specific Speed (N_s) expressed by

$$N_s = \frac{N\sqrt{Q}}{H_m^{3/4}} \tag{7.107}$$

where Q is the discharge, H_m the total head and N rotational speed (rev/min).

Multiple Pump Systems

In many cases it is not possible to obtain the desired discharge or head with a single pump. So, we have to use more than one pump. But according to the needs we use pumps as connected parallel or series.

a) Parallel Operation: Pumping stations frequently contain several pumps in a 'parallel' arrangement. In this configuration (Fig. 7.35) any number of the pumps can be operated simultaneously, the objective being to deliver a range of discharges. This is a common feature of sewage pumping stations where the inflow rate varies during the day. By automatic switching according to the level in the suction well any number of the pumps can be brought into operation (Featherstone and Nalluri, 1984).

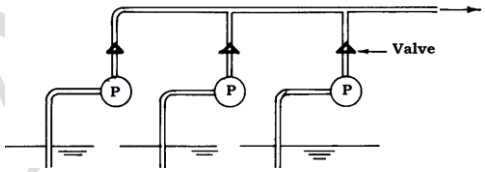


Figure 7.35 Pumps operating in parallel

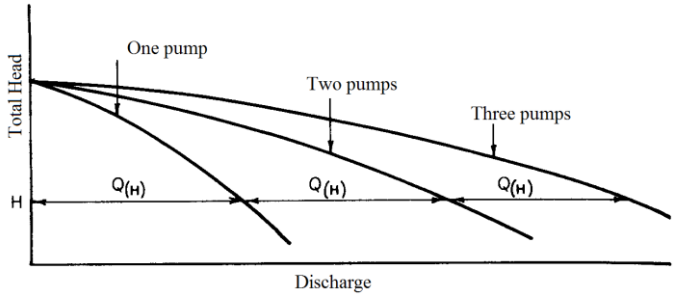


Figure 7.36 Characteristic curves for identical pumps operating in parallel.

b) Series operation: This configuration is the basis of multistage and borehole pumps; the discharge from the first pump (or stage) is delivered to the inlet of the second pump, and so on. The same discharge passes through each pump receiving a pressure boost in doing so. Fig. 7.37 shows the series configuration together with the resulting head V discharge characteristics, which are clearly obtained

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by adding the individual pump manometric heads at any arbitrary discharges. Note that, of course, all pumps in a series system must be operating simultaneously (Featherstone and Nalluri, 1984).

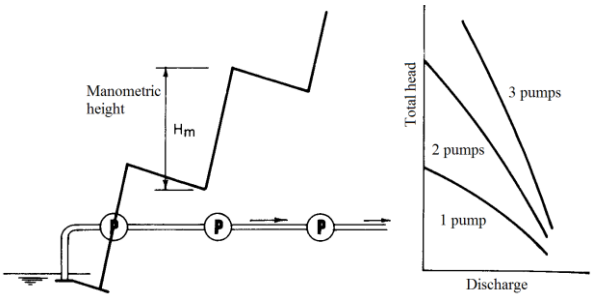


Figure 7.37 Pumps operating in series

Variable Speed Pump Operation

By the use of variable speed motors the discharge of a single pump can be varied to suit the operating requirements of the system (Featherstone and Nalluri, 1984).

Using dimensional analysis and dynamic similarity criteria it can be shown that if the pump delivers Q_1 at manometric head H_1 when running at speed N_1, the corresponding values when the pump is running at speed N_2 are given by relationships

$$Q_2 = Q_1\left(\frac{N_2}{N_1}\right) \tag{7.108}$$

$$H_2 = H_1\left(\frac{N_2}{N_1}\right)^2 \tag{7.109}$$

In constructing the characteristic curve for speed N_2, several pairs of values of Q_1, H_1 from the curve, N_1 can be obtained and transformed into homologous points Q_2, H_2 on the N_2 curve (Fig. 7.38).

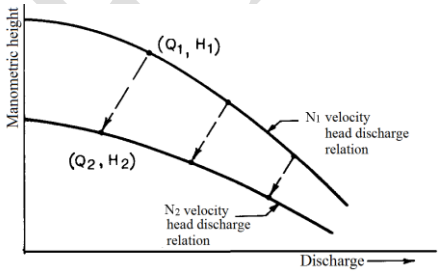


Figure 7.38 Effect of speed change on pump characteristics

Suction Lift Limitations

Cavitation, the phenomenon which consists of local vaporization of a liquid and which occurs when the absolute pressure falls to the vapor pressure of the liquid at the operating temperature, can occur at the inlet to a pump and on the impeller blades, particularly if the pump is mounted above the level in the suction well. Cavitation causes physical damage, reduction in discharge and noise, and to avoid it the pressure head at inlet should not fall below a certain minimum which is influenced by the further reduction in pressure within the pump impeller (Fig. 7.39).

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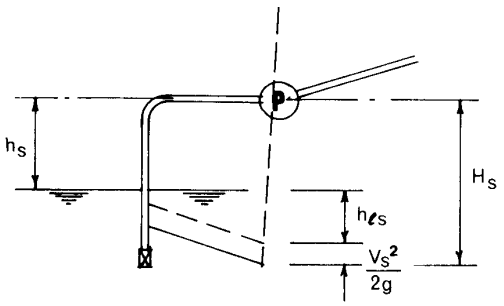


Figure 7.39 Head conditions in suction pipe

If p_s represents the pressure at inlet then $\frac{p_s - p_v}{\rho g}$ is the absolute head at the pump inlet above the vapor pressure (p_v) and is known as the net positive suction head, NPSH.

Thus $$NPSH = \left(\frac{p_s - p_v}{\rho g}\right) = \frac{p_0}{\rho g} - Hs - \frac{p_v}{\rho g} \quad (7.110)$$

Where

p_0 =ambient atmospheric pressure

H_s =manometric suction head ($= h_s + h_{1s} + \frac{V_s^2}{2g}$),

h_s =suction lift,

h_{1s} =total head loss in suction pipe,

V_s =velocity head in suction pipe.

Values of NPSH (*Net Positive Suction Head*) can be obtained from the pump manufacturer and are derived from full- scale or model tests; these values must not be exceeded if cavitation is to be avoided.

Thoma introduced a cavitation number (index) $\sigma \left(= \frac{NPSH}{H_m}\right)$ and from physical tests found this to be strongly related to specific speed.

7.5 Pipe Network Analysis – Hardy Cross Method

Interconnected pipes through which the flow to a given outlet may come from several circuits are called a *network of pipes*, in many ways analogous to flow through electric networks. Problems on these in general are complicated and require trial solutions in which the elementary circuits are balanced in turn until all conditions for the flow are satisfied. A point at which more than one pipe meet is called a node. A closed conduit consisting of more than one pipe is called a loop. A loop starts from a node and terminates on the same node (see Fig.40).

The followings must be satisfied in a network of pipes:

1. The algebraic sum of discharges entering and leaving a node is zero.
2. The algebraic sum of the total head loss around each circuit must be zero (continuity equation).
3. The Darchy-Weisbach equation, or equivalent exponential friction formula, must be satisfied for each pipe; i.e., the relation between head loss and discharge must be maintained for each pipe.

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Hardy-Cross method uses successive approximations based on conditions stated above. The head loss in each pipe can be calculated using either the Darchy-Weisbach equation or the Hazen-Williams formula. The Hazen-Williams equation is good for water only. Minor losses should be included as equivalent lengths in each pipe. If the Hazen-Williams equation is used, a formula for calculating the correction can be obtained as follows:

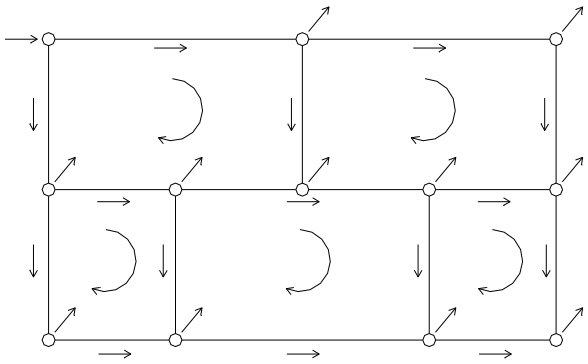


Figure 7.40 Network

Steps in an arithmetic procedure may be itemized as follows:

1– Assume the best distribution of flows that satisfies continuity $\Sigma Q = 0$ at each node by careful examination of the network.

2– For each pipe head loss is calculated $\Delta h_L = r.Q^n$. The value of r is constant in each pipeline (for Darcy–Weisbach $n = 2$, for Williams – Hazen $n = 1.85$).

3– For each circuit sum the net head loss $\left(\Sigma\Delta h_L = \Sigma r.Q^n\right)$.

If $\sum \Delta h_L = 0$ the assumed discharges are correct.

If $\sum \Delta h_L \neq 0$ the assumed discharges are not correct, and for each circuit the discharges should be corrected by ΔQ .

4– ΔQ correction discharge is computed as below;

For each pipeline the head loss

$$\Delta h_L = r.(Q_0 + \Delta Q)^n \tag{7.111}$$

and

$$\Delta h_L = r.\left(Q_0^n + n.Q_0^{n-1}.\Delta Q + n.(n-1).Q_0^{n-2}.\Delta Q^2 + ...\right) \tag{7.112}$$

If ΔQ is small compared with Q_0, all terms of the series after the second may be dropped. Now for a circuit,

$$\Sigma\Delta h_L = \Sigma r.Q^n = \Sigma r.Q_0^n + \Delta Q.\Sigma r.n.Q_0^{n-1} = 0 \tag{7.113}$$

ΔQ is obtained from Eq. (7.116)

$$\Delta Q = -\frac{\Sigma r.Q_0^n}{\Sigma r.n.Q_0^{n-1}} \tag{7.114}$$

or

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$$\Delta Q = -\frac{\Sigma \Delta h_L}{n.\Sigma \Delta h_L / Q_0} \tag{7.115}$$

is obtained.

In this expression sign is important in the numerator but it is not important in denominator. When ΔQ is applied to each pipe in a circuit, the directional sense is important; i.e., it adds to flows in the clockwise direction and subtracts from flows in the counterclockwise direction.

$$Q = Q_0 + \Delta Q \tag{7.116}$$

Q_0 ; Assumed initial discharge, Q; Corrected discharge

For the pipelines of the neighboring circuits, the correction is to make by taking the difference of each correction discharge.

5– Procedures are repeated until the correction ΔQ 's are arbitrarily small.

7.6 Flow Measurement through Pipes

Application of continuity, energy and momentum equations to a given system of fluid flow makes velocity and volume measurements possible (Featherstone and Nalluri, 1984).

1- Venturi metre and Orifice meter

A pressure difference is created along the flow by providing either a gradual (venturimeter) or sudden (orifice plate meter) constriction in the pipeline, and is related to flow velocities and discharge by the energy and continuity principles (Fig.7.41). The pressure differences are obtained by the help of manometers and piezometers.

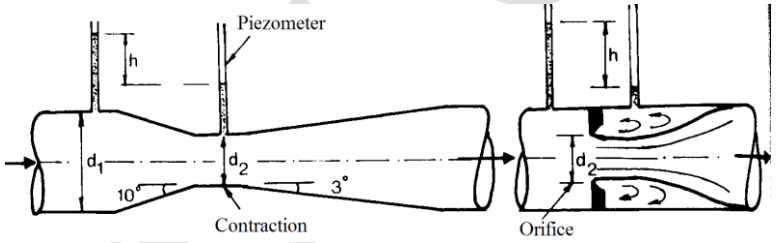


Figure 7.41 Discharge measurement through pipes

Writing Bernoulli's equation and continuity equation between inlet section and constriction Eq. (4.21) is obtained (local losses are ignored). This equation can only be written as below;

$$Q = V_1 A_1 = \frac{A_1 A_2 \sqrt{2gh}}{\sqrt{A_1^2 - A_2^2}} \tag{7.117}$$

or

$$Q = A_1 \sqrt{\frac{2gh}{k^2 - 1}} \tag{7.118}$$

where $k = A_1/A_2$

Equation (7.120) is an ideal equation obtained by neglecting all losses. The actual discharge is, therefore, written by introducing a coefficient C_D in equation (7.120)

$$Q = C_D A_1 \sqrt{\frac{2gh}{k^2 - 1}} \tag{7.119}$$

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The numerical value of C_D, the coefficient of discharge, will depend upon the ratio A_1/A_2, type of transition, velocity and viscosity of the flowing fluid.

The gradual transition of the venturi meter between its inlet and outlet induce least amount of losses and the value of CD lies between 0.96 and 0.99 for turbulent flows (Featherstone and Nalluri, 1984).

The transition in the case of an orifice plate meter is sudden and hence the flow within the meter experiences greater losses due to contraction and expansion of the jet through the orifice. Its discharge coefficient has a much lower value (0.6 to 0.63) as the area A_2 in Equation (7.119) refers to the orifice and not to the contracted jet.

The reduction in the constriction diameter causes velocity to increase, and correspondingly a large pressure differential is created between inlet and constriction, thus enabling greater accuracy in its measurement. High velocities at the constriction cause low pressures in the system and if these fall below the vapor pressure limit of the fluid, cavitation sets in which is highly undesirable. Therefore, the selection of the ratio D_2/D_1 is to be considered carefully. This ratio may be kept between 1/3 and 3/4 and a more common value is ½.

2- Pitot Tube

A Pitot tube in its simplest form is an L-shaped tube (tube is bent by 90°) held against the flow as shown in Fig 7.42, creating a stagnation point in the flow. By using Eq. (7.119a) flow velocity is obtained but the actual velocity will be slightly less than the velocity given Eq. (7.119a) and introducing a coefficient, K, modifies the actual velocity

$$V = K\sqrt{2gh} \tag{7.119a}$$

K is usually between 0.95 and 1.0.

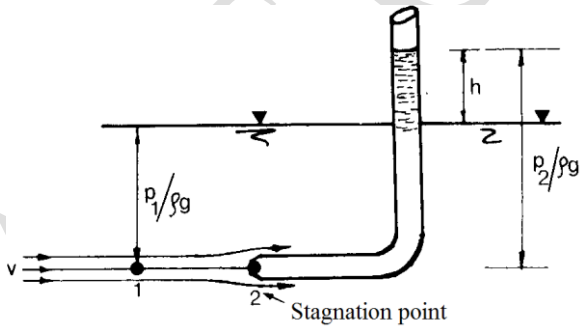


Figure 7.42 Pitot tube

3- Flow Measurement Through Orifices and Mouthpieces

(a) Small Orifice

If the head, h, causing flow through an orifice of diameter, d, is constant (small orifice: h>>d) as shown (Fig. 7.43), by Bernoulli's equation (ignoring losses)

$$h + \frac{p_1}{\rho g} + \frac{V_1^2}{2g} = 0 + \frac{p_2}{\rho g} + \frac{V_2^2}{2g} \tag{7.120}$$

$V_2^2/2g = h$

Or the velocity through the orifice

$$V_2 = \sqrt{2gh} \tag{7.121}$$

Equation 7.123 is called Toricelli's theorem and the velocity is called the theoretical velocity

$$V_2 = C_V\sqrt{2gh} \tag{7.122}$$

where C_v is the coefficient of velocity defined as:

$$C_V = \frac{\text{Actual velocity}}{\text{Theoretical velocity}} \tag{7.123}$$

The jet area is much less than the area of the orifice due to contraction, and the corresponding coefficient of contraction is defined as

$$C_C = \frac{\text{Area of jet}}{\text{Area of orifice(a)}} \tag{7.124}$$

At a section very close to the orifice, known as the v*ena-contracta*, the velocity is normal to the cross-section of the jet and hence the discharge can be written as:

Q = Area of jet ×velocity of jet (at *vena contracta*) (7.125)

$$Q = C_C \times a \times C_V \times \sqrt{2gh} \tag{7.126}$$

$$Q = C_D \times a \times \sqrt{2gh} \tag{7.127}$$

where C_D is called the coefficient of discharge and defined as:

$$C_D = \frac{\text{actual discharge}}{\text{theoretical discharge}} = C_C \times C_D \tag{7.128}$$

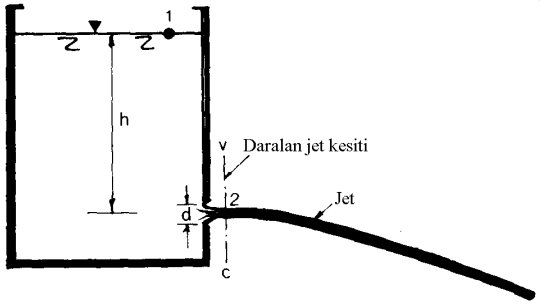


Figure 7.43 Small orifice (h>>d)

Some typical orifices and mouthpieces (short pipe lengths attached to orifice) and their coefficients, C_C,C_V and C_D are shown in Fig. 7.44.

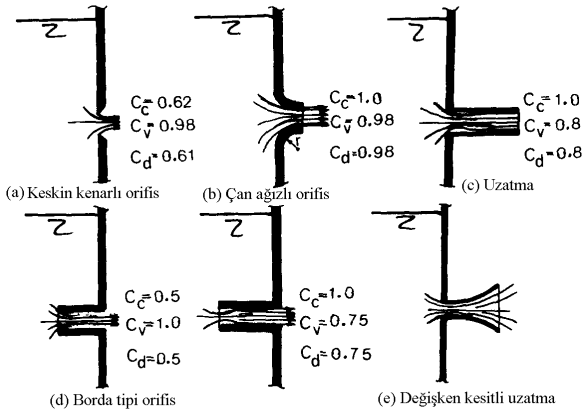


Figure 7.44 Hydraulic coefficients for some typical orifices and mouthpieces

(b) Large Rectangular Orifice

As the orifice is large the velocity across the jet is no longer constant; however, if we consider a small area bdh, at a depth, h,

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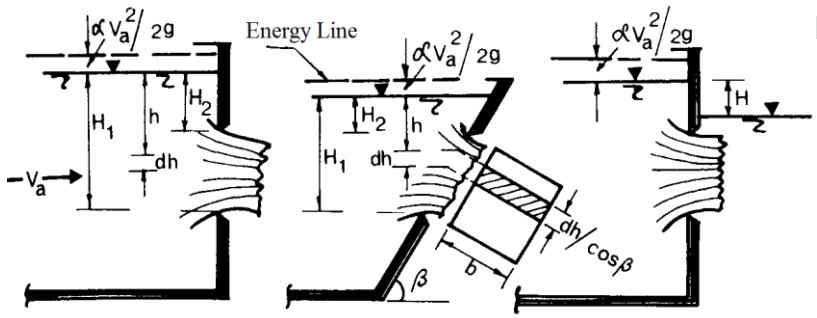
dq = C_d×area of strip × velocity through the strip

$$= C_d b dh\sqrt{2gh} \tag{7.129}$$

Total discharge through the entire opening

$$Q = \int dq = C_d b\sqrt{2g}\int_{H_1}^{H_2} h^{1/2} dh$$

$$Q = \frac{2}{3} Cd\sqrt{2g} \times b \times \left(H_1^{3/2} - H_2^{3/2}\right) \tag{7.130}$$



a. Orifice with approach velocity b. Orifice in inclined wall c. Submerged orifice

Figure 7.45 Velocity of approach- large rectangular orifice

If V_a is the velocity of approach, the head responsible for the strip velocity is

$$h + \alpha\frac{V_a^2}{2g}$$

and hence the strip velocity is

$$\sqrt{2g\left(h + \alpha V_a^2/2g\right)}$$

α is the kinetic energy correction factor (Coriolis coefficient)

Discharge through the strip, $dq = C_d b dh\sqrt{2g\left(h + \alpha V_a^2/2g\right)}$

And the total discharge

$$Q = \frac{2}{3} C_d\sqrt{2g} \times b \times \left[\left(H_1 + \alpha V_a^2/2g\right)^{3/2} - \left(H_2 + \alpha V_a^2/2g\right)^{3/2}\right] \tag{7.131}$$

Side wall of the tank inclined at an angle β

The effective strip area=bdh/cosβ

$$Q = dq = \int_{H_2}^{H_1} C_d \frac{bdh}{\cos\beta}\sqrt{2g\left(h + \alpha V_a^2/2g\right)}$$

$$Q = \frac{2}{3} C_d\sqrt{2g}\frac{b}{\cos\beta}\left[\left(H_1 + \alpha V_a^2/2g\right)^{3/2} - \left(H_2 + \alpha V_a^2/2g\right)^{3/2}\right] \tag{7.132}$$

In case of submerged orifice (Fig 7.45c):

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It can be shown by the Bernoulli's equation that the velocity across the jet is constant and equal to $\sqrt{2gH}$ or $\sqrt{2g\left(h+\alpha V_a^2/2g\right)}$ if V_a is considered.

$$Q = C_d A\sqrt{2g\left(H+\alpha V_a^2/2g\right)} \tag{7.133}$$

Note: Discharge under varying head: Since the head causing flow is varying, the discharge through the orifice varies with time.

If h is the head at any instant t, the velocity through the orifice at that instant $\sqrt{2gH}$

Volume reduced = Volume escaped through the orifice

$$-Adh = C_d a\sqrt{2gh}dt$$

$$dt = -\frac{A}{C_d a\sqrt{2g}}\frac{dh}{h^{1/2}} \tag{7.134}$$

Time taken to lower the water level from H_1 to H_2

$$T = \int dt = \frac{2\times A\times\left(H_1^{1/2}-H_2^{1/2}\right)}{C_d\times a\times\sqrt{2g}} \tag{7.135}$$

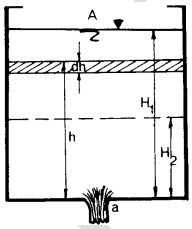


Figure 7.45 Time of emptying a tank

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8
Steady Flows in Open Channel

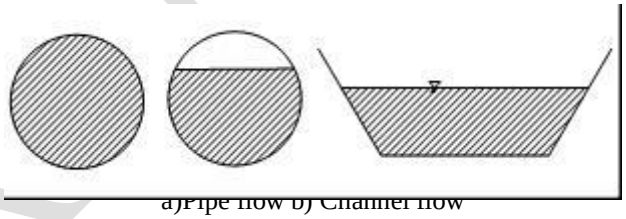
8.1 Introduction

The flow of water in a conduit may be either ***open-channel flow*** or ***pipe flow.*** The two kinds of flow are similar in many ways but differ in one important respect. Open-channel flow involves the flow of a liquid in a channel or conduit that is not completely filled which means open-channel flow must have a free surface whereas pipe flow has none, since the water must fill the whole conduit. There exists a free surface between the flowing fluid (usually water) and fluid above it (usually the atmosphere) in an open-channel flow. The main driving force for such flows is the fluid weight-gravity forces the fluid to flow downhill. Pipe flow, being confined in a closed conduit, exerts no direct atmospheric pressure but hydraulic pressure only. Therefore, unlike full pipe flow, where the pressure is normally above atmosphere pressure, but sometimes below it, the pressure on the surface of the liquid in open channel flow is always that of the ambient atmosphere.

Flow in rivers, drainage ditches; gutters along roads, channels and sewers not flowing full are examples to open channel flow.

There can be no pressure force driving the fluid through the channel or conduit. Any attempt to impose a pressure gradient in the flow direction is met with failure because of the negligible inertial and viscous effects of the gas (atmosphere) above the flowing fluid. For steady, fully developed channel flow, the pressure distribution within the fluid is merely hydrostatic.

The main characteristic of open channel flow is the ability of change of free surface, this characteristic causes hydraulic drop, hydraulic jump. Hence, examination of open channel flow is quite different and more complex.



a)Pipe flow b) Channel flow

Figure 8.1

8.1.1 Channel Geometry and Elements

The definitions of most geometric elements of basic importance of Fig. 8.2 are given below.

The depth of flow y is the vertical distance of the lowest point of a channel section from the free surface. In most cases, this terminology is used interchangeably with the terminology *depth of flow section* h, which is the depth measured perpendicular to the channel section. The relation between y and h is $y = \frac{h}{\cos\theta}$ where θ is the angle of the channel bottom with a horizontal line. If θ is small y~h.

The ***stage*** is the elevation or vertical distance of free surface above a datum. If the lowest point of the channel is chosen as the datum, the stage is identical with the depth of flow.

The water (flow) area A is the cross-sectional area of the flow normal to the direction of flow. But in some regions of channel flow, if the cross-section expands suddenly the kinetic energy of flow

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decreases and because of the pressure the flow turns back in expansion regions. This expansion region is called as eddy zone.

The top width B is the width of channel section at the free surface.

The wetted perimeter, P is the length of the line of intersection of the channel surface with a cross-sectional plane normal to the direction of flow.

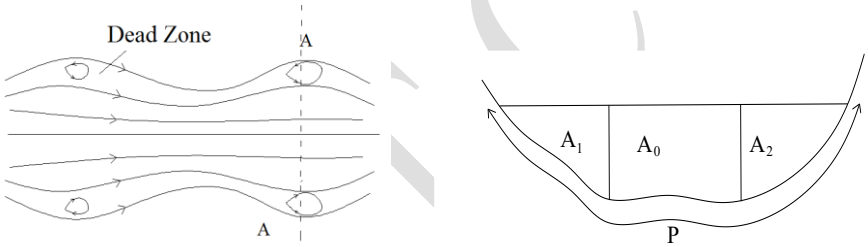
The hydraulic radius R is the ratio of water area to its wetted perimeter, ($R = \frac{A_0}{P}$).

The hydraulic depth y_m is the ratio of the water area to the top width, ($y_m = \frac{A_0}{B}$).

The average velocity V is the ratio of discharge to the water area ($V_{ort} = \frac{Q}{A_0}$).

The section factor Z is the product of the water area and the square root of the hydraulic depth, ($Z = A\sqrt{y_m} = A\sqrt{A/B}$).

The section factor for uniform-flow computation is the product of the water area and the two-thirds power of the hydraulic radius.



(a) (b)

Figure 8.2 (a) Plan, (b) A-A cross section

8.1.2 Comparison of Pipe Flow and Open-Channel Flow

1- Physical condition of open channels varies much more widely than that of pipes. In pipes the cross-section of flow is fixed, since it is completely defined by the geometry of the conduit. The cross-section of a pipe is generally round, but that of an open channel may be of any shape –from the circular to the irregular forms of natural streams.

2- In pipe flow since the cross-section is fixed, the average velocity can be obtained easily, but in channel flow it is very difficult to obtain the average velocity.

3- In pipe flow, the slope of the pipe is not an effective parameter on the average velocity the slope of the energy grade line is important. But in channel flow, the slope of the channel is an effective parameter on the average velocity.

4- In pipe flow, the roughness of the interior surface depends on the type of pipe and can be obtained easily on the other hand the roughness may either change in a cross section or through flow in an open channel flow. Therefore, the selection of friction coefficient is attended by greater uncertainty for open channels than for pipes.

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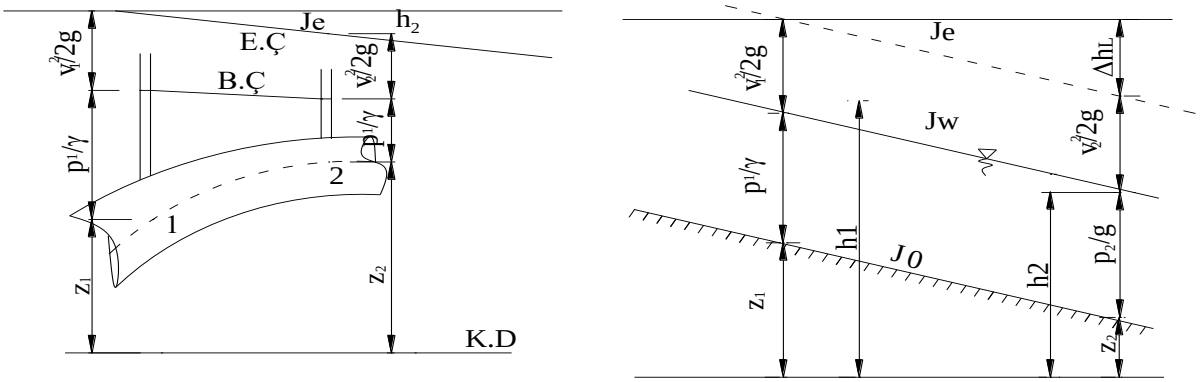


Figure 8.3 Comparison between pipe flow and open-channel flow

8.1.3 Classification of Open-Channel Flow

The classification of open-channel flow is summarized as follows;

1- Steady flow

a) Uniform flow
b) Varied flow

i) Gradually varied flow
ii) Rapidly varied flow

2- Unsteady flow

a) Uniform flow
b) Varied flow

i) Gradually varied flow
ii) Rapidly varied flow

8.1.4 Velocity Distribution in a Channel Section

The velocity of flow varies at different points of the cross-section. The frictional resistance of the sides causes the water to slow down toward the sides of the channel, and the frictional resistance between the water surface and the atmosphere causes a slight reduction of velocity at the free surface.

The velocity distribution in a channel section depends also on other factors, such as the arbitrary shape of the section, the roughness of the channel, and the presence of bends. In a broad, rapid and shallow stream or in a very smooth channel, the maximum velocity may often be found at the free surface. The roughness of the channel will cause the curvature of the vertical-velocity- distribution curve to increase.

The velocity at a solid boundary must be zero, and in open channel flow it generally increases with distance from the boundaries. The maximum velocity does not occur at the free surface but it usually below the free surface a distance of 0.05 to 0.25 of the depth. The mean velocity on any vertical section occurs at a depth of approximately 0.6 of the total depth; it varies with the type of channel and the friction of the sides. But a more reliable method is to take the average of the velocities at 0.2 and 0.8 of the depth.

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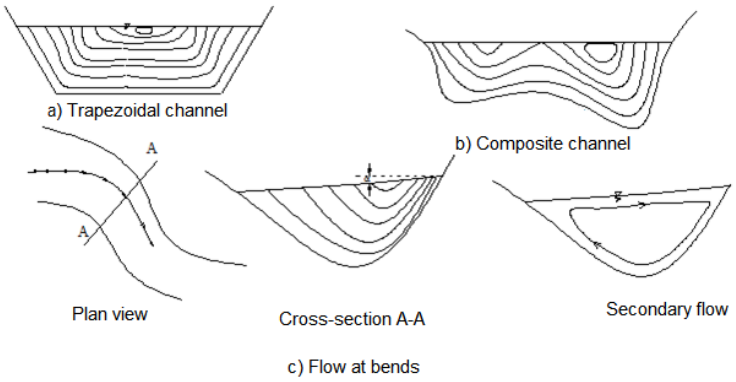


Figure 8.4 Typical curves of equal velocity in various channel sections

Observations in very wide-open channels have shown that the velocity distribution in the central region of the section is essentially the same as it would be a rectangular channel of infinite width. A ***wide open channel*** can safely be defined as a rectangular channel whose width is greater than 10 times the depth of flow.

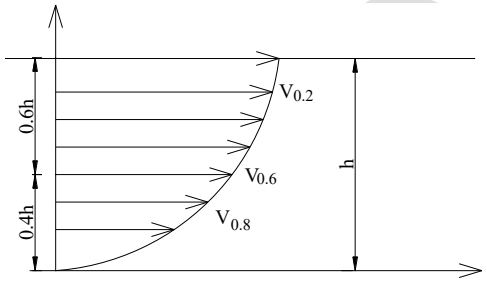


Figure 8.5 Vertical velocity distribution

As it seen in Fig 8.4c in a curved channel, the spiral flow is induced by the centrifugal force. Spiral flow refers to movement of water particles along a helical path in the general direction of the flow. Thus, in addition to the major velocity component normal to the channel cross-section, there are transverse velocity components on the cross section these transverse components will create so-called secondary flow in the plane of the cross section.

It is believed that the spiral flow is due mainly to 1- friction on the channel walls, which causes higher filamental velocities near the center of the channel than near the walls; 2- centrifugal force, which deflects the particles of water from straight line motion; and 3- a vertical velocity distribution which exists in the approach channel and thus initiates spiral flow in the flow. The centrifugal force is responsible also for the super elevation in the flow surface.

It is generally known trial, looking downstream; a channel curve to the right causes a counter clockwise spiral, whereas a channel curve to the left causes a clockwise spiral. This lateral current will be direct to the outer wall at surface and inner wall at the bottom. This current causes erosion at the outer wall and deposition at the inner wall. So, intake structures are always built in the outer wall of the channels.

The actual pattern of spiral flow is complicated and three-dimensional.

As a result of non-uniform distribution of velocities over a channel section, the velocity head of an open channel is generally greater than the value computed according to the expression $V^2/2g$, where V is the mean velocity. When the energy principle is used in computation, the true velocity head may be expressed as $\alpha V^2/2g$, where α is known as the energy coefficient or Coriolis coefficient. Experimental data indicate that the value of α varies from about 1.03 to 1.36 for fairly straight prismatic channels. The value is generally higher for small channels and lower for large streams of considerable depth.

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The non-uniform distribution of velocities also affects the computation of momentum in open-channel flow. From the principle of mechanics, the momentum of the fluid passing through a channel-section per unit time is expressed by $\beta\rho QV/g$ where β is known as the momentum coefficient or Boussinesq coefficient.

It is generally found that the value of β for fairly straight prismatic channels varies approximately from 1.01 to 1.12.

8.2 Steady Uniform Open Channel Flows

The uniform flow to be considered has the following main features:

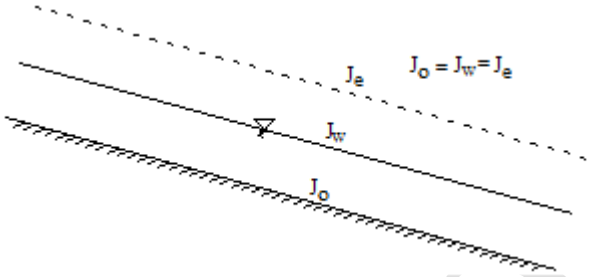


Figure 8.6 Uniform channel flow

1-The depth, velocity, water area and the discharge at every section of the channel reach are constant during the time interval under consideration.

2-The energy line, the channel bottom and the water surface are all parallel each other, that is their slopes are equal, or $J_0 = J_w = J_e$.

3-The orbits of the fluid particles are straight and perpendicular to the cross section.

8.2.1 Pressure Distribution

Let's write the equation's of motion of an element in a uniform flow in the η axis, which has a volume $dy \times 1 \times 1$,

$$\sum F_\eta = m\, a_\eta$$

Since the streamlines are straight and parallel to each other, the acceleration, a_η, will be zero. Hence

$$\sum F_\eta = p \times 1 - (p + dp) \times 1 + (\rho g dy \times 1 \times 1) \times \cos\theta = 0$$

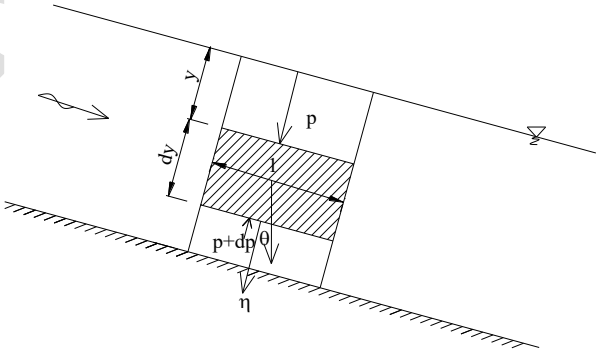


Figure 8.7 The effective forces on a fluid element

$$dp = \rho g dy \cos\theta$$

If the above equation is integrated

$$p = \rho g y \cos\theta + C$$

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is obtained.

With the boundary condition the integration constant is obtained

For $y = 0$ $p = p_0$ and $C = p_0$

For small angles $\theta \approx 0$ and $\cos\theta \cong 1$

$$p = p_0 + \rho g y \quad \text{or}$$

$$p = p_0 + \gamma y \tag{8.1}$$

is obtained. This means ***the pressure distribution obeys the hydrostatic pressure distribution in uniform open channel flow*** (in case of the slope of the channel is very small).

The pressure at any point on the cross section of the flow in a channel of small slope can be measured by the height of the water column in a piezometer tube installed at a point. Ignoring minor disturbances due to turbulence, etc., it is apparent that this water column should rise from the point of measurement up to the hydraulic grade line or the water surface. Therefore, the pressure at any point on the section is directly proportional to the depth of the point below the free surface and equal to the hydrostatic pressure corresponding to this depth. In other words, the distribution of pressure over the cross section of the channel is the same as the distribution of hydrostatic pressure; that is the distribution is linear and can be represented by a straight line.

8.2.2 Velocity Distribution

The mean velocity is obtained from the slope of the hydraulic grade line by two steps:

1) By using the slope of the energy grade line the shear stress between the fluid and the surface,
2) Obtaining the mean velocity from the shear stress.

Shear Stress

Let's consider a wide-open flume of constant depth (Figure 8.8);

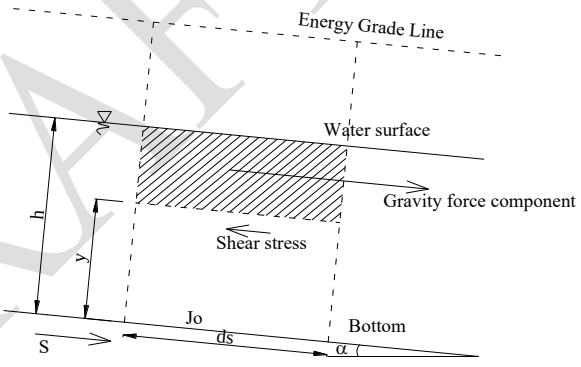


Figure 8.8 A fluid element

Let's consider a fluid element in this channel, the volume of the unit length of this element is $1 \times (h - y)\, ds$ and the mass is $\rho(h - y)ds$. Since the acceleration is zero in the uniform flow there is equilibrium between the forces acting on the element. Pressure forces acting against forefront and back front are equal. Because of the lower boundary shear stress, the force opposite to the s direction is $\tau_y ds$.

The gravity force is another effective force acting on this element, and the component of this force on the s direction is

$$\text{mass} \times g \times J_0 = \rho g (h - y) J_0 ds$$

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$\mathrm{Sin}J_0 = \sin\alpha$ may be changed into J_0 , since the slope is always small (the error is less than $\frac{1}{6}J_0^2$)

So equilibrium between the forces yields:

$$\tau_y \, ds = \rho g (h - y) J_0 \, ds$$

$$\tau_y = \rho g (h - y) J_0 \tag{8.2}$$

and the shear along the bottom

$$\tau_0 = \rho g h J_0 \tag{8.3}$$

The shear stress for an open flume of the arbitrary cross-section is shown in Fig. 8.9;

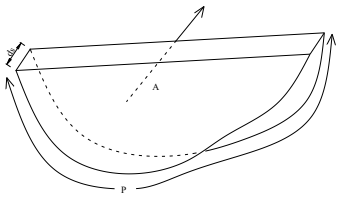


Figure 8.9 A flume with an arbitrary cross section

If the equilibrium equation is written for the whole cross section A, for a length of ds

$$\text{gravity in the flow direction} = \rho g A \, ds J_0$$

The tangential shear along wall

$$\left(\int_P \tau_0 dP\right) ds = P\bar{\tau}_0 . ds$$

By equating these two equations

$$\rho A g J_0 ds = P \bar{\tau}_0 ds$$

$$\bar{\tau}_0 = \rho g \frac{A}{P} J_0$$

$$\bar{\tau}_0 = \rho g R J_0 \tag{8.4}$$

In the very wide flat cross-section the depth h is equal to R (hydraulic radius). Eq. (8.4) is the general equation of the shear stress. In this case Eq. (8.4) is equal to Eq. (8.3).

Average Flow Velocity

A relation between shear stress inside a fluid and gradient of velocity is given by definition of kinematic viscosity, ν;

$$\nu = (\tau_y / \rho) / (du / dy) \tag{8.5}$$

This is valid for a Newtonian fluid and for laminar flow. Contrary to the first step, it is different if the flow is laminar or turbulent.

Re number is the criterion to differentiate if the flow is laminar or turbulent. Re number is a velocity times a length over kinematic viscosity.

$$R_e = VR / \nu \tag{8.6}$$

It is obvious that the velocity may be the mean velocity. In the case of a flat cross-section the depth may be used as length in Reynold's number; in the case of a circular pipeline flowing full, it is customary to take the diameter. Instead, we now take the hydraulic radius R.

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Re > 500 flow is turbulent

Re < 500 flow is laminar

According to French (1986)

Re ≤ 500 flow is laminar

500 ≤ Re ≤ 12500 transient flow

12500 ≤ Re flow is turbulent

8.2.2.1 Turbulent Open Channel Flow

For turbulent flow the Eq. (8.5) is not valid anymore. Because by mixing of adjacent layers of fluids, momentum is transported from one layer to another. This leads to the conclusion that the shear will not be proportional to the gradient of velocity but the square of that gradient $\left(\frac{du}{dy}\right)^2$ (Fig. 8.10).

$$\sqrt{\frac{\tau_y}{\rho}} = L\left(\frac{du}{dy}\right) \tag{8.7}$$

The length L has a physical meaning. The greater L the smaller velocity gradient for a given value of $\sqrt{\frac{\tau_y}{\rho}}$ in other words, the greater L the stronger the mixing. Therefore L is called as ***mixing length.***

Karman assumed that L will be proportional to the distance from the wall

$$L = \chi y \tag{8.8}$$

χ, Karman's parameter is considered as a constant, at least for normal fluids $\chi = 0.4$. In suspensions, where turbulence is suppressed by the suspended particles may perhaps be less than 0.4.

The mixing length can be defined for wide-open flume with a constant depth as below.

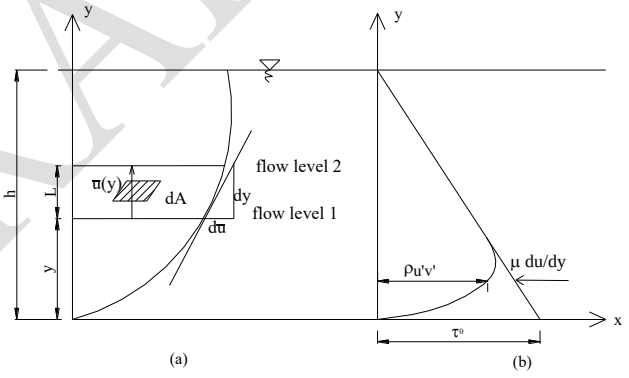


Figure 8.10 Free surface flow (a) Variation of velocity, b) Variation of shear stress

Near the bottom mixing length will be χy in the top layers the sidewise excursions will be opposed by the presence of the surface. Here the mixing length has to come back to zero when y approaches h (Fig. 8.11). Hence the mixing length can be expressed as

$$L = \chi y\left(1 - \frac{y}{h}\right)^n \tag{8.9}$$

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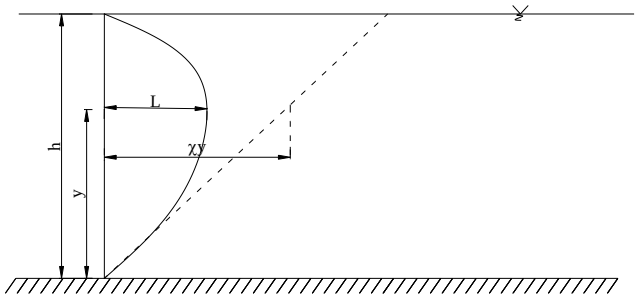


Figure 8.11 Variation of mixing length

In view of the fact that the surface will allow more motion than the bottom, the exponent n should be less than unity. Experiments show that the best value is ½, so

$$L = \chi y(1 - y/h)^{\frac{1}{2}} \tag{8.10}$$

If we substitute this equation into Eq. (8.7)

$$\sqrt{\frac{\tau_y}{\rho}} = \chi y\sqrt{1-\frac{y}{h}}\left(\frac{du}{dy}\right)$$

and taking τ_y from Eq. (8.2)

$$\sqrt{g(h-y)J_0} = \chi y\sqrt{1-\frac{y}{h}}\left(\frac{du}{dy}\right)$$

In this equation g, h, J_0 and χ are constants this is an equation in v and y, which can be integrated

$$du = \frac{\sqrt{g\,h\,J_0}}{\chi}\,\frac{dy}{y}$$

where $\sqrt{g\,h\,J_0} = \sqrt{\frac{\tau}{\rho}}$ has the dimension of velocity (like $\sqrt{K/g}$, which is the speed of propagation of sound in a fluid, K = modulus of compressibility). This velocity is called ***shear velocity (friction velocity), (u_*)***, but is not a real velocity, which is a measure of intensity of turbulent eddying of the transfer of momentum due to fluctuations.

$$u_* = \sqrt{\frac{\tau_0}{\rho}} = \sqrt{g\,h\,J_0} \tag{8.11}$$

If the equation is written as below and by integrating

$$du = \frac{u_*}{\chi}\,\frac{dy}{y}$$

$$u_y = \frac{u_*}{\chi}\ln y + \text{sabit} \tag{8.12}$$

is obtained. The integration constant must depend on the condition of the bottom. This equation cannot be true for y=0, because for y=0, lny= - ∞ and as the constant must be finite. This of course is impossible and above equation will be true for only down to some (very small) distance above the bottom. The constant can be written as

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$$\text{sabit} = -\frac{u_*}{\chi}\ln y_0$$

where y_0 is the new integration constant. It is the distance from the bottom where the velocity of the fluid is zero. The equation is valid for only above y_0

The usual way of writing Eq. (8.12) is:

$$u_y = \frac{u_*}{\chi}\ln\frac{y}{y_0} \tag{8.13}$$

This equation shows that for turbulent flows the velocity distribution is logarithmic.

It is clear that something must happen near the bottom and it is the laminar sub-layer defined by Prandtl. It exists when the bottom is quite smooth. Along the bottom, turbulence is completely suppressed then Eq. (8.12) cannot be used but Eq. (8.5) is used.

It is evident that the thickness of the sub-layer which is denoted by δ_* has to be more than y_0, at the top of the layer, where there is the transition from laminar to turbulent flow and Eq. (8.12) must yield a positive velocity.

$$u_{\delta_*} = \frac{u_*}{\chi}\ln\frac{\delta_*}{y_0} > 0 \tag{8.14}$$

In the sub-layer, which is very thin, the shear stress, τ, should be considered, so

$$\frac{du_y}{dy} = \frac{\tau_0/\rho}{\nu} = \frac{u_*^2}{\nu}$$

or

$$du_y = \frac{u_*^2}{\nu}dy$$

$$u_y = \frac{u_*^2}{\nu}y + \text{sabit}$$

In a laminar flow even a smooth bottom is rough compared with the nearest layer of fluid, therefore u = 0 and constant is zero too:

$$u_y = \left(\frac{u_*^2}{\nu}\right)y \qquad \text{for} \qquad (0 < y < \delta) \tag{8.15}$$

$$u_{\delta_*} = \left(\frac{u_*^2}{\nu}\right)\delta_* \qquad \text{for} \quad y = \delta_* \text{ and as there can be one value of u. This must be the same Eq. (8.13).}$$

$$u_{\delta_*} = \frac{u_*}{\chi}\ln\left(\frac{\delta_*}{y_0}\right) = \left(\frac{u_*^2}{\nu}\right)\delta_* \tag{8.16}$$

If we know δ_* the problem is solved and Eq. (8.16) will enable to calculate y_0.

The laminar layer will be thick when the viscosity is great. In this layer a kind of Re number may be constructed from the shear velocity, u_*

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$$\mathrm{Re} = \frac{u_* \delta_*}{\nu} \tag{8.17}$$

When this dimensionless parameter were examined experimentally, it has found that it is really a constant

Re = 11.6

The viscous sub-layer is obtained as

$$\delta_* = 11.6 \frac{\nu}{u_*} \tag{8.18}$$

If this equation is substituted into Eq. (8.16)

$$\left(\frac{u_*}{\chi}\right) \ln\left(\frac{\delta_*}{y_0}\right) = 11.6 u_*$$

$$y_0 = \frac{\delta_*}{104} \tag{8.19}$$

This equation is substituted into Eq. (8.13) and velocity for *Turbulent Hydraulically Smooth Regime* is obtained

$$u_y = \left(\frac{u_*}{\chi}\right) \ln\left(\frac{104y}{\delta_*}\right) \tag{8.20}$$

The equation can also be written as below:

$$\frac{u_y}{u_*} = 2.5 \ln\left(\frac{104y}{\delta_*}\right)$$

$$\frac{u_y}{u_*} = 5.75 \log\left(\frac{104}{11.6} \frac{u_* R}{\nu}\right)$$

$$\frac{u_y}{u_*} = 5.75 \log\left(8.97 \frac{u_* R}{\nu}\right)$$

$$\frac{u_y}{u_*} = 5.75 . \log\left(\frac{u_* R}{\nu}\right) + 5.5 \tag{8.21}$$

Let's calculate the discharge per unit width of the channel, q, which is

$$q = \int_{\text{bottom}}^{\text{surface}} u_y dy$$

For the turbulent part of the flow by using Eq. (8.20),

$$q = \left(\frac{u_*}{\chi}\right) . \int_{y=\delta}^{y=h} (\ln 104 + \ln y - \ln \delta_*) dy$$

Remembering that $$\int \ln y dy = y \ln y - y = y \ln\left(\frac{y}{e}\right)$$

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$$q = \left[(h - \delta_*)(\ln 104 - 1) + h \ln h / \delta_*\right]\left(\frac{u_*}{\chi}\right)$$

Since δ_* is very thin when compared with $q = h\left(\frac{u_*}{\chi}\right)\ln\frac{104h}{e\delta_*}$

or

$$\frac{q}{h} = V = \left(\frac{u_*}{\chi}\right)\ln\frac{104h}{e\delta_*}$$

The average velocity for *Turbulent Hydraulically Smooth Regime*

$$V = \left(\frac{u_*}{\chi}\right)\ln\left(\frac{38.1h}{\delta_*}\right) \tag{8.22}$$

or

$$\frac{V}{u_*} = 5.75.\log\left(\frac{38.1}{11.6}.\frac{h\,u_*}{\nu}\right)$$

$$\frac{V}{u_*} = 5.75.\log\left(\frac{u_* h}{2\nu}\right) + 4.7 \tag{8.23}$$

The error made to pass from Eq. (8.20) to Eq. (8.23) is one over million.

What happens if the wall is not smooth enough to allow the laminar sub-layer to develop (which we express by saying that the wall is hydraulically rough instead of hydraulically smooth)?

The well-known experiments of Nikuradse give an answer to this question. Nikuradse covered the walls of his experimental conduit by a layer of grains of sand of approximately uniform diameter, k (Fig. 8.12). By using the same principle the velocity equations for hydraulic rough open channel flow can also be expressed. Here Eq. (8.12) is used again. But the integration constant y_0 will depend on the roughness of the wall instead of sub-layer. A relation between y_0 and a=k/2 should be given in Eq. (8.13), this can only be done by experiments.

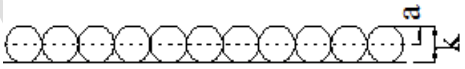


Figure 8.12 Roughened surface

According to the experimental results,

$$y_0 = \frac{a}{15} = \frac{k}{30}$$

The velocity for *Turbulent Hydraulically Rough Regime*

$$u_y = \left(\frac{u_*}{\chi}\right)\ln\left(\frac{15y}{a}\right) \tag{8.24}$$

or

$$\frac{u_y}{u_*} = 5.75\log\left(\frac{R}{k}\right) + 8.5 \tag{8.25}$$

The average velocity is obtained in the same way of hydraulically smooth regime

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$$V = \left(\frac{u_*}{\chi}\right)\ln\left(\frac{15R}{ea}\right)$$

$$V = \left(\frac{u_*}{\chi}\right)\ln\left(\frac{5.5R}{a}\right) \tag{8.26}$$

Again the same expression

$$\frac{V}{u_*} = 5.75.\log\left(\frac{5.5R}{a}\right)$$

$$\frac{V}{u_*} = 5.75.\log\left(\frac{11R}{k}\right)$$

$$\frac{V}{u_*} = 5.75.\log\left(\frac{22R}{2k}\right)$$

$$\frac{V}{u_*} = 5.75.\log\left(\frac{R}{2k}\right) + 7.7 \tag{8.27}$$

Comments: If the equations for both channel and pipe flows are examined, the velocity expressions are almost same but only the constants are different.

	Channel flow	Pipe flow
Hydraulically smooth	38.1	46.2
Hydraulically rough	5.5	6.7

The average velocities for both pipe and channel flows approximately are given as below:

For hydraulically smooth walls

$$V = (\frac{u_*}{\chi})\ln(\frac{42R}{\delta_*}) \tag{8.28}$$

or

$$\frac{V}{u_*} = 5.75.\log(\frac{Ru_*}{2\nu}) + 4.94 \tag{8.29}$$

For hydraulically rough walls

$$V = \left(\frac{u_*}{\chi}\right)\ln\left(\frac{6R}{a}\right) \tag{8.30}$$

or

$$\frac{V}{u_*} = 5.75.\log\left(\frac{R}{2k}\right) + 7.93 \tag{8.31}$$

With these approximations there is no important error

"Colebrook-White" have given the equations for the transition from hydraulically smooth to hydraulically rough for both pipe and channel flows.

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$$V = \left(\frac{u_*}{\chi}\right) \ln\left(\frac{6R}{a + \frac{\delta_*}{7}}\right) \tag{8.32}$$

or

$$V = 5.75u_* \log\left(\frac{6R}{a + \frac{\delta}{7}}\right) \tag{8.33}$$

If the shear velocity is inserted into this equation;

$$V = 5.75\sqrt{g}\sqrt{RJ} \log\left(\frac{6R}{a + \frac{\delta_*}{7}}\right)$$

$$V = 18\sqrt{RJ} \log\left(\frac{6R}{a + \frac{\delta_*}{7}}\right) \tag{8.34}$$

This equation can be written as;

$$V = C\sqrt{RJ} \tag{8.35}$$

where C is Chézy coefficient.

$$C = 18\log\left(\frac{6R}{a + \delta_* / 7}\right) \tag{8.36}$$

Table 8.1 Velocity distributions for turbulent flow

HYDRAULICALLY SMOOTH	**HYDRAULICALLY ROUGH**
$u(y) = 5.75u_* \log(104R/\delta_*)$ $V = 5.75u_* \log(42R/\delta_*)$	$u(y) = 5.75u_* \log(15R/a)$ $V = 5.75u_* \log(6R/a)$
TRANSITION	
$V = 5.75u_* \log\left(\frac{6R}{a + \delta_*/7}\right)$ or $V = 18\sqrt{RJ} \log\left(\frac{6R}{a + \delta_*/7}\right)$ or $V = C\sqrt{RJ}$	
$a < \delta_*/7$ hydraulically smooth (or $\frac{u_* a}{\nu} < 0.5$) $a \approx \delta_*/7$ transition $a > \delta_*/7$ hydraulically rough (or $\frac{u_* a}{\nu} > 30$)	

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8.2.2.2 Laminar Open Channel Flow

Eq. (8.2) is $\tau_y = \rho g(h - y)J$, if this is substituted into Eq. (8.5) $\nu = \left(\frac{\tau_y}{\rho}\right) / \left(\frac{du}{dy}\right)$

$$g(h - y).J = \nu du / dy$$

or

$$du = (gJ / \nu)(h - y)dy$$

Integrating

$$u_y = \frac{gJ}{\nu}(hy - \frac{y^2}{2}) + \text{constant} \tag{8.37}$$

This expression shows that the velocity distribution is logarithmic

For y=0, since u = 0 the constant is zero

For y=h the velocity at the free surface

$$u_h = \frac{gJh^2}{2\nu} = \frac{gJR^2}{2\nu}$$

The average velocity is 2/3 of this equation

$$V = \frac{gJh^2}{3\nu} \tag{8.38}$$

or

$$V = \frac{gJR^2}{3\nu} \tag{8.39}$$

The development of boundary layer for free surface flow is shown in Fig. 8.13.

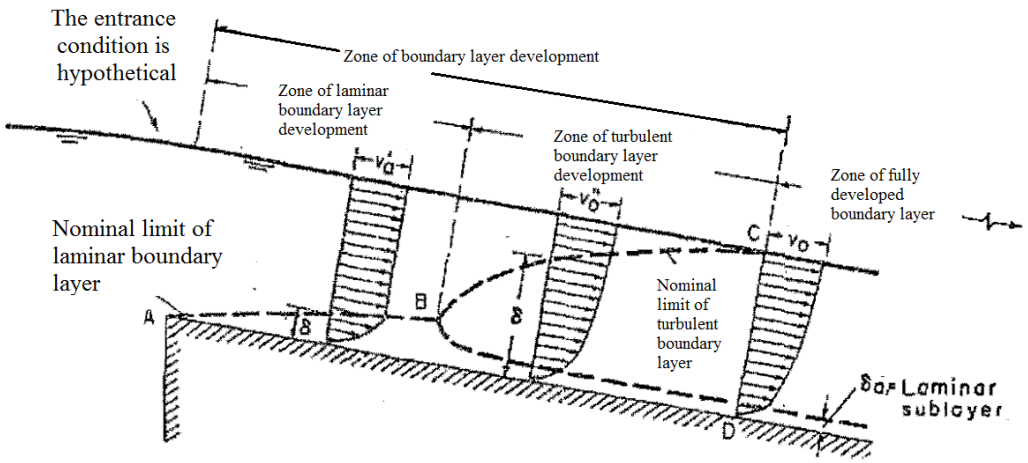


Figure 8.13 Development of boundary layer for free surface flow

8.2.3 Expressing the Velocity of a Uniform Flow

For hydraulic computations the mean velocity of a turbulent uniform flow in open channel is usually expressed approximately by a so-called uniform flow formula. Most practical uniform-flow formulas can be expressed in the following general form:

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$$V = CR^{\alpha}J^{\beta}$$

where V is the mean velocity, R is the hydraulic radius, J is the energy slope, α and β are exponents; and C is factor of flow resistance varying with the mean velocity, hydraulic radius, channel roughness, and many other factors.

For practical purposes, the flow in a natural channel may be assumed uniform under normal conditions, that is, if there are no flood flows or markedly varied flows caused by channel irregularities. In applying the uniform flow formula to a natural stream, it is understood that the result is very approximate since the flow condition is subject to more uncertain factors than would be involved in a regular artificial channel.

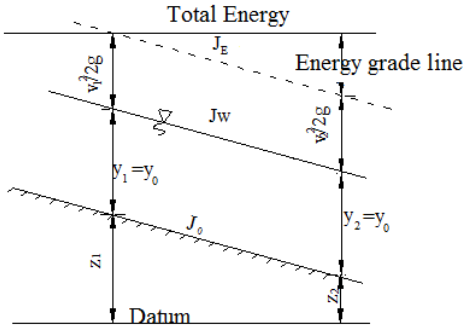


Figure 8.14 Uniform channel flow

1– Chézy formula

The first uniform flow formula, the ***Chézy*** formula which is usually expressed as follows:

$$V = C\sqrt{RJ} \tag{8.40}$$

where C is Chézy constant (factor of flow resistance) $(L^{1/2}T^{-1})$. Bazin gave C constant as below:

$$C = \frac{87\sqrt{R}}{\gamma_b + \sqrt{R}} \tag{8.41}$$

The Bazin's coefficients, γ_b, for different channels are listed in Table 8.2.

Table 8.2 Proposed values of Bazin's γ_b (Karahan, 1986)

Description of channel	γ_b
Very smooth cement	0.06
Roubble masonry	0.46
Earth channel	1.30
For mixed sides	0.85
Rubble masonry and lined channels	> 1.75

2– Manning's formula

$$V = \frac{1}{n}R^{2/3}J^{1/2} \tag{8.42}$$

where n (coefficient of friction) is Manning coefficient $(L^{1/3}/T)$

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Table 8.3 Typical values of Manning's n (Karahan, 1986)

Type of channel	min	max
Smooth cement surface	0.010	0.013
Smooth metal channels	0.011	0.015
Concrete	0.011	0.013
Brick work with cement mortar	0.012	0.017
Earth channels in perfect condition	0.025	0.033
Earth channels in rough condition	0.045	0.060
Natural streams with high grass	0.075	0.150

3- Darcy–Weisbach formula

$$J = \lambda \frac{1}{D} \frac{V^2}{2g} \tag{8.43}$$

$R = \frac{D}{4}$ for any arbitrary cross-section

$$J = \lambda \frac{1}{4R} \frac{V^2}{2g} \quad (8.44)$$

λ is Darcy–Weisbach coefficient, if the velocity is drawn from the equation

$$V = \sqrt{\frac{8g}{\lambda}} R^{1/2} J^{1/2} \tag{8.45}$$

The relations between these coefficients (Manning, Chézy and Darcy–Weisbach) are

$$C = \sqrt{\frac{8g}{\lambda}} = \frac{R^{1/6}}{n} = \frac{R^{1/6}}{0.04 d^{1/6}} = 18 \log \frac{6R}{a + \delta_* / 7} \tag{8.46}$$

where d diameter of the sand. λ Darcy–Weisbach coefficient is expressed in different ways for laminar and turbulent flow. For open channel flows;

A) For laminar flow

$$\lambda = \frac{96}{Re} \tag{8.47}$$

where Re = Reynolds number = $\frac{V\,4R}{\nu}$

B) For turbulent flow

For hydraulically smooth wall;

$$\frac{1}{\sqrt{\lambda}} = 2 \log \frac{Re\sqrt{\lambda}}{3.4} \tag{8.48}$$

For hydraulically rough wall;

$$\frac{1}{\sqrt{\lambda}} = 2 \log 3.1 \frac{4R}{k_S} = 2 \log 12.4 \frac{y}{k_S} \tag{8.49}$$

Colebrook-White formula is used for transition regime (Steady Uniform Flow in Open Channels, No: 60, Stockholm Inst.).

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$$\frac{1}{\sqrt{\lambda}} = -2\log\left(\frac{k_S}{12.4R} + \frac{3.4}{Re\sqrt{\lambda}}\right) \tag{8.50}$$

Channels of Composite Roughness

In simple channels, the roughness along the wetted perimeter may be distinctly different from part to part of the perimeter, but the mean velocity can still be computed by a uniform-flow formula without actually subdividing the section. In applying the Manning formula to channels having different n values for the bed and sides it is necessary to compute an equivalent n value to be used for the whole section. The water area is divided into N parts having wetted perimeters P_1, P_2,…,P_N with associated roughness coefficients n_1, n_2,..n_N. On the basis of this assumption, the *equivalent coefficient of roughness* may be obtained by the following equations;

1 - $$n = \left[\frac{\sum_1^N (P_N n_N^{1.5})}{P}\right]^{\frac{2}{3}} = \frac{(P_1 n_1^{1.5} + P_2 n_2^{1.5} + ... + P_N n_N^{1.5})^{\frac{2}{3}}}{P^{\frac{2}{3}}} \tag{8.51}$$

2 - $$n = \left[\frac{\sum_1^N (P_N n_N^{2})}{P^{\frac{1}{2}}}\right]^{\frac{1}{2}} = \frac{(P_1 n_1^{2} + P_2 n_2^{2} + ... + P_N n_N^{2})^{\frac{1}{2}}}{P^{\frac{1}{2}}} \tag{8.52}$$

3- $$n = \frac{PR^{5/3}}{\sum_1^N (\frac{P_N R^{5/3}}{n_N})} = \frac{PR^{5/3}}{\frac{P_1 R_1^{5/3}}{n_1} + \frac{P_2 R_2^{5/3}}{n_2} + ... + \frac{P_N R_N^{5/3}}{n_N})} \tag{8.53}$$

where P total wetted perimeter, R_N hydraulic radius for different rough regions, P_N wetted perimeter for different rough region.

8.2.4 Computation of Uniform Open Channel Flow

The discharge of uniform flow in a channel may be expressed as the product of the velocity and the water area, or by considering Manning's formula;

$$Q = VA = \frac{1}{n} R^{2/3} J_0^{1/2} A \tag{8.54}$$

where

$$\frac{1}{n} R^{2/3} A = K \tag{8.55}$$

The term K is known as the ***conveyance*** of the channel section; it is a measure of the carrying capacity of the channel section, since it is directly proportional to Q.

$$Q = K J_0^{1/2}$$

and the conveyance is

$$K = \frac{Q}{J_0^{1/2}}$$

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$$K = \frac{1}{n}\frac{AR^{2/3}}{J_0^{1/2}}$$

The above equation is used to compute the conveyance when the geometry of the water area and the resistance factor or roughness coefficient are given

$$AR^{2/3} = nK = n\,\frac{Q}{J_0^{1/2}}$$

The above expression is called the ***section factor for uniform flow computation***; it is an important element in the computation of uniform flow. The equation applies to a channel section when the flow is uniform. The right side of the equation contains the values of n, Q and J_0; but the left side depends only on the geometry of the water area. Therefore, it shows that, for a given condition of n, Q and J_0, there is only one possible depth for maintaining a uniform flow, provided that the value of $AR^{2/3}$ always increases with increase in depth, which is true for most cases.

$AR^{2/3}$ section factor $(L^{8/3})$ (8.56)

The geometric elements of a trapezoidal channel generally expressed as in Fig. 8.15;

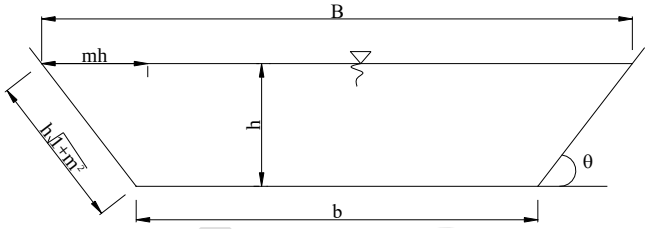


Figure 8.15 Trapezoidal channel

where B top width, b bottom width, h depth, θ side slope.

Water area $A = h(b + mh)$ (8.57)

$m = \cot g\theta$

Wetted perimeter $P = b + 2h\sqrt{1+m^2}$ (8.58)

Hydraulic radius $R = \frac{A}{P} = \frac{h(b+mh)}{b+2h\sqrt{1+m^2}}$ (8.59)

Top width $B = b + 2mh$ (8.60)

$$Q = \frac{1}{n}\left(\frac{h(b+mh)}{b+2h\sqrt{1+m^2}}\right)^{2/3} J_0^{1/2}(h(b+mh)) \quad (8.61)$$

- For very wide channels $B \rangle\rangle h$ $R \approx h$

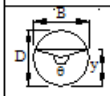
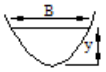
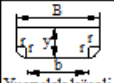
Hence,

$$Q = \frac{1}{n}h^{2/3}J_0^{1/2}(h(b+mh)$$

$$Q = \frac{1}{n}h^{5/3}J_0^{1/2}(b+mh) \quad (8.62)$$

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Tablo 8.4 Geometric elements of different channel sections

Kesit	Alan(A)	Islak çevre(P)	Hidrolik yarıçap(R)	Su yüzü(B)	Ortalama derinlik(ym)
 Dikdörgen	by	b+2y	by/(b+2y)	b	y
 Trapez	(b+my)y	$b+2y\sqrt{1+m^2}$	$\frac{y(b+my)}{b+2y\sqrt{1+m^2}}$	b+2my	$\frac{y(b+my)}{b+2my}$
 üçgen	my^2	$2y\sqrt{1+m}$	$\frac{my}{2\sqrt{1+m}}$	2my	1/2 y
 Dairesel	$1/8\,(\theta-\sin\theta)D^2$	1/2θD	1/4 (1-sinθ/θ)D	sin(1/2θ)D ve ya $2\sqrt{y(D-y)}$	$1/8\left(\frac{\theta-\sin\theta}{\sin 1/2\,\theta}\right)D$
 Parabol	2/3 By	$B+(8/3)\,y^2/B$	$\frac{2By^2}{3B^2+8y^2}$	$3/2\left(\frac{A}{y}\right)$	$2/3(y)$
 Yuvarlak köşeli Dikdörtgen	$\left(\frac{\pi}{2}-2\right)r^2+(b+2r)y$	$(\pi-2)r+b+2y$	$\frac{(\pi/2-2)r^2+(b+2r)y}{(\pi-2)r+b+2y}$	b+2r	$\frac{\pi/2-2)r^2}{b+2r}+y$

8.2.4.1 The Best Hydraulic Sections

It is known that the conveyance of a channel section increases with increase in the hydraulic radius or with decrease in the wetted perimeter. From the hydraulic point view, therefore, the channel section having the least wetted perimeter for a given area has the maximum conveyance; such a section is known as ***the best hydraulic section.*** Among all possible channel cross sections, the best hydraulic cross section is a semi circle since, for a given area, it has the minimum wetted perimeter.

It should be noted that from the point of view of application, the best hydraulic section is not necessarily the most economic section. In practice the following factors must be considered:

1. The best hydraulic section minimizes the area required to convey a specified flow; however, the area which must be excavated to achieve the flow area required by the best hydraulic section may be significantly larger if the over-burden which must be removed is considered.
2. It may not be possible to construct a stable best hydraulic section in the available natural material. If the channel must be lined, the cost of the lining may be comparable with the cost of excavation.
3. The cost of excavation depends not only on the amount of material, which must be removed, but also on the ease of access to the site and the cost of disposing of the material removed.
4. The slope of the channel in many cases must also be considered a variable since it is not necessarily completely defined by topographic considerations. For example, while a reduced channel slope may require a larger channel flow area to convey the specified flow, the cost of excavating the overburden may be reduced.

From Chézy's equation

$$Q = A\,C\sqrt{RJ}$$

$$Q = A\,C\left(\frac{A}{P}\right)^{1/2} J^{1/2}$$

If A, C and J are constant, the discharge Q will be maximum when the wetted perimeter P is minimum or when dP/dh is zero.

$R_{max} = A / P_{min}$ for Q_{max}

1- The most economical hydraulic section for trapezoidal channel

For maximum Q, P should be minimum. For a trapezoidal channel the wetted perimeter is;

$$P = b + 2h\sqrt{1+m^2}$$

and area

$A = h(b + mh)$ from which

$$b = \frac{A}{h} - mh$$

Substituting this into wetted perimeter equation

$$P = \frac{A}{h} - mh + 2h\sqrt{1+m^2}$$

For a most economical section, $\frac{dP}{dh} = 0$

$$\frac{dP}{dh} = -\frac{A}{h^2} - m + 2\sqrt{1+m^2} = 0$$

$$2\sqrt{1+m^2} - m = \frac{A}{h^2}$$

$$h = \left(\frac{A}{2\sqrt{1+m^2} - m}\right)^{1/2}$$

$$M = 2\sqrt{1+m^2} - m$$

$$h = \left(\frac{A}{M}\right)^{1/2}$$

or

$$A = Mh^2$$

The wetted perimeter

$$P = \frac{Mh^2}{h} - mh + 2h\sqrt{1+m^2}$$

$$P = Mh - mh + 2h\sqrt{1+m^2}$$

$$P = Mh + h(2\sqrt{1+m^2} - m)$$

$P = 2\,M\,h$

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$$R = \frac{A}{P} = \frac{Mh^2}{2Mh} = \frac{h}{2}$$

$$R = \frac{h}{2} \tag{8.63}$$

$$b = \frac{A}{h} - mh = \frac{Mh^2}{h} - mh = h(M - m) = h(2\sqrt{1 + m^2} - m - m)$$

$$b = 2h\left(\sqrt{1 + m^2} - m\right) \tag{8.64}$$

2– The most economical hydraulic section for rectangular channel

$$R = \frac{h}{2} \tag{8.65}$$

$$b = 2h \tag{8.66}$$

$$B = 2h\sqrt{1 + m^2} \tag{8.67}$$

Conclusions

1- For the most economical trapezoidal section a circle that has the center at the free surface could be drawn tangential to the sides of the section. With the same manner it is possible to draw many sections, but the most economical one among them has the minimum area. This section is that which has a side angle of $\theta = 60^\circ$ as in Fig. 8.16.

2- For the hydraulically most economical trapezoidal section; the summation of the lengths of the sides should be equal to the length of the free surface (Fig. 8.17).

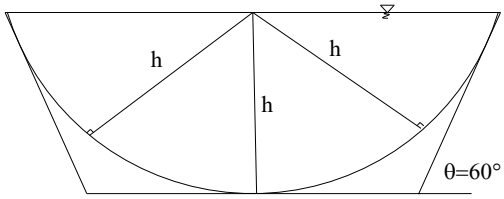


Figure 8.16

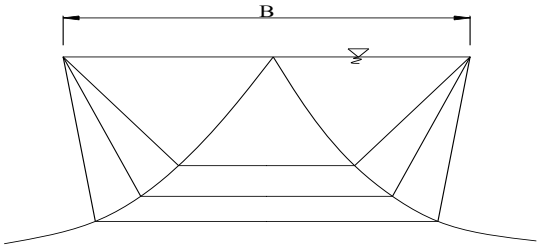


Figure 8.17

8.2.4.2 Channels of Compound Section

The cross section of a channel may be composed of several distinct subsections with each subsection different in roughness from the others. A typical example of a compound section is a river channel with flood plains. The roughness of the side channels will be different (generally rougher) than that of the main channel and the method of analysis is to consider the total discharge to be the sum of

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component discharges computed by the Manning equation. The mean velocity in the main channel is greater than the mean velocities in the side channels.

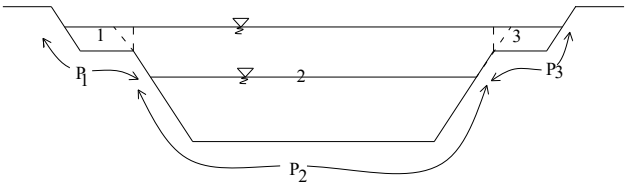


Figure 8.18 Compound channel section

Owing to the differences that exist among the velocities of the subsections, the velocity distribution coefficients of the whole section are different from those of the subsections. Thus in the channel shown in Fig. 8.16, assuming that the bed slope is the same for the three sub-areas:

$$Q = \left(\frac{A_1 R_1^{2/3}}{n_1} + \frac{A_2 R_2^{2/3}}{n_2} + \frac{A_3 R_3^{2/3}}{n_3} \right) J_0^{1/2}$$

The most important thing is to compute the wetted perimeter of each subsection. During the computation of the wetted perimeter only the solid boundaries should be considered. The water depths of the side sections are very small when compared with the main channel. In this case the wetted perimeter increases suddenly but the water area doesn't. In this case, it is better to compute the discharge by dividing the channel into sections instead of computing as a whole.

8.2.4.3 Design of Stable, Unlined, Earthen Channels

In comparison with the design process typically used for lined channels, the design of stable, unlined or erodible, earthen channels is a complex process involving numerous parameters, most of which cannot be accurately quantified. The complexity of the erodible channel design process results from the fact that in such channels stability is dependent not only on hydraulic parameters but also on the properties of the material which composes the bed and sides of the channel. In the context of this discussion, a stable channel section is one in which *neither objectionable scour nor deposition occurs*. In contrast there are three types of unstable channel.

1- The banks and bed of the channel are scoured but no deposition occurs.

2- There is deposition but no scour

3- Both scour and deposition occur.

Two methods of approach to proper design of erodible channels are described:

a) Method of tractive force

b) Method of permissible velocity

When water flows in a channel, a force is developed that acts in the direction of flow on the channel bed. This force, which is simply the pull of water on the wetted area, is known as the *tractive force.* In a uniform flow the tractive force is apparently equal to the effective component of the gravity force acting on the body of water, parallel to the channel bottom. The average value of the tractive force per unit wetted area, or the so-called unit tractive force (average shear stress) $\tau_0 = \rho g R J_0$

It should be noted that the unit tractive force in channels, except for wide-open channels, is not uniformly distributed along the wetted perimeter. The distribution varies with channel shape but not with size.

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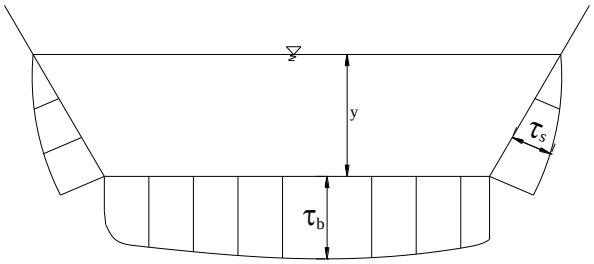


Figure 8.19 Distribution of tractive force in a trapezoidal channel section

If the shear stress can be kept below a value which will cause the material of the channel boundary to move, the channel will be stable. The critical tractive force of a particular material is the unit tractive force that will not cause erosion of the material on a horizontal surface. Material on the sides of the channel is subjected, in addition to the shear force due to flowing water, to a gravity force down to slope. It can be shown that if τ_{cb} is the critical tractive force the maximum critical shear stress due to water flow on the sides is

$$\tau_{CS} = \tau_{Cb}\sqrt{1-\frac{\sin^2\theta}{\sin^2\varphi}} \tag{8.68}$$

where θ is the slope of the sides to the horizontal, and ϕ is the angle of repose of the material. Table 8.5 gives some typical values of critical tractive force and permissible velocity.

Table 8.5 Critical tractive force and mean velocity for different bed materials

Material	Size (mm)	Critical tractive force (N/m^2)	Approximate mean velocity (m/s)	Manning's coefficient of roughness (n)
Sandy loam (non-colloidal)		2.0	0.50	0.020
Silt loam (non-colloidal)		2.5	0.60	0.020
Alluvial silt (non-colloidal)		2.5	0.60	0.020
Ordinary firm loam		3.7	0.75	0.020
Volcanic ash		3.7	0.75	0.020
Stiff clay (very colloidal)		1.22	1.15	0.025
Alluvial silts (colloidal)		12.2	1.15	0.025
Shales and hard-pans		31.8	1.85	0.025
Fine sand (non-colloidal)	0.062-0.025	1.2	0.45	0.020
Medium sand (non-colloidal)	0.25-0.50	1.7	0.50	0.020
Coarse sand (non-colloidal)	0.50-2.0	2.5	0.60	0.020
(fine gravel)	4-8	3.7	0.75	0.020
Coarse gravel	8-64	14.7	1.25	0.025
Cobbles and shingles	64-256	44.0	1.55	0.035
Graded loam and cobbles (non-colloidal)	0.004-64	19.6	1.15	0.030
Graded silts to cobbles (colloidal)	0-64	22.0	1.25	0.030

The maximum and minimum permissible velocities should be considered depending on the channel material to avoid deformation of the channel section. The *minimum permissible velocity* refers to the lowest velocity which will prevent both sedimentation and vegetation growth. In general, an average velocity of 0.61 to 0.91 m/s will prevent sedimentation when the silt load of the flow is low. A velocity of 0.76 m/s is usually sufficient to prevent the growth of vegetation which could significantly

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affect the conveyance of the channel. The *maximum permissible velocity*, or the non-erodible velocity, is the greatest velocity that will not cause erosion of the channel. This velocity is very uncertain and variable, and can be estimated only with experience and judgment. In general, old and well-seasoned channels will stand much higher velocities than the new ones, because the old channel bed is usually better stabilized, particularly with deposition of colloidal matter. When other conditions are the same, a deeper channel will convey water at a higher mean velocity without erosion than a shallower one. This is probably because the scouring is caused primarily by the bottom velocities and, for the same mean velocity; the bottom velocities are greater in the shallower channel. For the concrete covered channels the maximum permissible velocity could reach up to 4.5 m/sec.

The side slopes of the channel are also an important factor in designing of the channel. The side slope is the function of the material. If the side slope is steeper than a certain slope it can loss its stability. So some suitable side slopes for channels built in various types of material are given in Table 8.6.

Table 8.6 Side slope angles for different bed materials

Material	**cotgθ (m)**
Silt	$1/1 \approx 1.25/1$
Ordinary loam	2/1
Silty loam	1.5 /1
Loose earth	$2.5/1 \approx 3/1$
Concrete	1/4

8.3 Non-Uniform Open Channel Flow

8.3.1 Introduction

In most practical cases, the cross-section, depth, and velocity in a channel vary along the channel and the uniform flow conditions are not often reached. Flow resistance, and changes in bottom slope, channel shape, discharge and surface conditions are taken into account. If a structure is built in a channel or any changing in the channel will cause the non-uniform flow conditions. Hence, the water depth and velocity will vary with distance ($dy/dx \neq 0$). Therefore the streamlines will not be parallel to each other resulting a pressure distribution different than the hydrostatic pressure distribution (Fig.8.20).

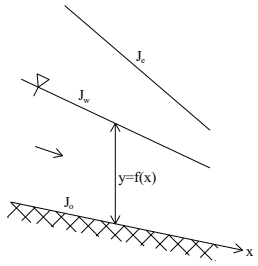
If the curvature of streamlines is substantial, the pressure gradient is not hydrostatic the flow is theoretically known as curvelinear flow. In the absence of additional information, the surface curvature and/or the bottom curvature can be used to give some indication of streamline curvature. The free surface and the channel bottom are both streamlines. Any change of flow stage is always accompanied with a change of streamline curvature. The effect of curvature is to produce appreciable acceleration components or centrifugal forces normal to the direction of flow. Thus, the pressure distribution over the section deviates from the hydrostatic if curvilinear flow occurs in the vertical plane. Such curvilinear flow may be either convex or concave. It is assumed that all streamlines are horizontal at the section under consideration. In concave flow the centrifugal forces are pointing downward to reinforce the gravity action; so the resulting pressure is greater than the hydrostatic pressure. In convex flow the centrifugal forces are acting upward against the gravity; consequently, the resulting pressure is less than the otherwise hydrostatic pressure of a parallel flow.

Non-uniform flow also termed, varied flow, is further classified as;

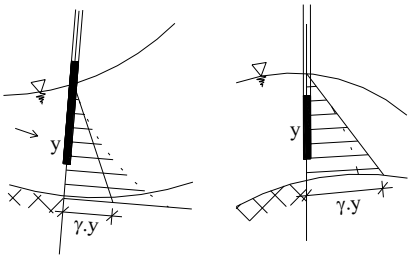
1) Non-uniform rapidly varied flow
2) Non-uniform gradually varied flow

The flow is rapidly varied if the depth changes abrubtly over a comperatively short distance such as is the case with a hydraulic jump and the hydraulic drop. A rapidly varied flow is also known as a local phenomenon. In the case of the gradually varied flow, the depth of flow changes rather slowly with distance such as is the case of a reservoir behind a dam (Fig. 8.21).

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(a)



Concave Convex

(b)

Figure 8.20 a) Non-uniform flow b) Pressure distribution

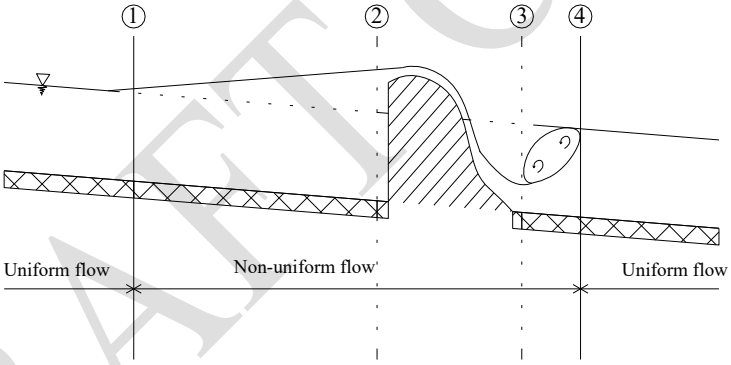


Figure 8.21 Rapidly and gradually varied flow

8.3.2 Specific Energy and Critical Depth

Specific energy in a channel section is defined as the energy per unit weight of water at any section of a channel measured with respect to the channel bottom. The energy per unit of liquid at a section of channel above some horizontal datum is (Fig. 8.22);

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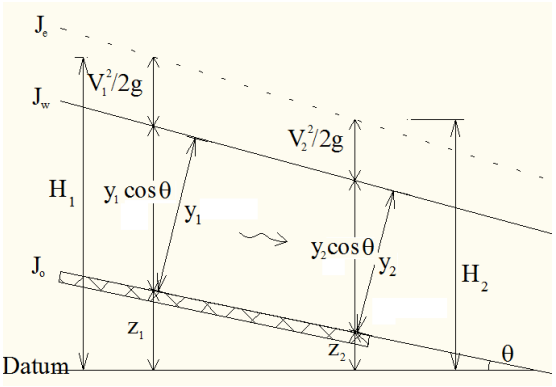


Figure 8.22 Energy in gradually varied open-channel flow

$$H = z + y\cos\theta + \alpha\frac{V^2}{2g}$$

For channels of small slope, $\cos\theta = 1$

$$H = z + y + \alpha\frac{V^2}{2g}$$

Specific energy becomes

$$E = y + \alpha\frac{V^2}{2g}$$

where α is kinetic energy correction factor or Coriolis coefficient that is used to correct for the non-uniformity of the velocity profile, ($1<\alpha<1.1$). The velocity head of an open channel is generally greater than the value computed according to the expression $V^2/2g$, where V is the mean velocity. Above equation indicates that the specific energy is equal to the sum of the depth of water and the velocity head. For simplicity, since V=Q/A,

$$E = y + \alpha\frac{Q^2}{2gA^2} \qquad (8.69)$$

for a given channel and discharge, Q, the specific energy in a channel section is a function of the depth of flow only.

As it is seen the specific energy is a function of two variables as water depth and discharge;

$$E = f(y, Q) \qquad (8.70)$$

In order to examine above equation, the variation of depth y will be studied by taking the specific energy and discharge separately.

1- Q=constant; E=f (y)

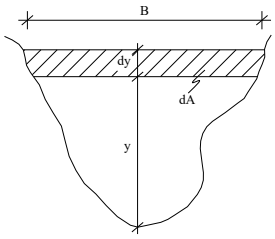


Figure 8.23 Arbitrary cross section of a channel

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Consider a cross-section of a channel (Fig. 8.23) through which a fluid is flowing at a constant rate Q, and examine the variation of energy with depth;

$$E = y + \frac{Q^2}{2gA^2} \qquad \text{and} \qquad A = f(y)$$

In the specific energy equation

If $y \to \infty$ $\quad \frac{Q^2}{2gA^2} \to 0$ and $E \to \infty$

If $y \to 0$ $\quad \frac{Q^2}{2gA^2} \to \infty$ and $E \to \infty$

Hence, $E \to \infty$ for both $y \to \infty$ and $y \to 0$. This shows the specific energy has a minimum value of $E=E_{min}$. For $E=E_{min}$ there is only one solution for y. This condition is obtained by differentiating and than equating zero, $\frac{dE}{dy} = 0$

$$\frac{dE}{dy} = 1 - \frac{2Q^2}{2gA^3}\frac{dA}{dy} \qquad \text{and} \qquad dA = B\,dy$$

$$\frac{dE}{dy} = 1 - \frac{Q^2}{gA^3}B$$

$$1 - \frac{Q^2 B}{gA^3} = 0 \Rightarrow \frac{Q^2 B}{gA^3} = 1 \tag{8.71}$$

The solution of the above equation will give only one y value; this value makes the specific energy minimum. This depth is called the ***critical depth***, y_c, and the corresponding velocity of flow is called the ***critical velocity,*** V_c. The flow is called the ***critical state of flow***.

$$\frac{Q^2}{g} = \frac{A^3}{B} \tag{8.72}$$

This expression is known as the general equation of the critical state and is valid for any type of cross-section. We obtain the below consequences;

1– For a given discharge, the specific energy must be at least E_{min} or $E>E_{min}$

2– When the depth of flow is plotted against the specific energy for a given channel section and discharge, a specific energy curve (Fig. 8.24) is obtained. This curve has two limbs AC and BC. The limb AC approaches the horizontal axis asymptotically toward the right. The limb BC approaches the line OD as it extends upward and to the right. Line OD is a line that passes through the origin and has an angle of inclination equal to 45°. At any point on this curve, the ordinate represent the depth, and the abscissa represents the specific energy, which is equal to the sum of the pressure head y and the velocity head $V^2/2g$.

The curve shows that for a given specific energy ($E>E_{min}$) there are two possible depths, for instance, the *low stage* y_2 and *high stage* y_1. When the depth of flow is greater than the critical depth, the velocity of flow is less than the critical velocity for a given discharge, and, hence, the flow is subcritical. When the depth of flow is less than the critical depth, the flow is supercritical. Hence, y_2 is the depth of a supercritical flow and y_1 is the depth of subcritical flow.

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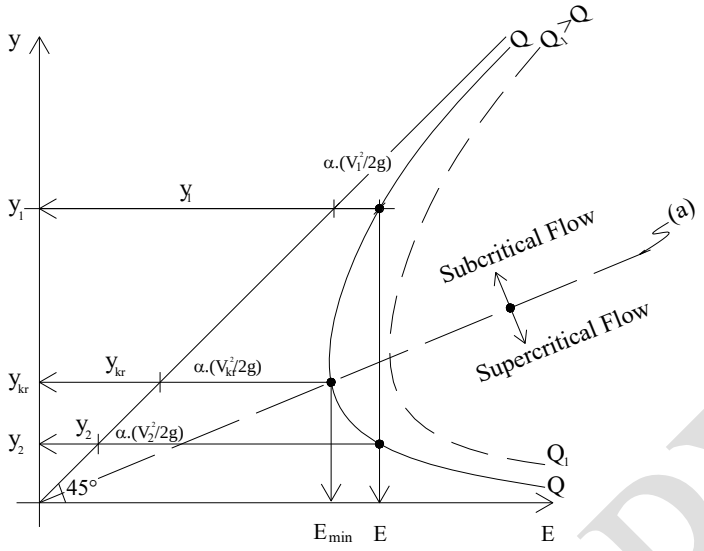


Figure 8.24 Specific energy curve

For a given discharge and specific energy $E > E_{min}$ the flow may have two different depths.

$y > y_c$ Subcritical flow

$y < y_c$ Supercritical flow

3– The dimensionless expression $\frac{Q^2 B_c}{g A_c^3} = 1$ is equal to the square of the Froude Number,

$$Fr^2 = \frac{V_c^2}{g\frac{A_c}{B_c}} = \frac{V_c^2}{g\, y_{m_c}} = 1 \qquad \text{or} \qquad Fr = 1$$

$Fr < 1$ Subcritical flow (where $Fr = \frac{V}{\sqrt{g\, y_m}}$)

$Fr > 1$ Supercritical flow

4–If the discharge changes, the specific energy will be changed accordingly. Q and Q′ represent positions of the specific energy curve when the discharge is less and greater respectively. The straight line "a" separates the supercritical and subcritical regions.

5– At the critical flow, the critical velocity could be obtained from the critical state expression:

$$\frac{(A_c V_c)^2}{g} = \frac{A_c^3}{B_c};$$

$$V_c = \sqrt{g\frac{A_c}{B_c}}$$

$\frac{A}{B} = y_m$ hydraulic depth;

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$$V_c = \sqrt{g\, y_{m_c}} \tag{8.73}$$

$$V_2 = \frac{Q}{A_2} > V_c = \frac{Q}{A_c} > V_1 = \frac{Q}{A_1}$$

$V < V_c$ Subcritical flow

$V > V_c$ Supercritical flow

The minimum energy could be obtained as below;

$$E_{min} = y_c + \frac{V_c^2}{2g} = y_c + \frac{y_{mc}}{2} \tag{8.74}$$

6– Investigation of the critical slope at critical flow conditions

$$V_c^2 = \left(\frac{1}{n_c} R_c^{2/3} J_c^{1/2}\right)^2 = g\, y_{mc}$$

$$J_c = \frac{n_c^2\, g\, y_{mc}}{R_c^{4/3}} \quad \text{or} \quad J_c = \frac{g\, y_{mc}}{C_c\, R_c} \tag{8.75}$$

V↑, J↑ or V↓, J↓

$J < J_c$ Subcritical flow

$J > J_c$ Supercritical flow

Special Case: Rectangular cross section

$$\frac{Q^2}{g} = \frac{(B_c\, y_c)^3}{B_c}; \qquad y_c = \sqrt[3]{\frac{Q^2}{B_c^2 g}}$$

Since B will be constant $B = B_c$ and $q = \frac{Q}{B}$ unit discharge of constant width;

$$y_{cr} = \sqrt[3]{\frac{q^2}{g}} \tag{8.76}$$

$$y_{m_c} = \frac{A_c}{B_c} = \frac{B.y_c}{B} = y_c; \qquad y_{m_c} = y_c \quad \text{and} \quad y_m = y$$

Hence the minimum energy will be as below;

$$E_{min} = y_c + \frac{y_c}{2} = \frac{3}{2} y_c; \qquad E_{min} = \frac{3}{2} y_c \tag{8.77}$$

The slope of the a line will be 2/3

Note: For a wide rectangular cross section $B >> h$

$$J_{kr} = g^{10/9} n_c^2 q^{-2/9} \tag{8.78}$$

2–E=constant; Q=f (y)

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If Q is obtained from specific energy equation;

$$Q = A\sqrt{2g(E-y)} \tag{8.79}$$

If $y \to 0$ $A \to 0$ and $Q \to 0$

If $y \to E$ $E\text{-}y \to 0$ and $Q \to 0$

The graphic of this equation will be as in Fig 8.25. And this expression shows that for a y value Q will be maximum; for $Q = Q_{max}$, $y = y_c$;

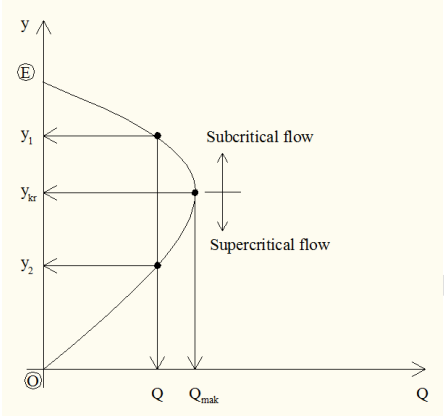


Figure 8.25 Koch parabola

$$\frac{dQ}{dy} = \frac{2g(E-y)\frac{dA}{dy} - Ag}{\sqrt{2g(E-y)}} = 0 \qquad \text{and} \qquad dA \cong B\,dy$$

$$\frac{2(E-y)B}{A} = 1 \qquad \text{or}$$

$$E - y_c = \frac{A_c}{2B_c}$$

is obtained.

If $(E-y_c)$ is obtained from $Q_{max} = A_c\sqrt{2g(E-y_c)}$ and substituted into the below expression;

$$E - y_c = \frac{Q_{max}^2}{A_c^2 2g}$$

$$\frac{Q_{max}^2}{A_c^2 2g} = \frac{A_c}{2B_c}$$

$$\frac{Q_{max}^2 B_c}{g A_c^3} = 1$$

The above equation is not different from the general equation of the critical flow. Hence, if a given energy is critical energy, the discharge will be maximum, $Q = Q_{max}$. In a channel of any cross-section, by using the general equation of the critical flow for a discharge the critical depth can be obtained, this could be explained with Fig. 8.26.

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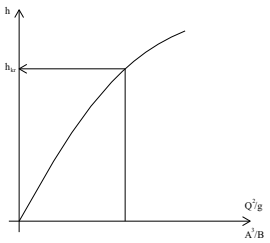


Figure 8.26 Critical flow curve

8.3.3 Momentum Function

Let's use momentum equation for the steady non-uniform open channel flow shown in Fig. 8.27;

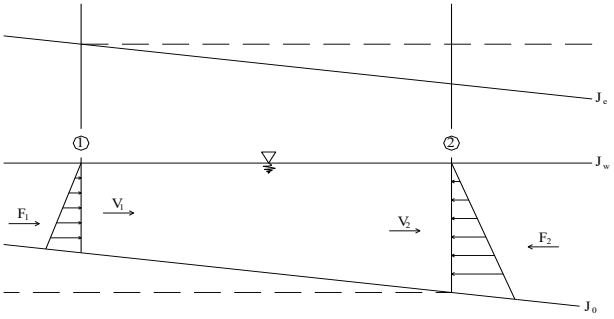


Figure 8.27 Non-uniform flow

In examining the application of Newton's second law of motion to the basic problems of steady open channel flow, it is convenient to begin with a general case problem such as the one shown schematically in Fig. 8.28. Within the control volume defined in this figure, there is an unknown energy loss and/or force acting on the flow between sections 1 and 2; the result is a change in the linear momentum of the flow. In many cases, the change in momentum is accompanied by a change in the depth of flow. The application of Newton's second law, in a one-dimensional form, to this control volume yields,

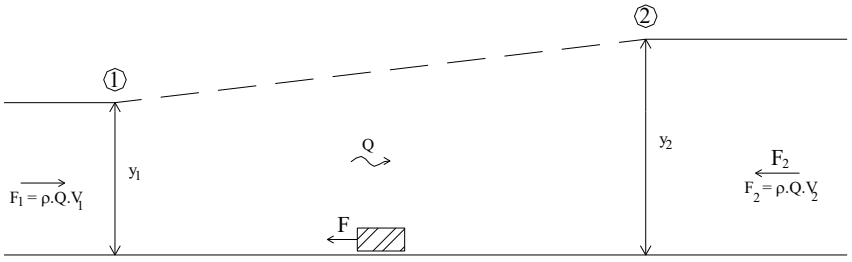


Figure 8.28 Application of momentum principle

$$F_1 - F - F_2 = \rho Q(\beta_2 V_2 - \beta_1 V_1) \tag{8.80}$$

where β Boussinesq coefficient.

For a parallel or gradually varied flow, the pressure forces may be computed by assuming hydrostatic pressure distribution. For a curvilinear or rapidly varied flow, however, the pressure distribution is no longer hydrostatic.

For $1 < \beta < \alpha$; $\left(\beta = \frac{1}{A V^2}\int_0^A V^2 dA\right)$

$$\rho g A_1 y_{1G} - F - \rho g A_2 y_{2G} = \rho\beta_2 Q^2/A_2 - \rho\beta_1 Q^2/A_1$$

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$$\frac{F}{\rho g}=\left(A_1 y_{1G}+\frac{\beta_1 Q^2}{A_1 g}\right)-\left(A_2 y_{2G}+\frac{\beta_2 Q^2}{A_2 g}\right) \tag{8.81}$$

In this equation the terms in parenthesis are shown with M and the equation can be written as below:

$$\frac{F}{\rho g}=M_1-M_2 \tag{8.82}$$

In applying the momentum principle to a short horizontal reach of a prismatic channel, the external force of friction (the weight of water if sloping channel) can be ignored. Thus the above momentum equation may be written

$$\left(A_1 y_{1G}+\frac{\beta_1 Q^2}{A_1 g}\right)=\left(A_2 y_{2G}+\frac{\beta_2 Q^2}{A_2 g}\right) \tag{8.83}$$

The two sides of the equation are analogous, hence may be expressed for any channel section by a general function

$$M=A_1 y_{1G}+\frac{\beta_1 Q^2}{A_1 g} \tag{8.84}$$

This function consists of two terms. The first term is the force per unit weight of water; the second term is the momentum of the flow passing through the channel section per unit time per unit weight of water. Since both terms are essentially force per unit weight of water, their summation may be called the specific force. Accordingly, Eq. (8.83) may be expressed as $M_1=M_2$. This means that the specific force of sections 1 and 2 are equal, provided that the external forces and the weight effect of the water in the reach between the two sections can be ignored.

By plotting the depth against the specific force for a given channel section and discharge, a specific-force curve is obtained (Fig. 8.29). For a given value of the specific force, the flow has two possible depths y_1 and y_2. As will be shown later, the two depths constitute the initial and sequent depths of a hydraulic jump. At a point on the curve the two depths become one, and the specific force is a minimum.

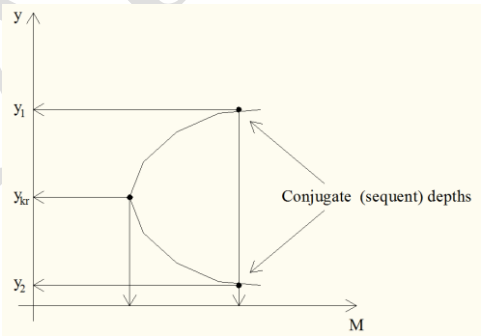


Figure 8.29 Specific force curve (Variation of momentum function)

If the momentum equation is applied to rectangular sections for unit width $y_G=\frac{y}{2}$, $\beta_1=\beta_2=1$;

$$M=\frac{y^2}{2}+\frac{q^2}{g\,y};$$

For $y\rightarrow 0$ and $y\rightarrow\infty$ momentum will be infinite. In this case the momentum function should have a minimum value. For a minimum value of the specific force, the first derivative of M with respect to y should be zero;

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For $\dfrac{dM}{dy} = 0$ $\qquad y = y_c = \sqrt[3]{\dfrac{q^2}{g}}$ (8.85)

This is the criterion for the critical state of flow derived earlier. Therefore, it is proved that the depth at the minimum value of the specific force is the critical depth. ***At the critical state of flow the specific force is a minimum for the given discharge***.

When the rapid change in the depth of flow is from a low stage (supercritical flow) to a high stage (subcritical flow), the result is usually an abrupt rise of water surface. This local phenomenon is known as the ***hydraulic jump (standing wave).*** It occurs frequently in a channel below a regulating sluice, at the foot of a spillway, or at the place where a steep channel slope suddenly turns flat.

The relation between the initial depth and the sequent depth of a hydraulic jump on a horizontal floor in a rectangular channel will be obtained. The external forces of friction and the weight effect of water in the hydraulic jump on a horizontal floor are negligible, because the jump takes place in a relatively short distance and the slope angle of the horizontal floor is zero. The specific forces of section 1 and 2, respectively, before and after the jump, can therefore be considered equal; that is

$$A_1 y_{1G} + \frac{Q}{g}(\beta_1 V_1 - \beta_2 V_2) - A_2 y_{2G} = 0 \tag{8.86}$$

For $(\beta_1 = \beta_2)$

$$\left(A_1 y_{1G} + \frac{Q^2}{A_1 g}\right) - \left(A_2 y_{2G} + \frac{Q^2}{A_2 g}\right) = 0$$

$$\left(b_1 y_1 \frac{y_1}{2} + \frac{q^2 b_1^2}{b_1 y_1 g}\right) - \left(b_2 y_2 \frac{y_2}{2} + \frac{q^2 b_2^2}{b_2 y_2 g}\right) = 0$$

$$b_1 = b_2 \Rightarrow \frac{y_1^2}{2} + \frac{q^2}{y_1 g} - \left(\frac{y_2^2}{2} + \frac{q^2}{y_2 g}\right) = 0$$

$$\frac{y_1^2}{2} + \frac{q^2}{y_1 g} = \frac{y_2^2}{2} + \frac{q^2}{y_2 g}$$

$$\frac{y_1^2}{2} + \frac{y_1^2 V_1^2}{y_1 g} = \frac{y_2^2}{2} + \frac{y_1^2 V_1^2}{y_2 g}$$

$$\frac{y_2^2 - y_1^2}{2} = \frac{V_1^2 y_1}{g} - \frac{V_1^2 y_1^2}{g y_2}$$

$$\frac{y_2^2 - y_1^2}{2} = \frac{V_1^2 y_1 y_2 - V_1^2 y_1^2}{g y_2}$$

$$\frac{y_2^2 - y_1^2}{2} = \frac{V_1^2 y_1 (y_2 - y_1)}{g y_2}$$

$$\frac{y_2 + y_1}{2} = \frac{V_1^2 y_1}{g y_2} \quad \text{or} \quad y_2 + y_1 = \frac{2 V_1^2 y_1}{g y_2}$$

$$y_2^2 + y_1 y_2 - 2\frac{V_1^2 y_1}{g} = 0$$

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + 2\frac{V_1^2 y_1}{g}} \tag{8.87}$$

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or

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + 2\frac{q^2}{g\,y_1}} \tag{8.88}$$

or

$$\frac{y_2}{y_1} = \frac{1}{2}\left(-1 + \sqrt{1 + 8\frac{V_1^2}{g.y_1}}\right)$$

$$\frac{y_2}{y_1} = \frac{1}{2}\left(-1 + \sqrt{\left(1 + 8Fr_1^2\right)}\right) \tag{8.89}$$

For a given Froude number Fr_1 of the approaching flow, the ratio of the sequent depth to the initial depth is given by the above equation, and or

$$\frac{y_1}{y_2} = \frac{1}{2}\left(-1 + \sqrt{\left(1 + 8Fr_2^2\right)}\right) \tag{8.90}$$

It should be understood that the momentum principle is used in this solution because the hydraulic jump involves a high amount of internal-energy losses that cannot be evaluated in the energy equation.

Conclusions: As obtained above the flow corresponding to the minimum energy and the maximum discharge is called as the critical flow. As seen from the specific energy curve, Koch parabola and the momentum function;

1– For a given specific energy the maximum discharge,
2– For a given discharge the minimum energy, and
3- For a given discharge the minimum momentum

is obtained only for the critical depth.

8.3.4 The Properties of the Subcritical and Supercritical Flow

1– Transition from the supercritical flow to subcritical flow is discontinuous. Why? This variation occurs with the hydraulic jump (standing wave) (Fig. 8.30).

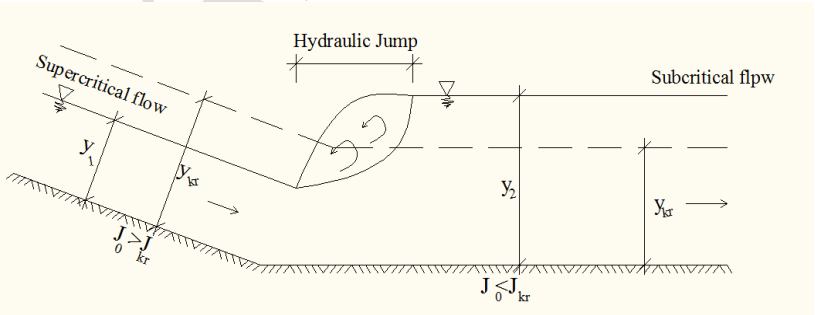


Figure 8.30 Transition from supercritical flow to subcritical flow

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2– Transition from subcritical flow to supercritical flow is continuous. Why? (Fig. 8.31).

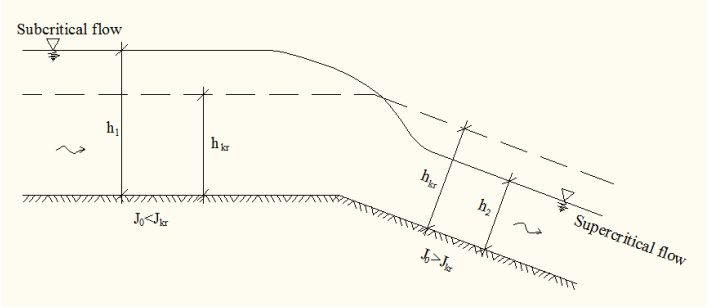


Figure 8.31 Transition from subcritical flow to supercritical flow

3– Wave celerity; the distinguishing feature of flows involving a free surface is the opportunity for the free surface to distort into various shapes. The surface of a lake or the ocean is seldom "smooth as a mirror". It is usually distorted into ever-changing patterns associated with surface waves. Some of these waves are very high, some barely ripple the surface; some waves are very long, some are short; some are breaking waves that form whitecaps, others are quite smooth. So the properties of these waves depend on the water depth, h, wave height, H, and wave length, L (Fig. 8.32).

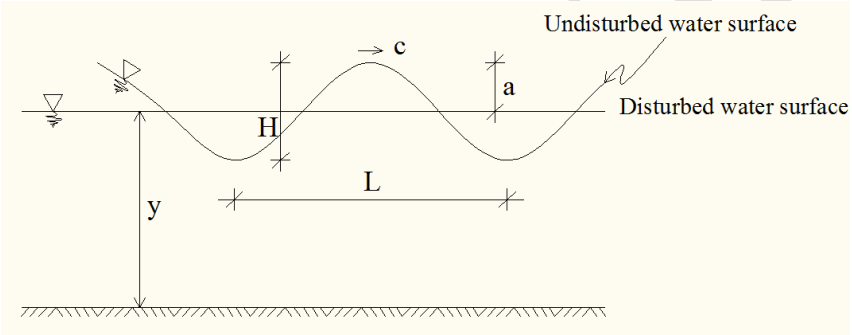


Figure 8.32 Wave characteristics

Celerity of small amplitude long waves is equal to the velocity of the critical flow $\sqrt{g\,y_{mc}}$. Since the wavelength of this type of waves is very big from water depth, depth of water is an important factor. These type of waves are called "Shallow Water Waves"

Let's consider the wave celerity for the small amplitude waves (H<<L).

$$c^2 = \frac{gL}{2\pi}\tanh\frac{2\pi y}{L} \tag{8.91}$$

If L>>y then $2\pi y/L$ will get smaller, therefore from the properties of the hyperbolic functions;

$$\tanh\frac{2\pi y}{L} \approx \frac{2\pi y}{L}$$

$$c^2 = \frac{gL}{2\pi}\frac{2\pi y}{L}$$

or

$$c^2 = gy \tag{8.92}$$

$$c = \sqrt{gy} \tag{8.92a}$$

This is the equation for the propagation of small waves in rectangular channels. It can be shown that the celerity of small waves in non-rectangular channel is

$$c = \sqrt{g y_m} \tag{8.92b}$$

where wave celerity, c, water depth, y, and hydraulic depth, y_m. The wave celerity depends on the water depth. The wave celerities depending on the depth were given in Table 8.6.

Table 8.6 Wave types and wave celerities

y/L	**Wave Type**	**Wave Celerity**
$< \frac{1}{20}$	Shallow water wave	$c \cong \sqrt{g.y}$
$\frac{1}{20} - \frac{1}{2}$	Transition wave	$c = \left(\frac{gL}{2\pi} \tanh \frac{2\pi y}{L} \right)^{1/2}$
$> \frac{1}{2}$	Deep water wave	$c = \sqrt{\frac{gL}{2\pi}}$

The equation of celerity may be used for studying the propagation of gravity waves. If a pebble (stone) is dropped into still water, the wave pattern can be represented by the concentric circles. The waves travel away from the source of the disturbance in all directions with a velocity or celerity equal to c.

If the Froude Number is considered

$$Fr = \frac{V}{\sqrt{gy}} = \frac{V}{c} \tag{8.93}$$

it is the ratio of the average velocity to wave celerity.

If the water is flowing, the wave pattern produced by a disturbance will be displaced in the direction of flow.

1– When the flow is subcritical, **Fr<1**, the velocity of flow is less than the celerity. Therefore, such a wave can propagate upstream against the flow, and upstream areas are in hydraulic communication with the downstream areas.

$$Fr = \frac{V}{c} < 1; \quad V < c \tag{8.93a}$$

As it is seen from Fig. 8.33 wave propagates to upstream with a velocity of $V_w = (V - c)$ and to downstream with a velocity of $V_w = (V + c)$. Any small change in the downstream flow conditions affects subcritical flows. Therefore, subcritical flows are controlled by downstream conditions. This is called a *downstream flow control.*

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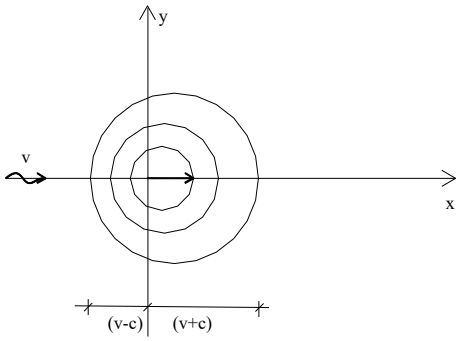


Figure 8.33 Subcritical flow

2–When the flow is critical flow, **Fr=1**, the velocity of water is equal to the celerity, the wave fronts in the upstream direction are stationary, or $V_w=0$, and those in the downstream direction have a velocity equal to $V_w=2c$ (Fig. 8.34);

$$Fr = \frac{V}{c} = 1; \quad V = c \tag{8.94}$$

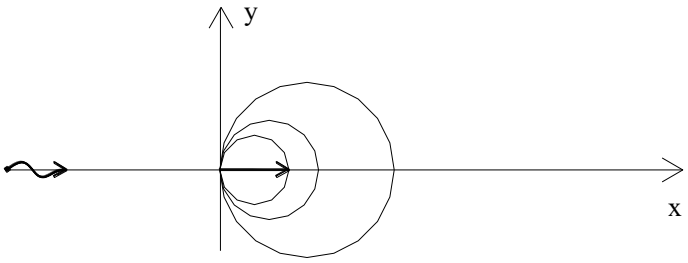


Figure 8.34 Critical flow

3– When the flow is supercritical, **F>1**, the velocity of flow is greater than the celerity. Therefore, such a wave cannot propagate upstream against the flow, the waves will travel downstream only, and the upstream areas of the channel are not in hydraulic communication with the downstream areas. Hence, supercritical flows can only be controlled from upstream. This is called *upstream flow contro*l.

$$Fr = \frac{V}{c} > 1; \quad V > c \tag{8.95}$$

So, any calculations should be made from the upstream to downstream end.

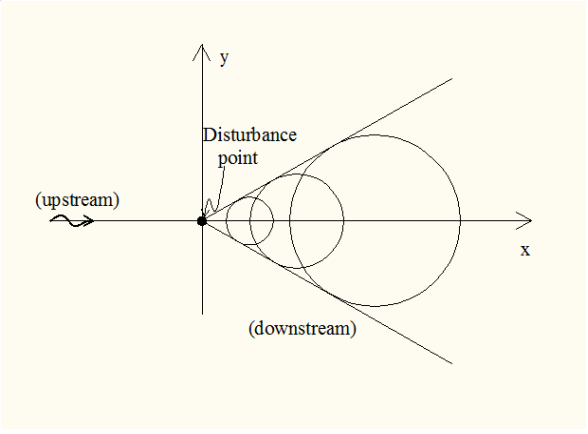


Figure 8.35 Supercritical flow

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The celerity c must be clearly distinguished from the absolute wave velocity V_w. The celerity is the velocity of a wave relative to the velocity of flow.

8.3.5 Classification of Channel Slopes

The character of a gradually varying flow is often classified in terms of the actual slope, J_0, compared with the slope required to produce uniform critical slope, J_{cr}. The slope of a channel that sustains a given discharge at a uniform and critical slope is called the ***critical slope*** J_c. A slope of a channel less than the critical slope will cause a slower flow of subcritical state for a given discharge, hence is called a subcritical slope. A slope greater than the critical slope will result in a faster flow of supercritical slope, and is called a steep or supercritical slope.

$J_0 > J_c$	Supercritical	flow
$J_0 = J_c$	Critical	flow
$J_0 < J_c$	Subcritical flow	

Because of the variation of the channel slope the flow depth changes from y_1 to y_2. But during the change the flow depth passes from a critical depth y_{cr}. Hence this section where critical state occur this section is called as control section. And the calculations start from this section (Fig. 8.36).

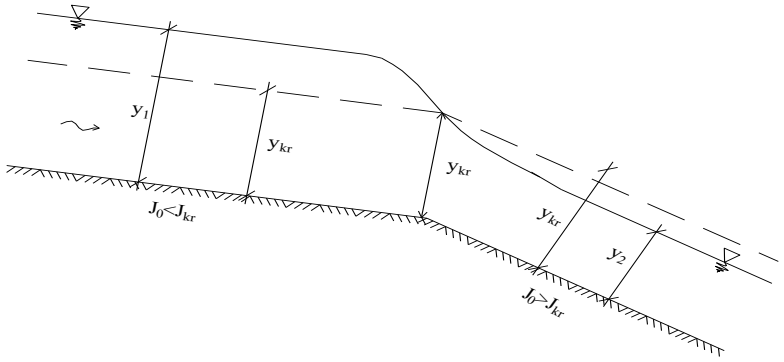


Figure 8.36 The variation of channel flow conditions

The surface profiles of non-uniform flows are classified by the channel slope as below;

$J_0 > J_c$	Steep Slope (the flow would be supercritical if it were uniform depth)
$J_0 = J_c$	Critical Slope (the flow would have Fr=1 if it were uniform depth)
$J_0 < J_c$	Mild Slope (the flow would be subcritical if it were uniform depth)
$J_0 < 0$	Adverse Slope (flow uphill)
$J_0 = 0$	Horizontal Slope

Let's write the Strickler Equation for both uniform flow and critical flow and than sum them up

$$q = k y_c^{5/3} J_c^{1/2}; \qquad q = k y_0^{5/3} J_0^{1/2};$$

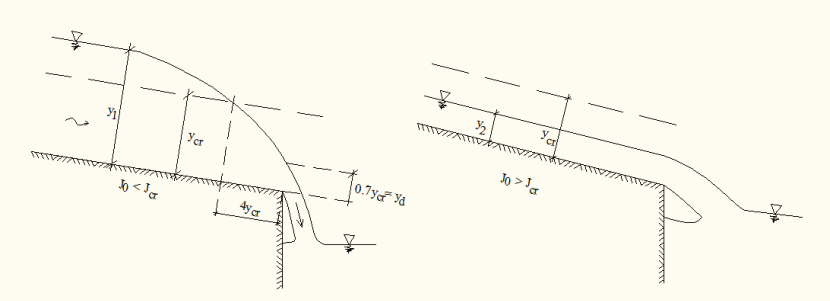
$$\frac{J_0}{J_c} = \left(\frac{y_c}{y_0}\right)^{10/3} \tag{8.96}$$

and from the expression

$$J_0 < J_c \rightarrow y_0 > y_c, \qquad J_0 = J_c \rightarrow y_0 = y_c, \qquad J_0 > J_c \rightarrow y_0 < y_c$$

At the overfall, the flow becomes suddenly accelerated. If the upstream flow is subcritical, critical depth is expected to take place at the transition. The specific energy will continue to be dissipated on the way downstream and finally reach a minimum energy content E_{min}. The flow at the brink is

actually curvilinear, for the curvature of flow is pronounced; hence, the method is invalid for determining the critical depth as the depth of brink. The actual situation is that the brink section is the true section of minimum energy, but it is not the critical section as computed by the principle based on parallel-flow assumption. It was found that for small slopes the computed critical depth is about 1.4 times the brink depth, y_c=1.4 y_d, and that is located about 3 y_c to 4 y_c behind the brink in the channel.



(a) Subcritical approaching flow (b) Supercritical approaching flow

Figure 8.37 Free overfall

8.3.6 Rapidly Varied Flow

8.3.6.1 Introduction

In many open channels, flow depth changes occur over a relatively short distance so that dx/dy~1. The flow changes are local. Then, friction loss and bottom slope effect may be neglected. Such rapidly varied flow conditions are often quite complex and difficult to analyze in a precise fashion. Fortunately, many useful approximate results can be obtained by using a simple one-dimensional model along with appropriate experimentally determined coefficients when necessary. In many cases the flow depth may change significantly in a short distance.

Some rapidly varied flows occur in constant area channels for reasons that are not immediately obvious. The hydraulic jump is one such case. In hydraulic jump flow, the flow may change from a relatively shallow, high-speed condition into a relatively deep, low-speed condition within a horizontal distance of just a few channel depths. Other rapidly varied flows may be due to a sudden change in the channel geometry such as the flow in an expansion or contraction section of a channel. For example, broad-crested weirs, sharp-crested weirs, bridge piers, channel bends and critical flow flumes may cause rapidly varied flow.

Flow regimes have different effects on flow structure because the variation of the free surface profiles depends on incident flow regimes.

The transition from supercritical to sub-critical flow occurs through the hydraulic jump (standing wave). Hydraulic jump is very important subject for the rapidly varied flow; so firstly, it will be explained in the following sections.

Basic concepts may be obtained using the specific energy. If the local loss is not known in the rapidly varied flow, the flow structure may be described using the momentum equation.

8.3.6.2 The Hydraulic Jump

A hydraulic jump is a step-like increase in fluid depth in an open channel. Hydraulic jumps result when there is a conflict between upstream and downstream controls which influence the same reach of channel. For example, if the upstream control causes supercritical flow while the downstream control dictates sub-critical flow, then there is a conflict which can be resolved only if there is some means for the flow to pass from one flow regime to the other. The transition from a supercritical flow to a sub-critical flow is called a *hydraulic jump*. The hydraulic jump is accompanied by significant turbulence and energy dissipation. It is extremely turbulent. It is characterized by the development of the large-scale turbulence region is usually called the *roller*. The length of the hydraulic jump is a function of the upstream flow conditions. The length of the roller (L_r) is between the toe of jump and the end of roller i.e. where the velocity is zero at free surface. While, the length of the jump (L_j) is between the

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toe of jump and the point where the velocity profile is uniform, i.e. jump length is the distance where free surface is flat (Fig.8.38). A hydraulic jump is a region of rapidly varied flow.

Generally, if a river or a channel has mild slope, the regime is sub-critical. If a dam or a gate is constructed in a river, the energy will increase because the friction loss decreases. The velocity increases at downstream of the dam and it flows with supercritical regime. Energy gradient increases due to increasing of velocity and hence friction loss (Fig. 8.38). When flow has transient conditions, the flow occurs in critical condition and the depth increases. Finally we may observe a hydraulic jump. Large amount of additional energy due to hydraulic jump is lost. Hence the energy gradient decreases after hydraulic jump.

The fact that there is an energy loss across a hydraulic jump is useful in many situations. For example, the relatively large amount of energy contained in the fluid flowing down the spillway of a dam could cause damage to the channel below the dam. By placing suitable flow control objects in the channel downstream of the spillway, it is possible (if the flow is supercritical) to produce a hydraulic jump on the apron of the spillway and thereby dissipate a considerable portion of the energy of the flow. That is, the dam spillway produces supercritical flow, and the channel downstream of the dam requires sub-critical flow. The resulting hydraulic jump provides the means to change the character of the flow.

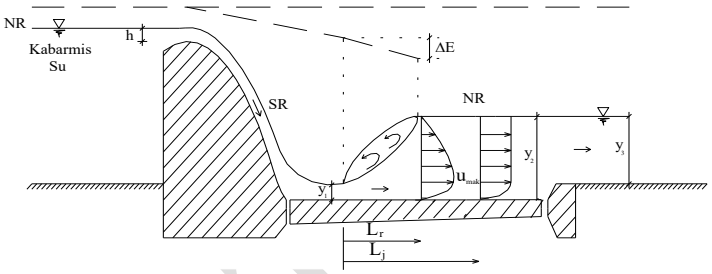


Fig. 8.38 Flow over a weir

The simplest type of hydraulic jump occurs in a horizontal channel, rectangular channel. It is assumed that the flow at section (1) and (2) is nearly uniform, steady, and one dimensional as is indicated in Fig. 8.39. In addition, the wall shear stress is neglected with in the relatively short segment between these two sections. Under these conditions, the downstream condition may be found using momentum and continuity equations by the contribution of the upstream condition.

If the loss of energy is known as shown in Fig 8.39, the energy equation may be written as follows,

$$E_1 = E_2 + \Delta E \tag{8.97}$$

Contrary, the momentum equation may be used;

$$M_1 = M_2 \tag{8.98}$$

In the previous section, the relation between the conjugate depths (or sequent depths) in a rectangular channel was found using the momentum equation (Belanger equation).

$$y_2 = -\frac{y_1}{2} + \sqrt{\frac{y_1^2}{4} + \frac{2q^2}{g y_1}} \qquad \text{or} \qquad \frac{y_2}{y_1} = \frac{1}{2}\left(-1 + \sqrt{\left(1 + 8Fr_1^2\right)}\right)$$

If the difference between the conjugate depths decreases, the hydraulic jump becomes weak and the free surface appears wavy which moves to downstream.

The energy equation gives the head loss;

$$\Delta E = E_1 - E_2$$

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$$\Delta E = y_1 + \frac{V_1^2}{2g} - y_2 - \frac{V_2^2}{2g} = y_1 - y_2 + \frac{1}{2g}\left(\frac{q^2}{y_1^2} - \frac{q^2}{y_2^2}\right)$$

$$\Delta E = y_1 - y_2 + \frac{q^2}{2g}\left(\frac{y_2^2 - y_1^2}{y_1^2 y_2^2}\right)$$

$$\Delta E = y_1 - y_2 + \frac{q^2}{2g}\frac{1}{y_1^2 y_2^2}(y_2 - y_1)(y_2 + y_1)$$

$$\Delta E = (y_1 - y_2)\left(1 - \frac{q^2(y_1 + y_2)}{2g y_1^2 y_2^2}\right)$$

$$\Delta E = (y_1 - y_2)\left(\frac{2g y_1^2 y_2^2 - q^2(y_1 + y_2)}{2g y_1^2 y_2^2}\right)$$

From previous momentum equation,

$$\frac{y_1 + y_2}{2} = \frac{V_1^2 y_1}{g y_2}$$

$$\frac{y_1 + y_2}{2} = \frac{q^2 y_1}{y_1^2 g y_2}$$

$$\frac{y_1 + y_2}{2} = \frac{q^2}{g y_1 y_2}$$

$$g y_1 y_2 = \frac{2q^2}{(y_1 + y_2)}$$

$$q^2 = \frac{g y_1 y_2 (y_1 + y_2)}{2}$$

$$\Delta E = (y_1 - y_2)\left(\frac{2g y_1^2 y_2^2 - \frac{g}{2} y_1 y_2 (y_1 + y_2)^2}{2g y_1^2 y_2^2}\right)$$

$$\Delta E = (y_1 - y_2)\left(\frac{4 y_1 y_2 - (y_1 + y_2)^2}{4 y_1 y_2}\right)$$

$$\Delta E = (y_1 - y_2)\left(\frac{4 y_1 y_2 - y_1^2 - 2 y_1 y_2 - y_2^2}{4 y_1 y_2}\right) = -\frac{(y_2 - y_1)^2}{4 y_1 y_2}(y_1 - y_2)$$

$$\Delta E = (y_2 - y_1)\frac{(y_2 - y_1)^2}{4 y_1 y_2}$$

$$\Delta E = \frac{(y_2 - y_1)^3}{4 y_1 y_2} \qquad (8.99)$$

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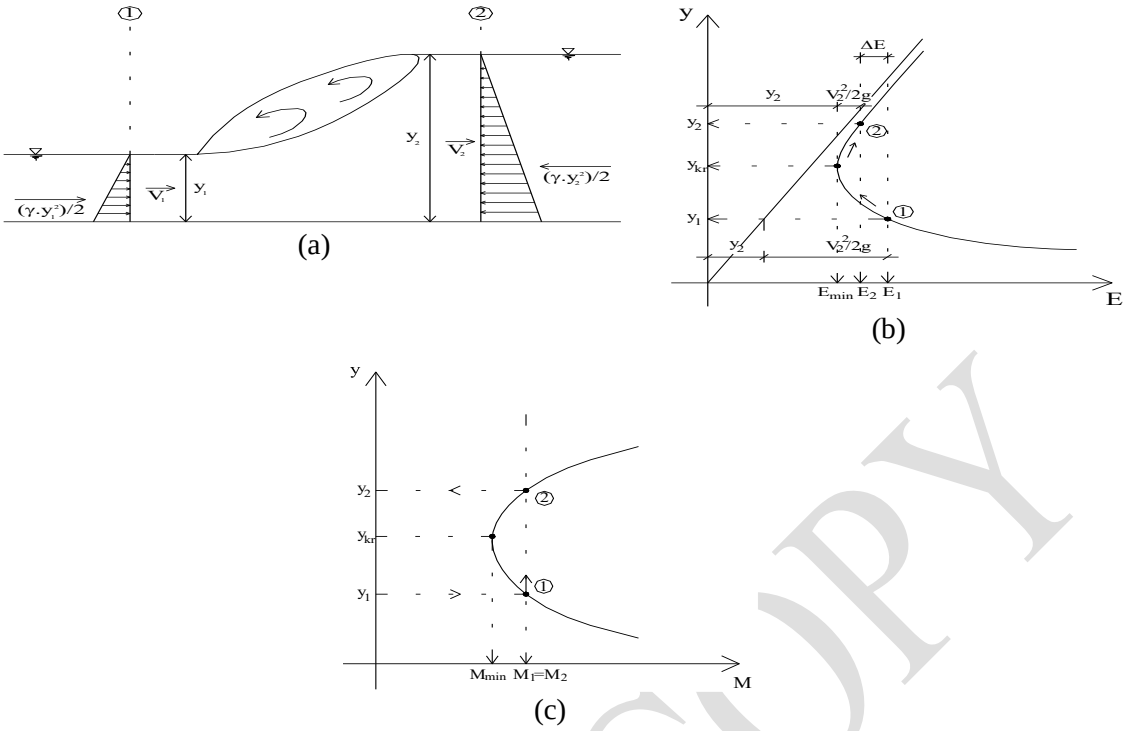


Fig. 8.39 Hydraulic jump-Specific energy and momentum function

Hydraulic jump may be classified as follows;

a) $Fr_1 = 1$ Jump impossible, critical depth

b) $Fr_1 = 1 - 1.7$ Standing wave or undulant jump (Fig. 8.40)

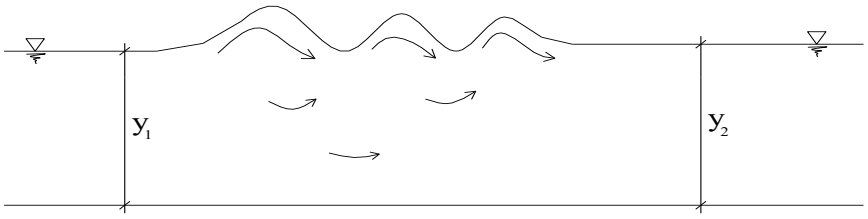


Fig 8.40 Undulant jump ($L_j \cong 3y_2$)

c) $Fr_1 = 1.7 - 2.5$ Weak jump (Fig. 8.41).

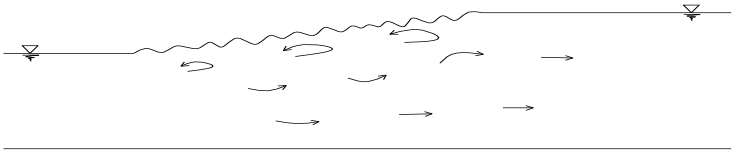


Fig. 8.41 Weak jump ($L_j \cong 4y_2$)

d) $Fr_1 = 2.5 - 4.5$ Oscillating jump (Fig. 8.42).

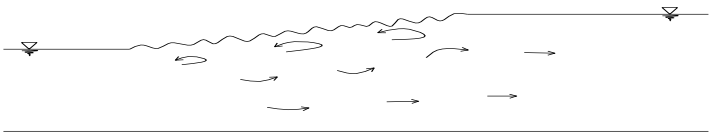


Fig. 8.42 Oscillating jump ($L_j \cong 5y_2$)

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e) $Fr_1 = 4.5 - 9$ Steady jump (Fig. 8.43).



Fig. 8.43 Steady jump ($L_j \cong 6y_2$)

f) $Fr_1 > 9$ Strong jump (Fig. 8.44).

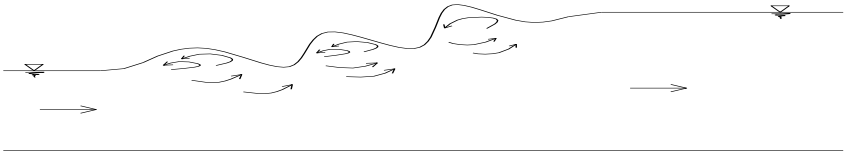


Fig. 8.44 Strong jump ($L_j \cong 6y_2$)

In the field of open channel flow, the applications of the hydraulic jump are many and include the following;

1-The dissipation of energy in flows is over dams, weirs, and other hydraulic structures.

2-The maintenance of high water levels in channels for water distribution purposes.

3-The increase of the discharge of a sluice gate by repelling the downstream tail water and thus increasing the effective head across the gate.

4-The reduction of uplift pressure under structures by raising the water depth on the apron of the structure.

5-The mixing of chemicals used for water purification or wastewater treatment

6-The aeration of flows and the dechlorination of wastewater.

7-Removal of air pockets from open channel flows in circular channels.

8-The identification of special flow conditions are such as the existence of supercritical flow or the presence of a control section for the cost effective measurement of flow.

Hydraulic jumps can and do occur in channels of such a slope that the gravitational forces acting on the flow must be considered. The six types of jumps defined in Fig.8.45.

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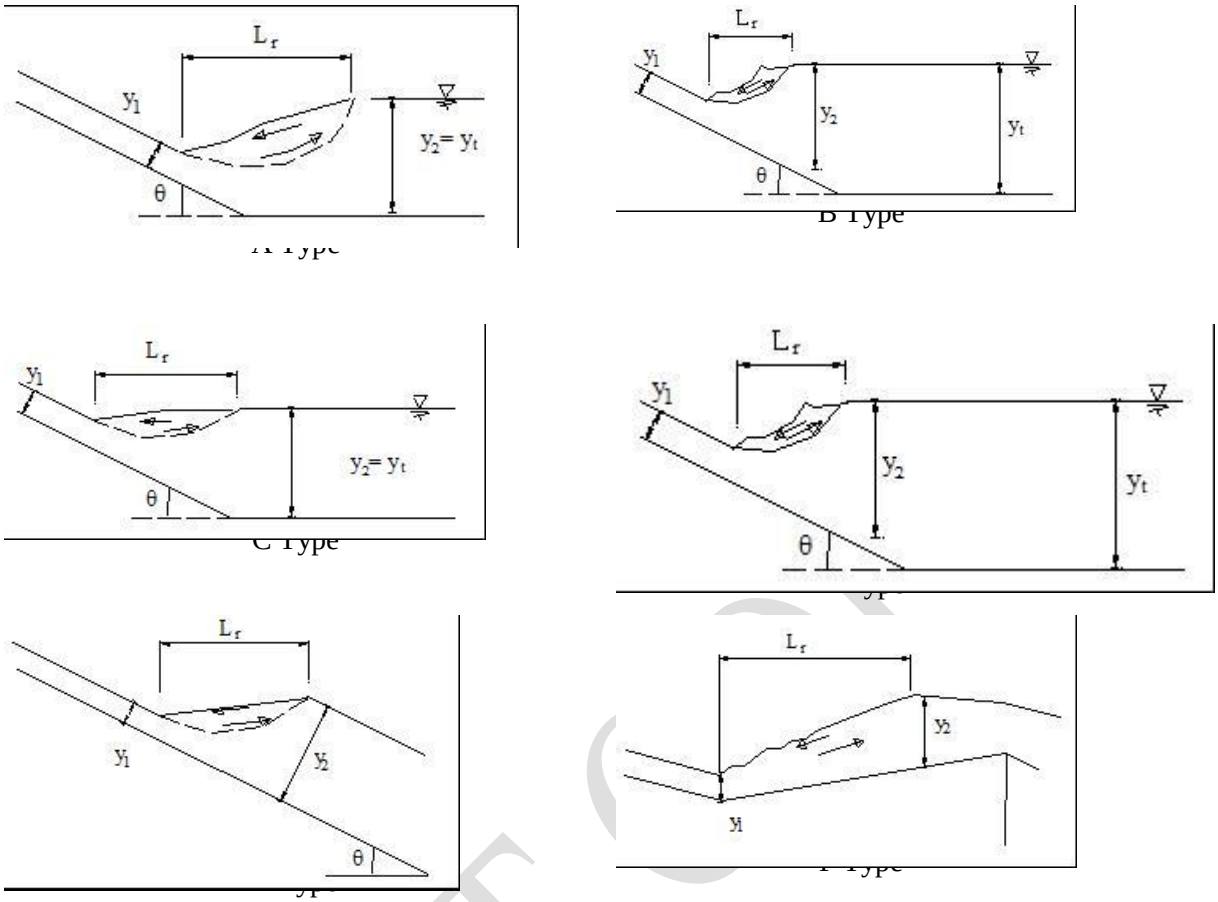


Fig. 8.45 Hydraulic jumps in sloping channels

In the previous section, methodologies for determining the downstream depth, which must prevail if a hydraulic jump is to form, given a specified supercritical upstream condition, were discussed. If the downstream depth is less than the sequent depth y_2, then a jump is not formed, and the supercritical flow maintains itself. If the downstream depth is greater than y_2, a submerged jump is formed (Fig. 8.46). Submerged jumps commonly form downstream of gates in irrigation systems, and the crucial unknown in such situations is the depth of submergence y_3.

Free jump for	$y_2=y_{downstream}$
Submerged jump for	$y_2<y_{downstream}$
No jump for	$y_2>y_{downstream}$
No jump for	$y_2<<y_{downstream}$

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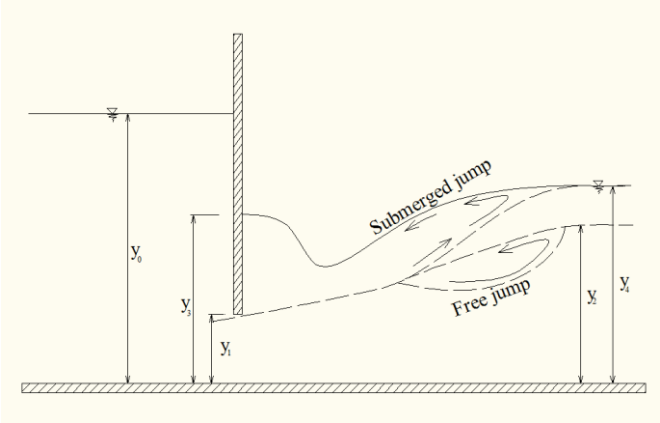


Fig 8.46 Definition sketch for submerged jump

Fig. 8.46 shows the typical submerged hydraulic jump formed in a horizontal channel. A submerged hydraulic jump is the jump where the toe is covered by water and the atmosphere has no direct access to the body of the jump (Hager,1992). Using the principles of conservation of momentum and mass, the relations for submerged jump are given in horizontal, rectangular channels. Chow (1959) gave,

$$\frac{y_3}{y_4} = \left[1 + 2\,Fr_4^2\left(1 - \frac{y_4}{y_1}\right)\right]^{1/2} \tag{8.100}$$

and Govinda Rao(1963) gave

$$\frac{y_3}{y_1} = \left[(1+S)^2\,\phi^2 - 2\,Fr_1^2 + \frac{2\,Fr_1^2}{(1+S)\,\phi}\right]^{1/2} \tag{8.101}$$

where

$S = \dfrac{y_4 - y_2}{y_2}$ is submergence factor

$$\phi = \frac{y_2}{y_1} = \frac{1}{2}\left(\sqrt{1 + 8Fr_1^2} - 1\right)$$

The occurrence of hydraulic jump in nonrectangular but prismatic channels, some relations are available;

Circular conduit;

$$y_2 = \frac{y_{kr}^2}{y_1} \quad \text{for} \quad Fr_1 < 1.7 \tag{8.102}$$

$$y_2 = \frac{y_{kr}^{1.8}}{y_1^{0.73}} \quad \text{for} \quad Fr_1 > 1.7 \tag{8.103}$$

$$Fr_1 = \left(\frac{y_{kr}}{y_1}\right)^{1.93}$$

$$y_{kr} = \frac{1.01}{D^{0.264}}\left(\frac{Q\sqrt{\alpha}}{\sqrt{g}}\right)^{0.506}$$

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Triangular conduit;

$$\left(\frac{y_2}{y_1}\right)^3 - 1 = 1.5\,Fr_1^2\left[1-\left(\frac{y_1}{y_2}\right)^2\right] \tag{8.104}$$

Parabolic conduit;

$$\left(\frac{y_2}{y_1}\right)^{2.5} - 1 = 1.67\,Fr_1^2\left[1-\left(\frac{y_1}{y_2}\right)^{1.5}\right] \tag{8.105}$$

Trapezoidal conduit;

$$k^2\frac{A_2}{A_1}\frac{y_2}{y_1} - k^2 = Fr_1^2\frac{A_1}{y_1 B_1}\left(1-\frac{A_1}{A_2}\right) \tag{8.106}$$

$$k = \frac{b}{m\,y_1}$$

The energy loss in nonrectangular channels;

$$\frac{\Delta E}{E_1} = \frac{2\,y_1/y_{m1}\left[1(y_2/y_1)\right] + Fr_1^2\left[1-(A_1/A_2)^2\right]}{(2\,y_1/y_{m1}) + Fr_1^2} \tag{8.107}$$

The jump lengths are for free jumps;

Rectangular channels;

$$\frac{L_j}{y_1} = 9.75(Fr_1 - 1)^{1.01} \tag{8.108}$$

Parabolic channels;

$$\frac{L_j}{y_1} = 11.7(Fr_1 - 1)^{0.832} \tag{8.109}$$

Triangular channels;

$$\frac{L_j}{y_1} = 4.26(Fr_1 - 1)^{0.695} \tag{8.110}$$

Trapezoidal channels;

$$k=16 \quad \text{and} \quad \frac{L_j}{y_1} = 5.36(Fr_1 - 1)^{0.905} \quad \text{for} \quad m=2 \tag{8.111}$$

$$k=8 \quad \text{and} \quad \frac{L_j}{y_1} = 7.01(Fr_1 - 1)^{0.885} \quad \text{for} \quad m=1 \tag{8.112}$$

$$k=4 \quad \text{and} \quad \frac{L_j}{y_1} = 10.67(Fr_1 - 1)^{0.836} \quad \text{for} \quad m=0.5 \tag{8.113}$$

The jump length and roller length are for submerged jump;

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$$\frac{L_j}{y_2} = 4.9S + 6.1 \tag{8.114}$$

$$S = \frac{y_4 - y_2}{y_2}$$

$$\frac{L_r}{y_{kr}} = \frac{3.31}{\left[\left(y_4 - y_3 / y_3\, Fr_1\right)\right]^{0.885}} \tag{8.115}$$

Notes (Chanson, 1999)

1-In a hydraulic jump, the downstream flow depth y_2 is always larger than the upstream flow depth y_1 .

2-The main parameter of a hydraulic jump is its upstream Froude number Fr_1.

3-If only the downstream flow conditions are known, the solution of the continuity and momentum equations is given by Belanger equation.

4-A hydraulic jump is a very effective way of dissipating energy. It is also an effective mixing device. Hydraulic jumps are commonly used at the end of a spillway or in a dissipation basin to destroy much of the kinetic energy of the flow. The hydraulic power dissipated in a jump equals; $\rho g Q \Delta E$, where ΔE is computed using equation (8.99).

5- Hydraulic jumps are characterized by air entrainment. Air is trapped at the impingement points of the supercritical inflow with the roller. The rate of air entrainment may be approximated as;

$$\frac{Q_{air}}{Q} = 0.018(Fr_1 - 1)^{1.245}$$

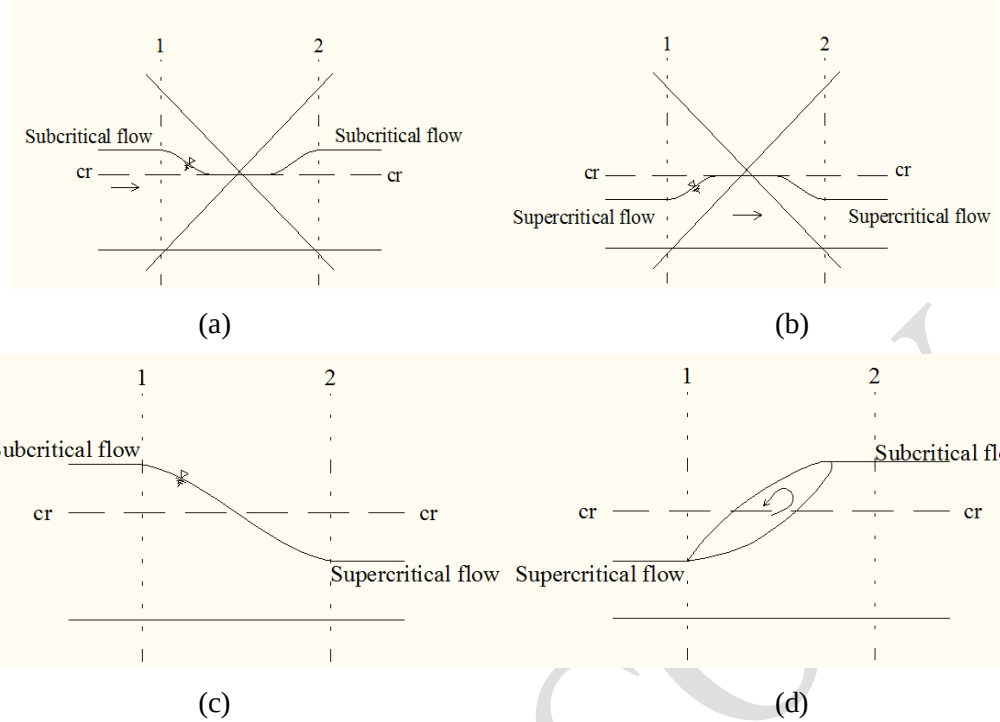
$$\frac{Q_{air}}{Q} = 0.014(Fr_1 - 1)^{1.4}$$

If the jump is in a closed duct then, as the air is released, it could accumulate on the roof of the duct. This phenomenon is called “blowback” and has caused failures in some cases.

6- Recent studies showed that the flow properties (including air entrainment) of hydraulic jumps are not only functions of the upstream Froude number but also of the upstream flow conditions; i.e. partially developed or fully developed inflow. The topic is currently under investigation.

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8.3.6.3 Variation of Flow Field at Critical Flow Sections



a-b) Impossible cases c-d) Possible cases

Fig. 8.47 The flows in the critical depths

Now, momentum equation applies to the control volume between sections (1) and (2) without considering the regimes in inlet and outlet sections in Fig. 8.47 in order to determine the conjugate depths.

$$\frac{q^2}{g y_1 y_2} = \frac{y_1 + y_2}{2} \qquad \text{or} \qquad 2q^2 = g y_1 y_2 (y_1 + y_2)$$

from critical depth equations in rectangular channel,

$$q^2 = g y_{kr}^3$$

it substitutes to the momentum equation;

$$2y_{kr}^3 = y_1 y_2 (y_1 + y_2)$$

if both sides of the equation are divided by y_{kr}^3, and we define the non-dimensional terms;

$$Y_1 = \frac{y_1}{y_{kr}} \text{ and } Y_2 = \frac{y_2}{y_{kr}}$$

a non-dimensional equation may be found,

$$Y_2 Y_1^2 + Y_1 Y_2^2 - 2 = 0 \qquad (8.116)$$

In order to examine the eq.(8.116), Fig.2.48 is plotted

As it is shown in Fig.8.48, we get the results as follows;

$Y_1 > 1$ or $y_1 > y_{kr}$ so $Y_2 < 1$ or $y_2 < y_{kr}$

$Y_1 < 1$ or $y_1 < y_{kr}$ so $Y_2 > 1$ or $y_2 > y_{kr}$

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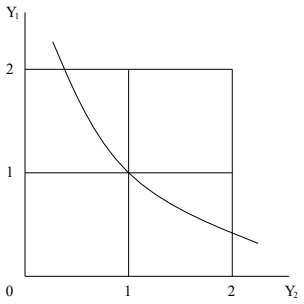


Fig. 8.48

If the flow is critical flow in any section, either " the flow must have sub-critical before critical section and supercritical after critical section" or " the flow must have supercritical before critical section and sub-critical after critical section".

8.3.6.4 Channel Transitions

A change in water surface occurs when the channel bottom is raised or lowered, when the width of the channel is increased (expansion) or decreased (contraction), or when a combination of both exists. The flow channel transitions are rapidly varied flow.

8.3.6.4.1 Decrease of Channel Cross-Section

- ***Case 1-Hump on the Channel Bottom;***

Consider a channel shown in Fig. 8.50. The bed of the channel is raised by amount e while the width remains constant. It is assumed that the energy loss is neglect. The specific energy is E_0 in upstream of the hump, and E_1 on the hump. Energy equation is between sections (1) and (2);

$$E_0 = E_1 + e_1 \tag{8.117}$$

where e_1 is the height of the hump. If we rearrange the eq.(8.117),

$$E_1 = E_0 - e_1 \tag{8.118}$$

The unit discharge, q, may flow over the hump for only $E \geq E_{min}$ as indicated in Fig. 8.49. So there are two cases for $E > E_{min}$. Decrease of the specific energy causes to decrease of the depth in sub-critical regime, but decrease of specific energy causes the increasing of depth in supercritical flow.

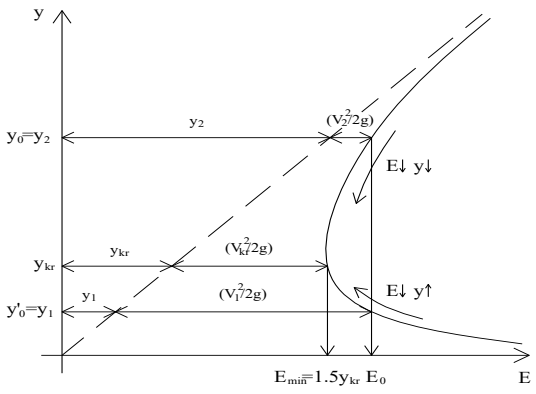


Fig. 8.49 Specific energy diagram for hump case

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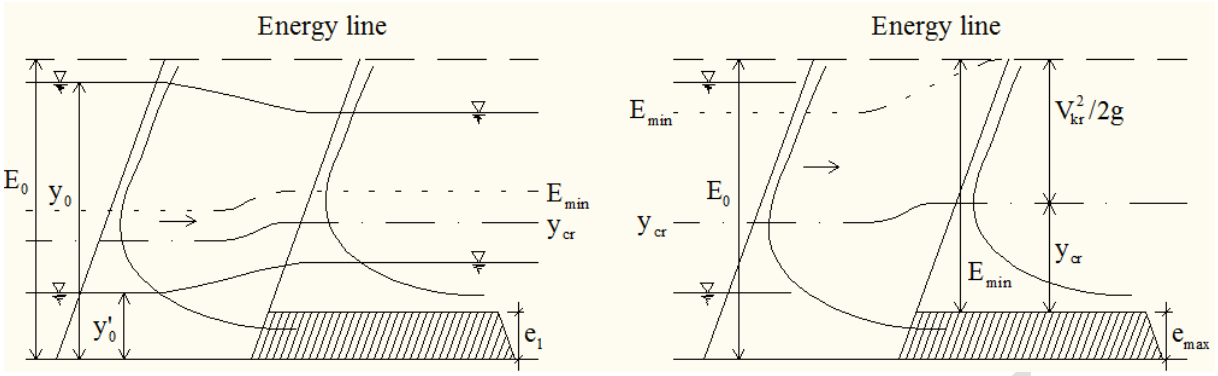


Fig. 8.50 Hump cases; (a) $E_1>E_{min}$, (b) $E_1\leq E_{min}$

Flow over the hump is examined for $E_1>E_{min}$ using energy diagram and Koch parabola in Fig. 8.51. Maximum hump height is obtained when the flow over the hump is critical and the regime will be change as shown in Fig.8.50.

$$E_{min} = E_0 - e_{max}$$

or

$$e_{max} = E_0 - E_{min} \tag{8.119}$$

If a hump which is higher than or equal to the maximum height $(e_2 \geq e_{max})$ located on the channel bottom, the specific energy over the hump $(E_2 \leq E_{min})$ is,

$$E_2 = E_0 - e_2 \tag{8.120}$$

In this case, the energy over the hump E_2 is less than E_{min}. But E_2 cannot be lowered below E_{min} at which the depth approaches the critical depth. Discharge does not flow over the hump with existing energy, E_2 , because it must equal to E_{min} to flow over the hump. In this case, water level will raise in the upstream of the hump and the new energy level is;

$$E_0^{'} = E_0 + \Delta E \tag{8.121}$$

Increasing of water level will be examined in the two steps;

a) Sub-critical flow: Energy level increases before the hump and energy level raise $E_0^{'}$. Water level decreases to the critical depth over the hump and the flow becomes supercritical regime in the downstream section. Since the depth decreases, the velocity and energy gradient increase, then its regime will be again changed to sub-critical with a hydraulic jump (Fig. 8.52). Where is the critical depth located between I and II section of the hump in Fig. 8.52.

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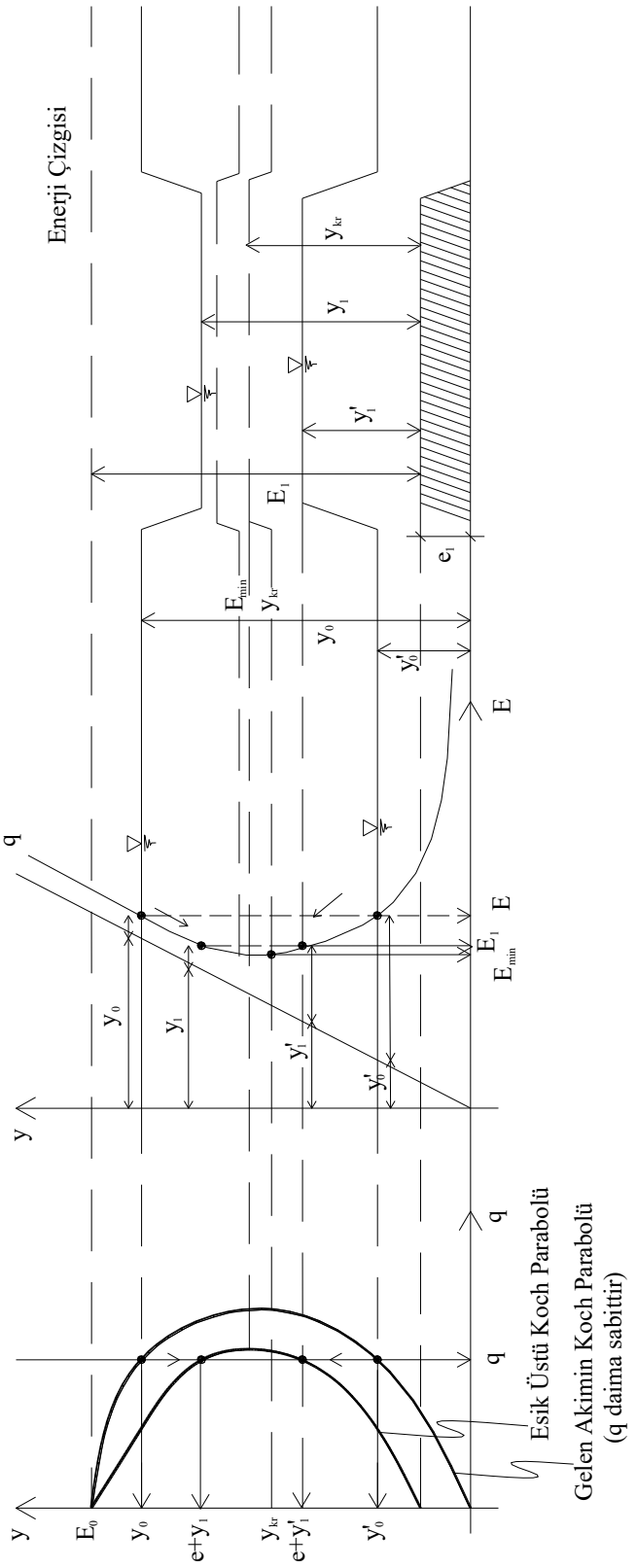
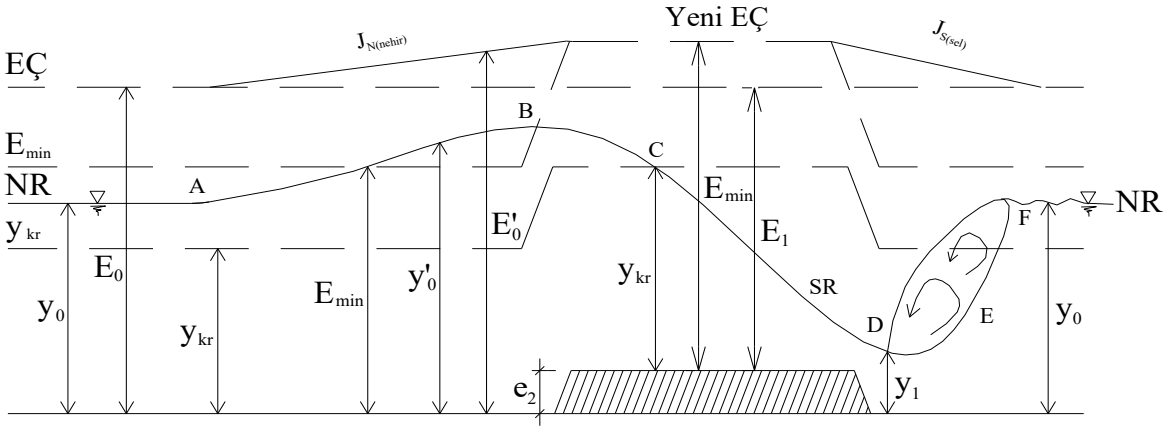
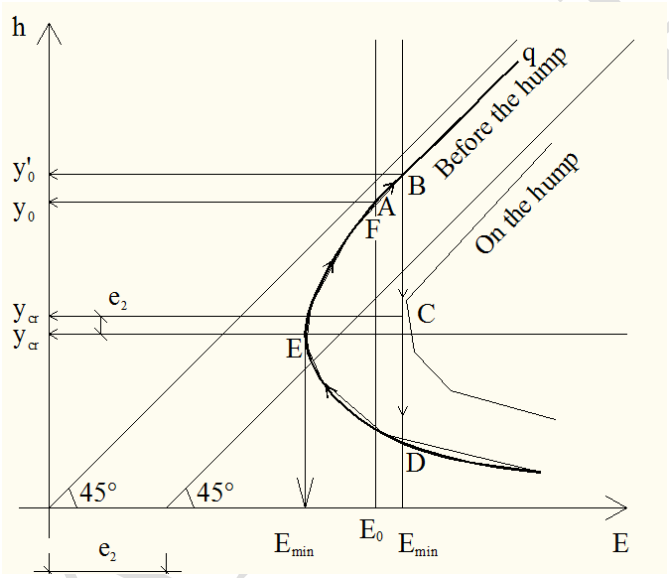


Fig. 8.51 Water surface profile over a hump

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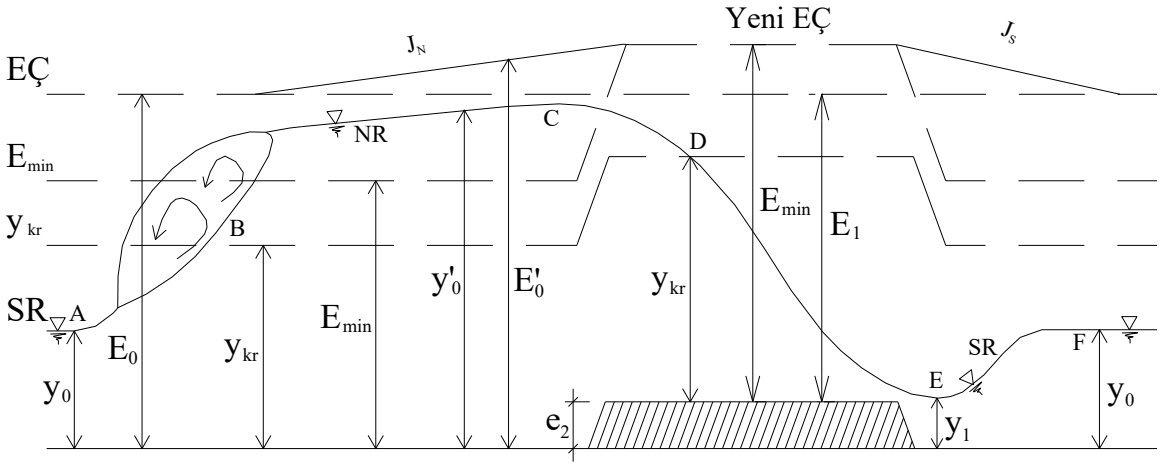
(a)



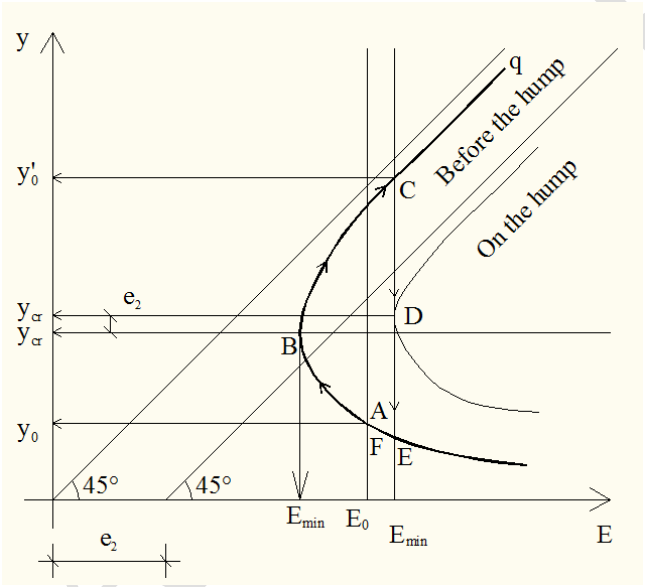
(b)

Fig. 8.52 Changing the flow regime for sub-critical flow at section (0)

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(a)



(b)

Fig.8.53 Changing the flow regime for supercritical flow at section (0)

b) Supercritical flow: Supercritical flow is upstream flow control, i.e. the hump does not affect the upstream flow region, the effect of any changing only appears in the downstream flow region. The water level raise in just upstream of the hump and a hydraulic jump is formed upstream of the hump and then the flow becomes sub-critical flow. The energy increases to $E_0^{'}=e+E_{min}$. The flow will be critical over the hump (E_{min}) and then the flow regime changes to supercritical flow. Energy gradient J increases due to excess energy ΔE and unit discharge flows with the depth y_0 which is less than y_1 until the energy equals to the initial specific energy E_0 .

- ***Case 2-Channel Contraction;***

Same concepts for the hump on the channel bottoms may apply to the channel contraction. If the rectangular channel width B_0 decreases to B_1 , unit discharge q and velocity V increase and the channel contraction induces a decrease of flow depth in sub-critical flow and while an increase of flow depth in supercritical flow (Fig. 8.54).

$$y_{kr,0} = \sqrt[3]{\frac{Q^2}{B_0^2 g}} \tag{8.122}$$

$$y_{kr,1} = \sqrt[3]{\frac{Q^2}{B_1^2 g}} \tag{8.123}$$

$$Q = B_0 q_0 = B_1 q_1 \tag{8.124}$$

$$B_0 > B_1 \Rightarrow q_1 > q_0$$

If the energy equation is written between the approaching channel and channel contraction, it should be $E_0 = E_1$. If the width of channel contraction is calculated by the condition of $E_1 > E_{min}$, the discharge q_1 should be exist for the present condition. The discharge equals to the maximum discharge, $q_1 = q_{max}$, for $E_1=E_{min}$ and the width of channel contraction is $B_1=B_{min}=Q/q_{max}$. If the width of channel contraction is $B_2 < B_1 \le B_{min}$ the discharge does not flow with exist energy $(E_2 < E_{min})$ and the energy should increase

$$E_0 = E_0^{'} + \Delta E \tag{8.125}$$

In this case, the water level increases to $y_0^{'}$. This case will be examined by two-step. The incident supercritical flow profile is shown in Fig. 8.55 for a channel contraction using energy diagram. A hydraulic jump occurs before the contraction when the flow moves from A to C. The flow is not possible from A to E because supercritical flow is upstream flow control and it does not effect to upstream region. The changes of flow will appear in the downstream flow region.

The channel contractions are used for the discharge measurements (e.g. Venturi channels, Parshall flumes). The discharge may be calculated;

$$E = y_0 + \frac{Q^2}{2gA_0^2} = y_1 + \frac{Q^2}{2gA_1}$$

$$y_0 - y_1 = \frac{Q^2}{2g}\left(\frac{1}{B_1^2 y_1^2} - \frac{1}{B_0^2 y_0^2}\right)$$

$$Q = \sqrt{2g} B_0 B_1 \frac{y_0 y_1 \sqrt{y_0 - y_1}}{\sqrt{B_0^2 y_0^2 - B_1^2 y_1^2}} \tag{8.126}$$

when y_0 and y_1 are measured in the channel, the discharge may be found.Notes (Chanson, 1999);

1-At the location of critical flow conditions, the flow is sometimes referred to as “choking”.

2-Considering a channel contraction such that the critical flow conditions are reached (i.e. $B=B_{min}$) the flow downstream of that contraction will tend to be sub-critical if there is downstream control, or supercritical in the absence of downstream control.

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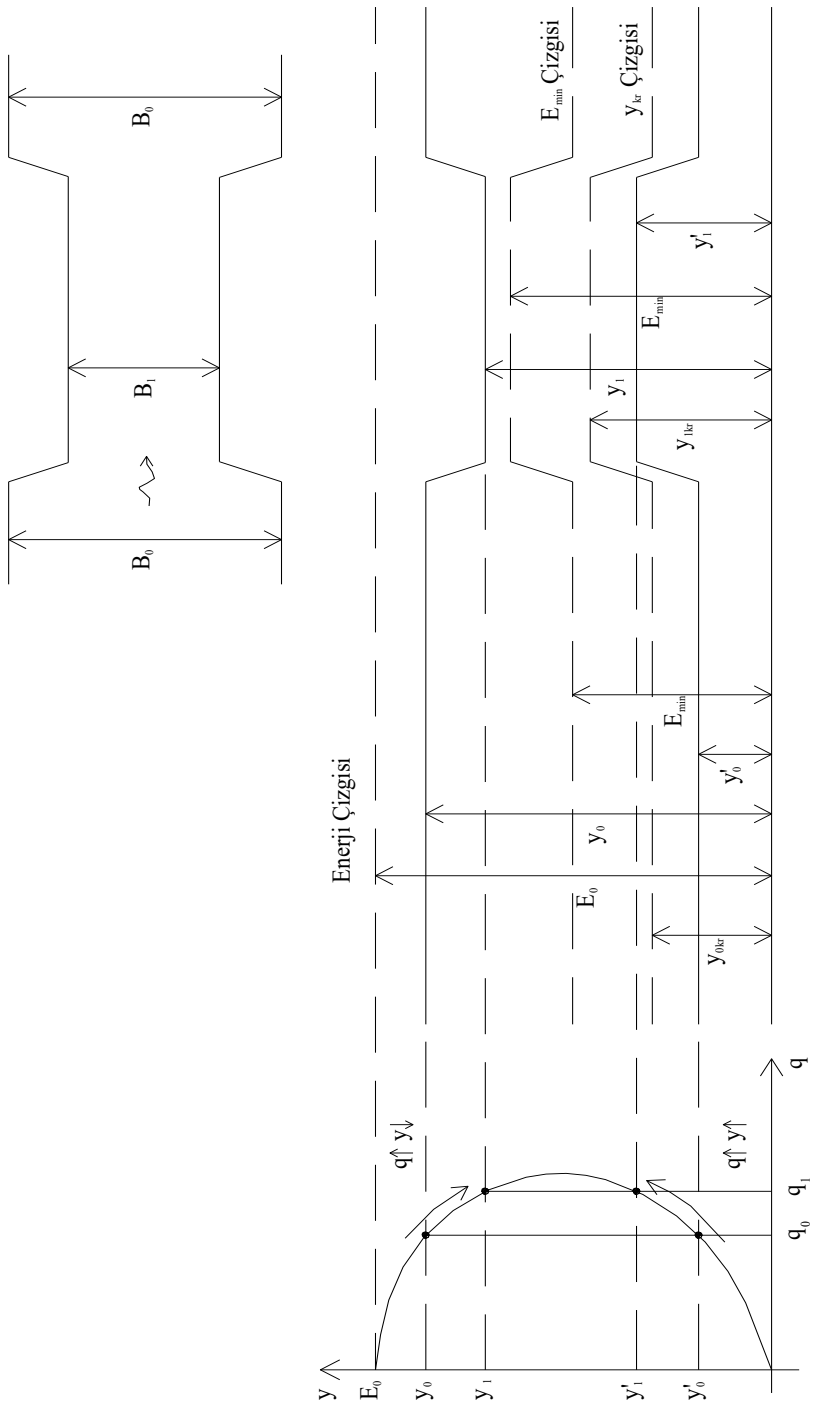


Fig. 8.54 Flow over channel contraction

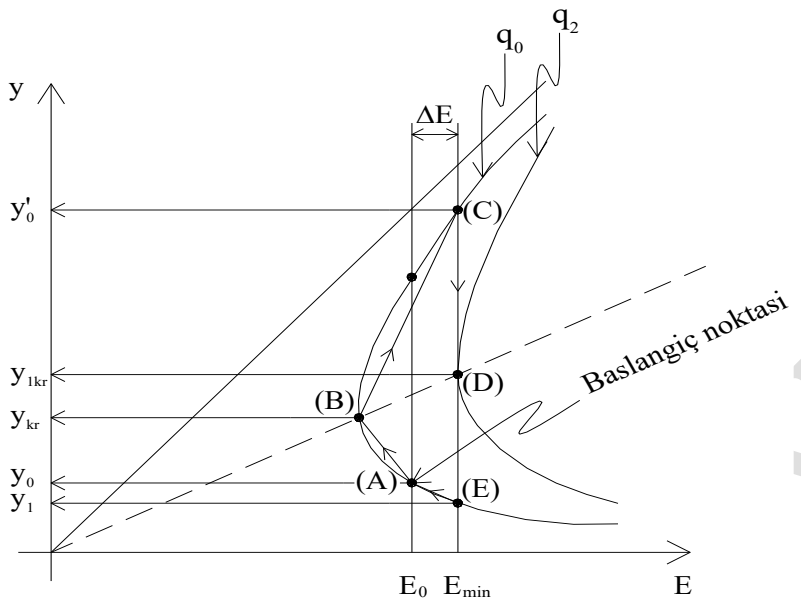


Fig. 8.55 Specific energy diagrams when the flow before the contraction is supercritical

8.3.6.4.2 Increasing of Channel Cross-Section

Increasing of the channel cross-sections are the lowering of the bottom or expansion of the channel width. In these cases, the energy over the transition is always higher than the minimum energy ($E>E_{min}$) and the flow always exist over the transition for present energy levels. So the water level does not increase before the upstream of the transition.

The depth increases in the transition for sub-critical incident flow condition but the depth decreases for supercritical flow conditions as shown in Fig. 8.56.

8.3.6.4.3 Theory

For the short and smooth transitions the energy losses are negligible and Bernoulli equation may be applied quite successfully. Short transitions may be gates (e.g. sluice or radial gates), weirs and steps (humps). The differential form the energy equation in the x direction, $dH/dx = 0$, is.

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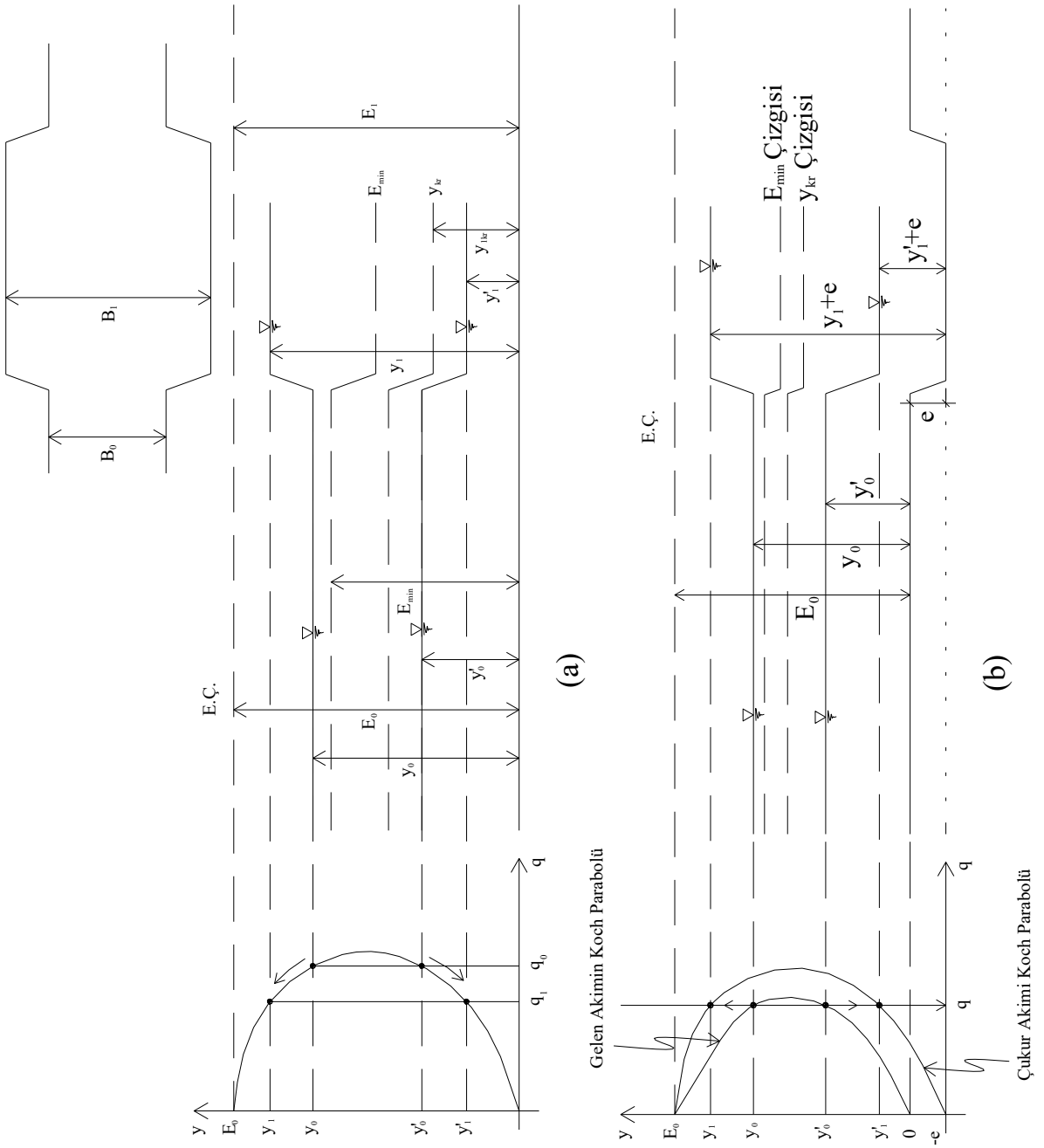


Fig. 8.56 a) Expansion, b) Lowering

$$H = E + z$$

$$\frac{dH}{dx} = \frac{dE}{dx} + \frac{dz}{dx} = 0 \qquad (8.127)$$

where

$$\frac{dE}{dx} = \frac{d}{dx}\left(y + \frac{q^2}{2gy^2}\right) = \frac{dy}{dx} - \frac{q^2}{gy^3}\frac{dy}{dx} + \frac{q}{gy^2}\frac{dq}{dx}$$

$$\frac{dE}{dx} = \frac{dy}{dx} - \frac{V^2 y^2}{gy^3}\frac{dy}{dx} + \frac{q}{gy^2}\frac{d}{dx}\left(\frac{Q}{B}\right)$$

$$\frac{dE}{dx} = \frac{dy}{dx} - \frac{V^2}{gy}\frac{dy}{dx} - \frac{q}{gy^2}\frac{Q}{B^2}\frac{dB}{dx}$$

$$\frac{dE}{dx} = \frac{dy}{dx}\left(1 - Fr^2\right) - \frac{q^2}{Bgy^2}\frac{dB}{dx} = 0$$

and if total energy head substitute in the above equation,

$$\frac{dz}{dx} + \frac{dy}{dx}\left(1 - Fr^2\right) - \frac{q^2}{Bgy^2}\frac{dB}{dx} = 0 \qquad (8.128)$$

the Eq. (8.128) is examined in the Table 8.7;

Table 8.7 Water profiles for the channel transitions

$dz/dx = 0$	Contraction	$dB/dx < 0$	$Fr < 1$	$dy/dx < 0$
			$Fr > 1$	$dy/dx > 0$
	Expansion	$dB/dx > 0$	$Fr < 1$	$dy/dx > 0$
			$Fr > 1$	$dy/dx < 0$
$dB/dx = 0$	Hump	$dz/dx > 0$	$Fr < 1$	$dy/dx < 0$
			$Fr > 1$	$dy/dx > 0$
	Lowering	$dz/dx < 0$	$Fr < 1$	$dy/dx > 0$
			$Fr > 1$	$dy/dx < 0$

$dz/dx = 0$	$dy/dx \neq 0$	$Fr = 1$	Accelerated flow over spillway and The depth over the crest i.e. Decreasing of water level
$dB/dx = 0$	$dy/dx = 0$	$Fr \neq 1$	Mostly occurs at this Section

8.3.7 Gradually Varied Flows

8.3.7.1 Introduction

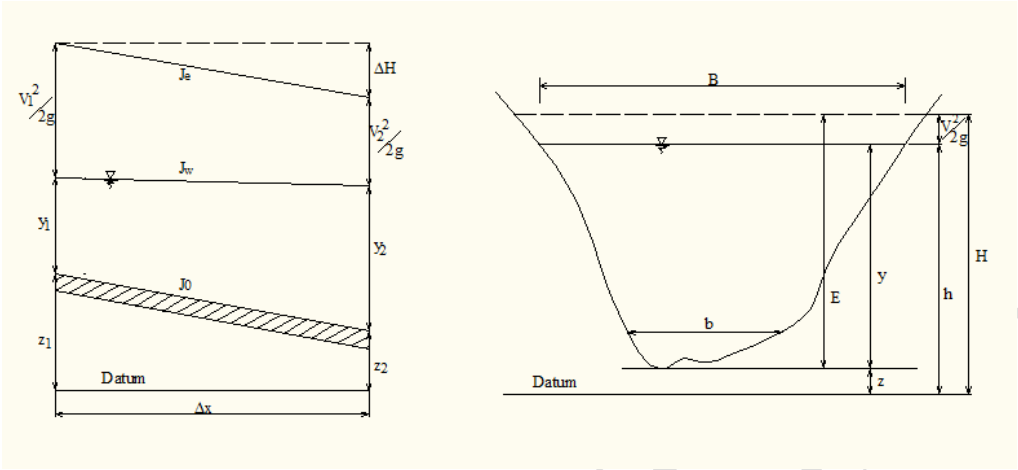
The flows discussed in the previous section were examples of rapidly varied non-uniform flow where the changes in depth and velocity occurred over a relating short distance. If these changes are gradually we refer to the flow as *gradually varied non-uniform flow* or shortly, when the depth of flow in an open channel flow varies with longitudinal distance, the flow is termed *gradually varied*. In both cases the flow does not vary with time and is thus steady. The equation for gradually varied flow and its solution is derived on the assumption (substantiated by experiment) that uniform flow equations may be used to evaluate the energy slope at a given channel section because the streamlines are almost

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parallel. So the slope of the channel is small, the pressure distribution may fits to the distribution in uniform flow.

The friction is an important factor in the gradually varied flows. The friction factors for gradually varied flows may be assumed to equal to uniform flow friction factors.

8.3.7.2 The Equation of the Gradually Varied Flow



(a) Longitudinal section (b) Cross-section

Figure 8.57 Gradually varied flow

Considering a control volume between sections (1) and (2) as shown in Fig. (8.57), energy equation is,

$$z_1 + y_1 + \frac{V_1^2}{2g} = z_2 + y_2 + \frac{V_2^2}{2g} + \Delta H \qquad (8.129)$$

$$(z_1 - z_2) + (y_1 - y_2) + \frac{(V_1^2 - V_2^2)}{2g} = \Delta H$$

$$\Delta z + \Delta y + \frac{\Delta V^2}{2g} = \Delta H \qquad , A = f(y, b)$$

the total derivative of velocity term may be written in partial derivative

$$d(V^2) = d\left(\frac{Q}{A}\right)^2 = -2\frac{Q^2}{A^3}\left(\frac{\partial A}{\partial y}dy + \frac{\partial A}{\partial b}db\right)$$

$$\Delta z = -J_0 \Delta x \qquad \Delta H = -J_e \Delta x$$

If we write the energy equation in differential form, we get

$$-J_0 dx + dy - 2\frac{Q^2}{A^3}\left(Bdy + \frac{\partial A}{\partial b}db\right)\frac{1}{2g} + Jedx = 0$$

$$-J_0 + \frac{dy}{dx} - \frac{Q^2}{gA^3}\left(B\frac{dy}{dx} + \frac{\partial A}{\partial b}\frac{db}{dx}\right) + J_e = 0$$

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$$\frac{dy}{dx} = \frac{J_0 - J_e + \frac{Q^2}{gA^3}\frac{\partial A}{\partial b}\frac{db}{dx}}{1 - \frac{BQ^2}{gA^3}} \tag{8.130}$$

it is known "the differential equation of the gradually varied flow", it is also known "the equation of water surface profile", where

$$\frac{BQ^2}{gA^3} = Fr^2$$

Assuming for a prismatic channel (i.e. known geometry does not vary with distance or along channel) is $\frac{db}{dx} = 0$ and the Eq.(8.130) reduces to

$$\frac{dy}{dx} = \frac{J_0 - Je}{1 - Fr^2} \tag{8.131}$$

If conveyance factor (K) is defined and using;

$$K = CA\sqrt{R} \quad ; \qquad K = f(y)$$

and from Chezy equation,

$$Q = CA\sqrt{RJ} \quad \text{veya} \quad Q = K\sqrt{J}$$

discharge is

$$Q = CAR^{1/2}J_e^{1/2} = KJ_e^{1/2} \quad \text{ve} \quad Q = C_0A_0R_0^{1/2}J_0^{1/2} = K_0J_0^{1/2}$$

and

$$\frac{dy}{dx} = J_0\frac{1 - \frac{J_e}{J_0}}{1 - Fr^2} = J_0\frac{1 - \left(\frac{K_0^2}{K^2}\right)}{1 - Fr^2} \quad \text{or}$$

$$\frac{dy}{dx} = J_0\frac{1 - \left(\frac{K_0}{K}\right)^2}{1 - Fr^2} \tag{8.132}$$

The Eq.(8.132) may be used to examine the variation of surface profiles with $K^2 = f(y^N)$, where N is hydraulic exponent.

8.3.7.3 Water Profiles

The surface profiles may be classified using the channel slope J_0 ;

1- $J_0 < J_{kr}$ Mild Slope – M type (M_1, M_2, M_3)

2- $J_0 = J_{kr}$ Critical Slope – C type (C_1, C_3)

3- $J_0 > J_{kr}$ Steep Slope – S type (S_1, S_2, S_3)

4- $J_0 < 0$ Adverse Slope – A type (A_2, A_3)

5- $J_0 = 0$ Horizontal – H type (H_2, H_3)

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In open channel flow the slope J_0 of the channel can be $J_0<J_{kr}$ (mild), $J_0>J_{kr}$ (steep), $J_0=J_{kr}$ (critical), $J_0=0$ (horizontal), $J_0<0$, (adverse slope). In the first three cases uniform flow is possible with $y_0>y_c$ in the mild channel, $y_0<y_c$ in the steep channel and $y_0=y_c$ in the case of the critical slope. Uniform flow is not possible for $J_0<0$ and $J_0=0$. J_c is function of q, so an open channel slope type depends on unit discharge. Since $q=V_0\ y_0 = Vy$ is valid for an unit discharge (q0constant) in a channel, velocity increases when depth decreases or vice versa ($y\ \overline{y_0}$ so $V\leftrightarrow V_0$) for a non-uniform flow condition in specified channel slope. We can thus have 12 surface profiles, M_1, M_2, M_3, S_1, S_2, S_3, C_1, C_3, A_2, A_3, H_2, H_3. Curves in zone 1 are called *backwater curves* (positive slope), curves in middle zone 2 are called *draw down curves* (they have a negative slope), (Fig.8.58). In all cases the uniform flow depth is approached asymptotically and the critical depth is intersected at right angles. At the upper limits of depth the curves become asymptotic to a horizontal water level, at the bed they intersect it at right angles. Fig. (8.59) summarized the surface profiles.

When using the surface profile curves we have to establish first of all the various *controls* i.e. sections where there is a definite relationship between depth and discharge such as dam, weirs, gates, channel entries and exits etc. In the transition from sub-critical to supercritical flow the critical depth is control ; in the reverse case the transition occurs through a hydraulic jump the location of which is generally not known in the first instance. Profile plotting and depth computation proceed from the control to upstream in sub-critical flow and to downstream in supercritical flow.

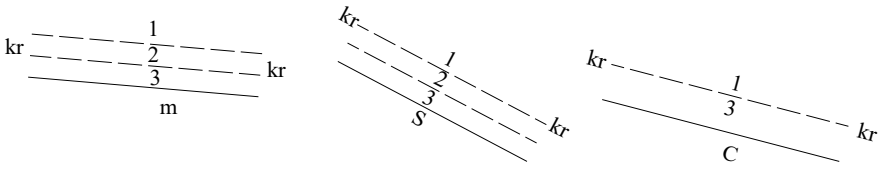


Figure 8.58 The flow regions for mild, steep and critical steeps

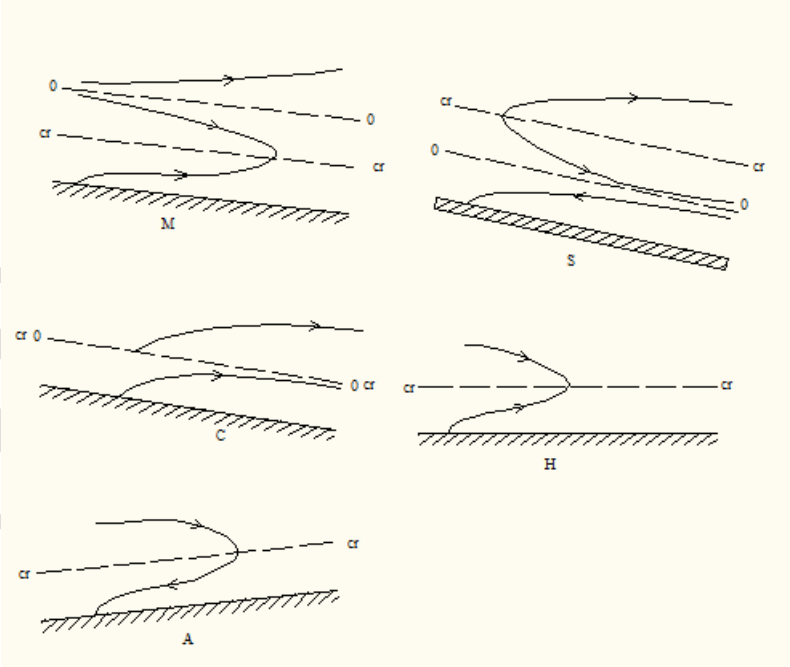


Figure 8.59 Water surface profiles

8.3.7.3.1. Examining of the Profile plotting

The following steps may be considered in rectangular channel;

$$\frac{dy}{dx} = \frac{J_0 - J_e}{1 - Fr^2} \tag{8.133}$$

or

$$\frac{dy}{dx} = J_0 \frac{1 - \left(\frac{K_0}{K}\right)^2}{1 - Fr^2}$$

$$\frac{K_0}{K} = \frac{C_0 A_0 R_0^{1/2}}{C.A.R^{1/2}} = \frac{B_0 y_0 y_0^{1/2}}{B.y.y^{1/2}} = \frac{y_0^{3/2}}{y^{3/2}}$$

$$\frac{K_0^2}{K^2} = \frac{y_0^3}{y^3}$$

$$Fr^2 = \frac{Q^2 B}{gA^3} = \frac{V_c^2 A_c^2 B}{gA^3} = \frac{g\frac{A_c}{B_c} A_c^2 .B}{gA^3} = \frac{A_c^3 B}{A^3 B_c}$$

$$Fr^2 = \frac{y_c^3 B_c^3}{y^3 B^3} = \frac{y_c^3}{y^3}$$

$$\frac{dy}{dx} = J_0 \frac{1 - \left(\frac{y_0}{y}\right)^3}{1 - \left(\frac{y_c}{y}\right)^3} \tag{8.134}$$

Eq (8.134) will be used to examine the profiles

a) determine the flow regime according to $y \gtrless y_c$, $Fr \gtrless 1$

b) Strickler equation may be used for non-uniform flow

$$V = kR^{2/3} J_e^{1/2}$$

continuity equation is

$q = V_0 y_0 = V.y$

$y \gtrless y_0$ so $V \leftrightarrow V_0$

$y \gtrless y_0$ so $J \leftrightarrow J_0$

Example: M_1 Profile

1) What is the regime of uniform flow

   $y_0 > y_c$ and $Fr < 1$

2) What type is the curve of non-uniform water surface i.e backwater curves or draw down curves

   $1-Fr^2 > 0$ is positive in $\frac{dy}{dx} = \frac{J_0 - J_e}{1 - Fr^2}$

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non-uniform flow is in region (1), so

$y>y_0$. Then $J_e < J_0$ and $J_0-J_e>0$ is positive.

3) $\frac{dy}{dx} = \frac{+}{+}$ or $\frac{dy}{dx} > 0$

So the function, y(x), increases through the channel.

4) What are the asymptotes;

y is in the region (1), so $\infty>y>y_0$ then

a) For $y\to\infty$, $V = \frac{Q}{A}$ so $V\to 0$, $Fr = \frac{V}{\sqrt{gy_m}}$ so $Fr \to 0$

$1 - Fr = 1$,

$V \to 0 \Rightarrow$ from Strickler $J_e \to 0$

$J_0 - J_e = J_0$, and we get J_0

Hence, it is arrived to $\lim_{y\to\infty} \frac{dy}{dx} = J_0$, the water surface asymptotically approaches a horizontal line for $y\to\infty$

b) For $y\to y_0$, $V \to V_0$, $J_e \to J_0$ then

$J_0 - J_e = 0$

Hence, it is arrived to $\lim_{y\to y_0} \frac{dy}{dx} = 0$, the water surface asymptotically approaches a water surface in the uniform flow. Fig. 8.60 shows M_1 type water surface.

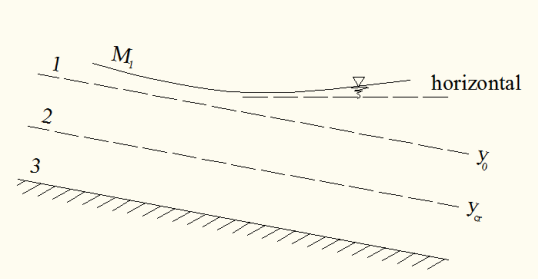


Figure 8.60 Water profile,M_1

Table 8.8 and 8.9 summarize properties of 12 water surfaces

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Table 8.8 Water surface profiles

Class	Slope	$y : y_0 : y_c$	Type	Symbol
Mild slope	$J_0>0$	$y > y_0 > yc$	1	M_1
	$J_0>0$	$y_0 > y > y_c$	2	M_2
	$J_0>0$	$y_0 > y_c > y$	3	M_3
Steep slope	$J_0>0$	$y > y_c > y_0$	1	S_1
	$J_0>0$	$y_c > y > y_0$	2	S_2
	$J_0>0$	$y_c > y_0 > y$	3	S_3
Critical slope	$J_0>0$	$y > y_0 = y_c$	1	C_1
	$J_0>0$	$y < y_0 = y_c$	3	C_3
Horizontal	$J_0=0$	$y > y_c$	2	H_2
	$J_0=0$	$y_c > y$	3	H_3
Adverse slope	$J_0<0$	$y > y_c$	2	A_2
	$J_0<0$	$y_c > y$	3	A_3

For $y>y_0$, $J_e < J_0$ and for $y_0 > y$, $J_e > J_0$ .The signs of dy/dx.

Sign table of $\frac{dy}{dx}$	$y < y_0$	$y > y_0$
Fr > 1 Supercritical	$\frac{-}{-} = +$	$\frac{+}{-} = -$
Fr < 1 Sub-critical	$\frac{-}{+} = -$	$\frac{+}{+} = +$

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Table 8.9 Properties of water surface profiles

		J/J_0	**Fr**	$\frac{dy}{dx}=\frac{J_0-J}{1-Fr^2}$	**Limit** (dy/dx)	**Results**	**Profile type**
Mild Slope $Y_0>Y_c$	$y>y_0>y_c$	<1	<1	+	$y\to\infty\Rightarrow J\to 0; Fr\to 1$ J_0 $y\to y_0\Rightarrow J\to J_0$ 0	y asymptotically approaches a horizontal line. y asymptotically approaches y_0	M_1
	$y_0>y>y_c$	>1	<1	-	$y\to y_0\Rightarrow J\to J_0$ 0 $y\to y_c\Rightarrow Fr\to 1$ ∞	y asymptotically approaches y_0 y becomes vertical.	M_2
	$y_0>y_c>y$	>1	>1	+	$y\to y_c\Rightarrow Fr\to 1$ ∞ $y\to 0\Rightarrow J\to\infty; Fr\to\infty$ $\frac{\infty}{\infty}$	y becomes vertical. Water surface makes an angle bottom with $J_0(y_0/y_c)^3$	M_3
Steep Slope $Y_0<Y_c$	$y>y_c>y_0$	<1	<1	+	$y\to\infty\Rightarrow J\to 0; Fr\to 0$ J_0 $y\to y_c\Rightarrow Fr\to 1$ ∞	y asymptotically approaches a horizontal line.	S_1
	$y_c>y>y_0$	<1	>1	-	$y\to y_c\Rightarrow Fr\to 1$ ∞ $y\to y_0\Rightarrow J\to J_0$ 0	y becomes vertical y asymptotically approaches y_0.	S_2
	$y_c>y_0>y$	>1	>1	+	$y\to y_0\Rightarrow J\to J_0$ 0 $y\to 0\Rightarrow J\to\infty; Fr\to\infty$ $\frac{\infty}{\infty}$	y asymptotically approaches y_0 Water surface makes an angle bottom with $J_0(y_0/y_c)^3$.	S_3
Critical Slope	$y>y_c=y_0$	<1	<1	+	$y\to\infty\Rightarrow J\to 0; Fr\to 0$ J_0 $y\to y_c\Rightarrow J\to J_0; Fr\to 1$ $\frac{0}{0}$	y asymptotically approaches a horizontal line.. Water surface makes a finite angle with normal depth.	C_1
	$y_c=y_0>y$	>1	>1	+	$y\to y_c\Rightarrow J\to J_0; Fr\to 1$ $\frac{0}{0}$ $y\to 0\Rightarrow J\to\infty; Fr\to\infty$ $\frac{\infty}{\infty}$	Water surface makes an finite angle with normal depth. Water surface makes an angle bottom with $J_0(y_0/y_c)^3$	C_3

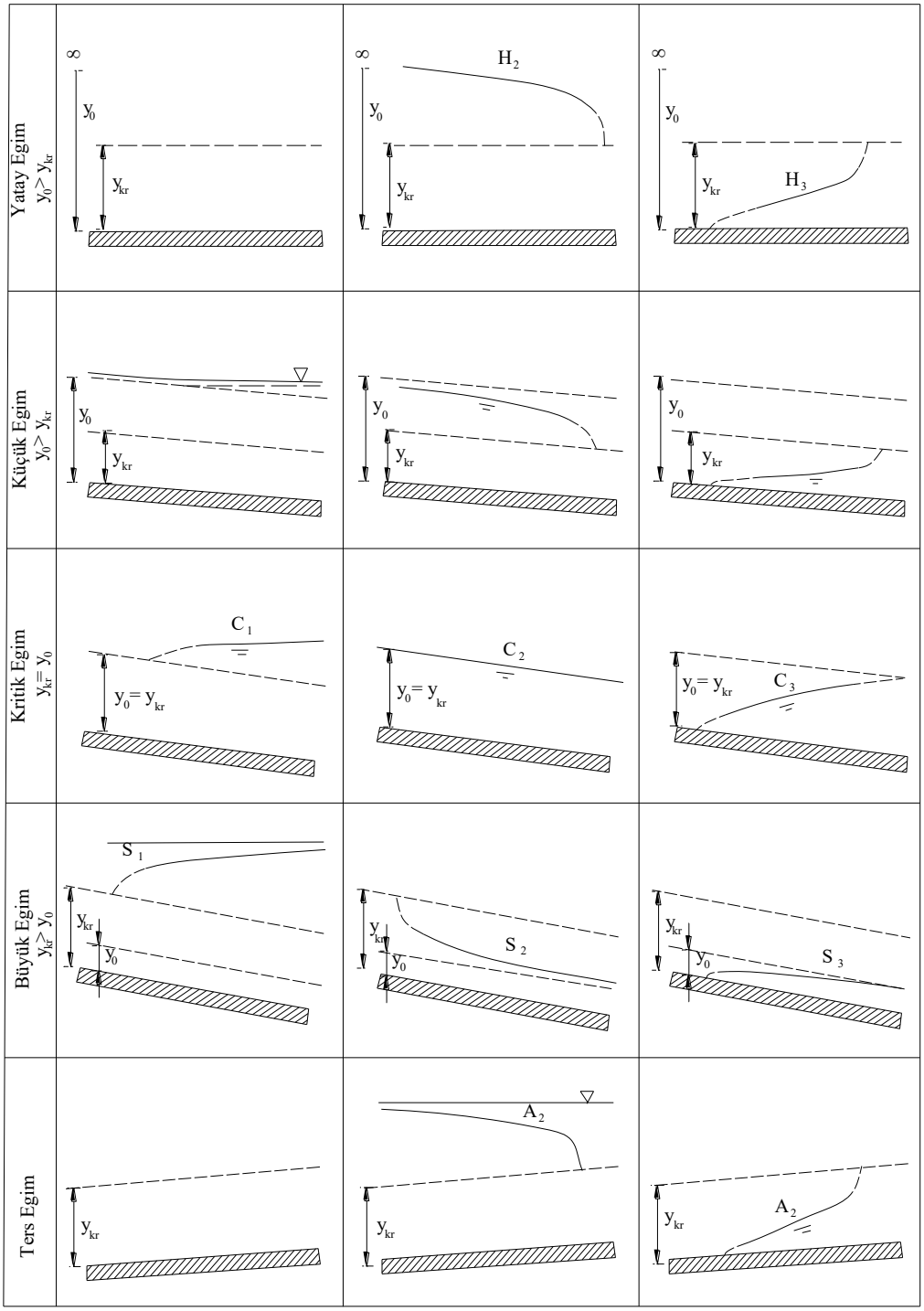
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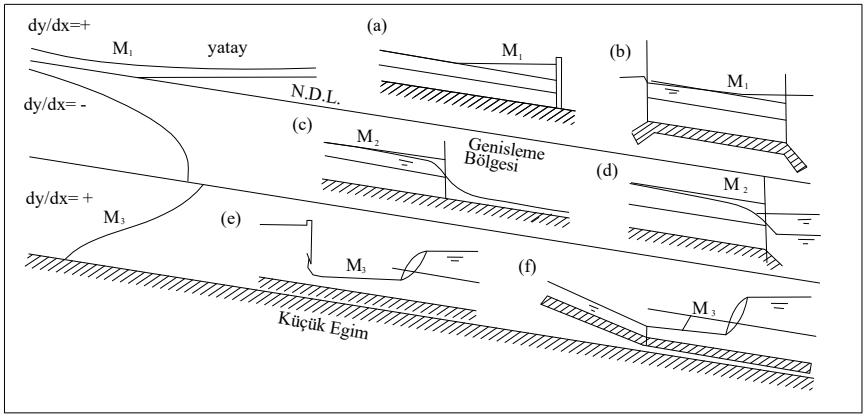
Table 8.9 Continued

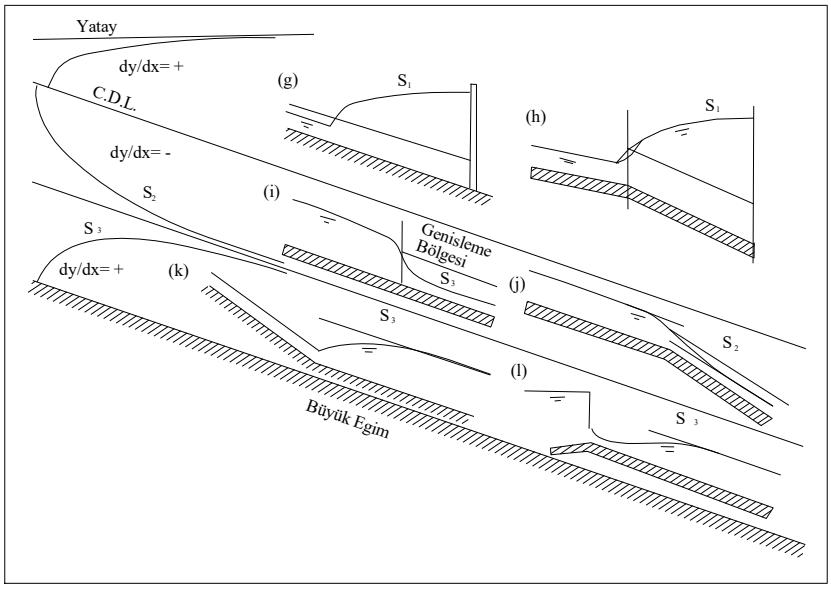
Yatay	$y>y_c$	>1	<1	-	$y \to \infty \Rightarrow J \to 0; Fr \to 0$ 0 $y \to y_c \Rightarrow Fr \to 1$ ∞	y asymptotically approaches a horizontal line. y becomes vertical	H_2
	$y<y_c$	>1	>1	+	$y \to y_c \Rightarrow Fr \to 1$ ∞ $y \to 0 \Rightarrow J \to \infty; Fr \to \infty$ $\frac{\infty}{\infty}$	y becomes vertical. Water surface makes an angle bottom with $J_0 (y_0/y_c)^3$.	H_3
Adverse Slope	$y>y_c$	>1	<1	-	$y \to \infty \Rightarrow J \to 0; Fr \to \infty$ J_0 $y \to y_c \Rightarrow Fr \to 1$ 0	y asymptotically approaches a horizontal line. y becomes vertical.	A_2
	$y<y_c$	>1	>1	+	$y \to y_c \Rightarrow Fr \to 1$ ∞ $y \to 0 \Rightarrow J \to \infty; Fr \to \infty$ $\frac{\infty}{\infty}$	y becomes vertical. Water surface makes an angle bottom with $J_0 (y_0/y_c)^3$	A_3

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Table 8.10 Water surface profiles







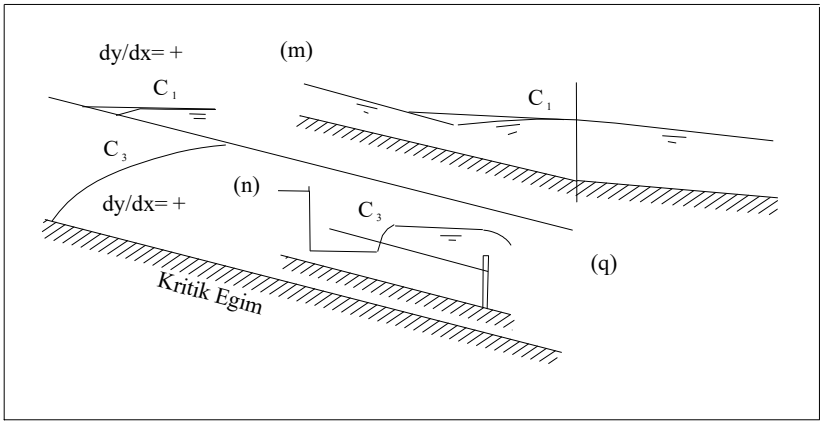
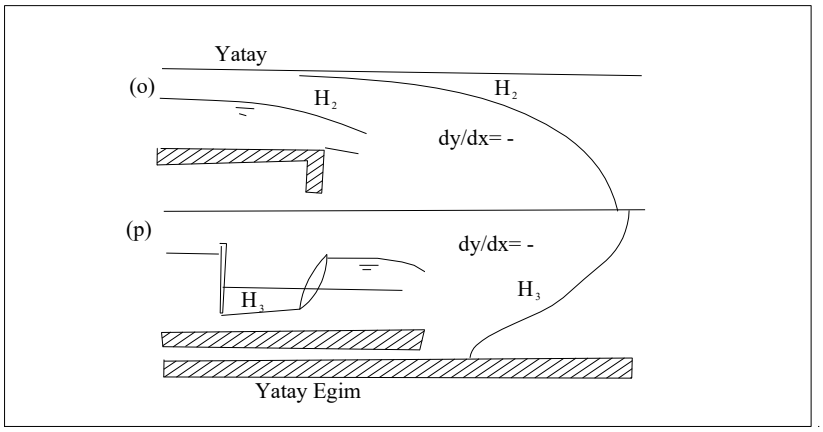


Figure 8.61 Examples for water surfaces



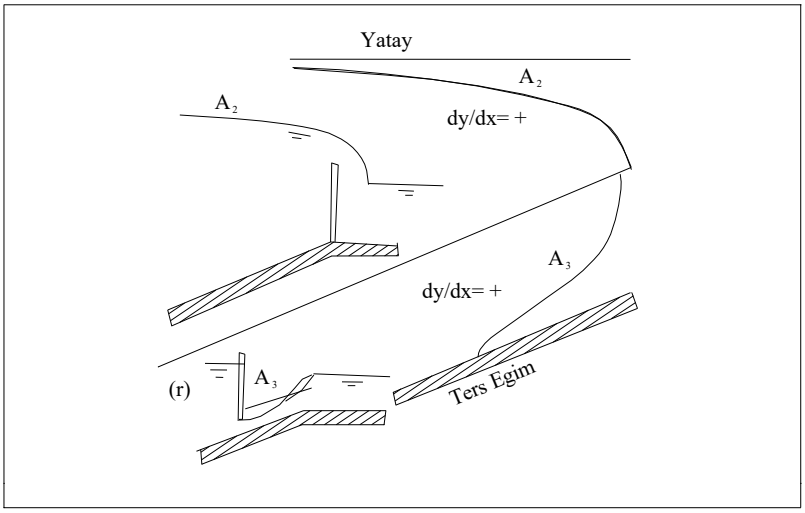


Figure 8.61 Continued

8.3.7.4 Computation of Gradually Varied Flow

Gradually varied flow equation for non-prismatic channel;

$$\frac{dy}{dx} = \frac{J_0 - J_e + \frac{\alpha Q^2}{gA^3}\frac{\partial A}{\partial b}\frac{db}{dx}}{1 - \frac{\alpha B Q^2}{gA^3}} \tag{8.135}$$

for prismatic channel;

$$\frac{dy}{dx} = J_0 \frac{1 - \left(\frac{K_0}{K}\right)^2}{1 - Fr^2} \tag{8.136}$$

computation of the differential equations will be given in two steps;

8.3.7.4.1 Prismatic Channels

$$\frac{dy}{dx} = J_0 \frac{1-\left(\frac{K_0}{K}\right)^2}{1-Fr^2}$$

There are number of solution methods which are direct integration and numerical methods.

1- Direct Integration

$$Fr^2 = \frac{\alpha Q^2 B}{gA^3} = \frac{A_c^3 B}{B_c A^3} = \left(\frac{y_c}{y}\right)^m \tag{8.137}$$

i.e. assuming that $\frac{A^3}{B} = f\left(y^m\right)$ varies as y^m and substituting for

$$\left(\frac{K_0}{K}\right)^2 = \left(\frac{y_0}{y}\right)^N \tag{8.138}$$

i.e. assuming that $K^2 = A^2 C^2 R = f\left(y^N\right)$ varies as y^N we get from Eq.(8.136);

$$\frac{dy}{dx} = J_0 \frac{1-\left(\frac{K_0}{K}\right)^2}{1-Fr^2} = J_0 \frac{1-\left(\frac{y_0}{y}\right)^N}{1-\left(\frac{y_{kr}}{y}\right)^M} = J_0 \frac{1-\left(\frac{y_0}{y}\right)^N}{1-\beta\left(\frac{y_0}{y}\right)^N} \tag{8.139}$$

where

$$\beta = \frac{Fr^2 J_0}{J_e} = \alpha \frac{Q^2 B}{gA^3} \frac{J_0 A^2 C^2 R}{Q^2} = \alpha \frac{C^2 J_0}{g} \frac{B}{P} = \frac{y_c^M}{y_0^N} \cdot y^{N-M} \tag{8.140}$$

from Eq. (8.139)

$$Fr^2 = \frac{\alpha Q^2 B}{gA^3} = \frac{\alpha C_0^2 A_0^2 R_0 J_0 B}{gA^2 RP} = \frac{\alpha K_0^2 J_0 B}{gA^2 RP} \cdot \frac{C^2}{C^2} = \frac{\alpha K_0^2 J_0 B}{gK^2 P} C^2$$

$$Fr^2 = \frac{\alpha C^2 J_0 B}{gP}\left(\frac{K_0}{K}\right)^2 = \frac{\alpha C^2 J_0 B}{gP}\left(\frac{y_0}{y}\right)^N$$

$$Fr^2 = \beta\left(\frac{y_0}{y}\right)^N \tag{8.141}$$

$$\beta = \frac{\alpha C^2 J_0 B}{gP} \tag{8.142}$$

assuming that $\beta = \frac{J_0}{J_c}$ for wide rectangular channels

Conveyance factor (K) is function of water depth, so

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$$\left.\begin{aligned} K^2 &= k.y^N \\ K_1^2 &= k.y^N \\ K_2^2 &= k.y_2^N \end{aligned}\right\} \qquad (8.143)$$

where k is a coefficient, N is hydraulic exponent for uniform flow. M and N values are given in Table 8.11.

Table 8.11 Values of M and N

Cross-Section	M	N
Rectangular	3.0	3 1/3
Trapezoidal	$\frac{3[1+2m(y/b)]^2 - 2m(y/b)[1+m(y/b)]}{[1+2m(y/b)][1+m(y/b)]}$	$\frac{10[1+2m(y/b)]}{3[1+m(y/b)]} - \frac{8(y/b)\sqrt{1+m^2}}{3\left[1+2(y/b)\sqrt{1+m^2}\right]}$
Triangular	5.0	16/3

Now the water surface equation is rearranged

$$J_0 \frac{dx}{dy} = \frac{1-\beta\left(\frac{y_0}{y}\right)^N}{1-\left(\frac{y_0}{y}\right)^N} = \frac{y^N - \beta y_0^N}{y^N - y_0^N}$$

$$J_0 \frac{dx}{dy} = 1 - \frac{y_0^N - \beta y_0^N}{y_0^N - y^N} = 1 - \frac{1-\beta}{1-\left(\frac{y}{y_0}\right)^N}$$

where

$\frac{y}{y_0} = \eta$ and the equation is integrated,

$$x_1 - x_2 = \frac{y_0}{J_0}\{(\eta_1 - \eta_2) - (1-\beta)[B(\eta_1) - B(\eta_2)]\} \qquad (8.144)$$

The function $\int \frac{d\eta}{1-\eta^N} = B(\eta)$ the Bakhmeteff function can be evaluated for various values of N and □□and tables of the function B( ) are available. For the ease of a wide rectangular channel using Chezy equation we get N=M=3 and the functions in equation (8.144) revert to the Bresse functions. Eq. (8.144) can be used to compute (x_1-x_2) for given values of η_1 and η_2 an explicit solution or to compute η_2 from given values of (x_1-x_2) and η_1 and implicit solution.

Bresse functions;

Water surface equation for Prismatic Channels (Belanger equation),

$$\frac{dy}{dx} = J_0 \frac{\left[y^3/y_0^3 - 1\right]}{\left[y^3/y_0^3 - y_c^3/y_0^3\right]} \qquad (8.145)$$

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denoting $y = \eta y_0$

$$y_0 \frac{d\eta}{dx} = \frac{(\eta^3 - 1) J_0}{\eta^3 - (C^2 J_0)/g}$$

$$dx = \frac{y_0}{J_0}\left[\frac{\eta^3 - (C^2 J_0)/g}{\eta^3 - 1}\right] d\eta$$

$$dx = \frac{y_0}{J_0}\left[\frac{\eta^3 - 1 + 1 - (C^2 J_0)/g}{\eta^3 - 1}\right] d\eta$$

$$dx = \frac{y_0}{J_0}\left[1 + \frac{1 - (C^2 J_0)/g}{\eta^3 - 1}\right] d\eta$$

$$dx = \frac{y_0}{J_0}\left[d\eta - \left(1 - \frac{C^2 J_0}{g}\right)\frac{1}{1 - \eta^3} d\eta\right]$$

integrated

$$x_2 - x_1 = \frac{y_0}{J_0}\left\{(\eta_2 - \eta_1) - \gamma \int_{\eta_1}^{\eta_2}\left(\frac{1}{1 - \eta^3}\right) d\eta\right\} \tag{8.146}$$

or

$$x_2 - x_1 = \frac{y_0}{J_0}\{(\eta_2 - \eta_1) - \gamma[\psi(\eta_2) - \psi(\eta_1)]\} \tag{8.147}$$

where ($\gamma = 1 - (C^2 J_0 / g)$ $\psi(\eta)$ Bresse function. Bresse functions are given in table 8.13. Chezy coefficient may assume constant. But $y = 1/2(y_1 + y_2)$ and $C = 18\log(12y/k)$ may be considered along $x_2 - x_1$.

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Table 8.12a Bakhmeteff function for η>1

η	N=2.8	N=3.0	N=3.2	N=3.4	N=3.6	N=3.8	N=4.0	N=4.2	N=4.6	N=5.0	N=5.4
1.001	2.399	2.184	2.008	1.856	1.725	1.610	1.508	1.417	1.264	1.138	1.033
1.005	1.818	1.649	1.506	1.384	1.279	1.188	1.107	1.036	0.915	0.817	0.737
1.010	1.572	1.419	1.291	1.182	1.089	1.007	0.936	0.873	0.766	0.681	0.610
1.015	1.428	1.286	1.166	1.065	0.978	1.902	0.836	0.778	0.680	0.602	0.537
1.02	1.327	1.401	1.078	0.982	0.900	0.828	0.766	0.711	0.620	0.546	0.486
1.03	1.186	1.060	0.955	0.866	0.790	0.725	0.668	0.618	0.535	0.469	0.415
1.04	1.086	0.967	0.868	0.785	0.714	0.653	0.600	0.554	0.477	0.415	0.365
1.05	1.010	0.806	0.802	0.723	0.656	0.593	0.548	0.504	0.432	0.374	0.328
1.06	0.948	0.838	0.748	0.672	0.608	0.553	0.506	0.464	0.396	0.342	0.298
1.07	0.896	0.790	0.703	0.630	0.569	0.516	0.471	0.431	0.366	0.315	0.273
1.08	0.851	0.749	0.695	0.595	0.535	0.485	0.441	0.403	0.341	0.292	0.252
1.09	0.812	0.713	0.631	0.563	0.506	0.457	0.415	0.379	0.319	0.272	0.234
1.10	0.777	0.681	0.601	0.536	0.480	0.433	0.392	0.357	0.299	0.254	0.218
1.11	0.746	0.652	0.575	0.511	0.457	0.411	0.372	0.338	0.282	0.239	0.204
1.12	0.718	0.626	0.551	0.488	0.436	0.392	0.354	0.321	0.267	0.225	0.192
1.13	0.692	0.602	0.529	0.468	0.417	0.374	0.337	0.305	0.253	0.212	0.181
1.14	0.669	0.581	0.509	0.450	0.400	0.358	0.322	0.291	0.240	0.201	0.170
1.15	0.647	0.561	0.490	0.432	0.384	0.343	0.308	0.278	0.229	0.191	0.161
1.16	0.627	0.542	0.473	0.417	0.369	0.329	0.295	0.266	0.218	0.181	0.153
1.17	0.608	0.525	0.458	0.402	0.356	0.317	0.283	0.255	0.208	0.173	0.145
1.18	0.591	0.509	0.443	0.388	0.343	0.305	0.272	0.244	0.199	0.165	0.138
1.19	0.574	0.494	0.429	0.375	0.331	0.294	0.262	0.235	0.191	0.157	0.131
1.20	0.559	0.480	0.416	0.363	0.320	0.283	0.252	0.226	0.183	0.150	0.125
1.22	0.531	0.454	0.392	0.341	0.299	0.264	0.235	0.209	0.168	0.138	0.114
1.24	0.505	0.431	0.371	0.322	0.281	0.248	0.219	0.195	0.156	0.127	0.104
1.26	0.482	0.410	0.351	0.304	0.265	0.233	0.205	0.182	0.145	0.117	0.095
1.28	0.461	0.391	0.334	0.288	0.250	0.219	0.193	0.170	0.135	0.108	0.088
1.30	0.442	0.373	0.318	0.274	0.237	0.207	0.181	0.160	0.126	0.100	0.081
1.32	0.424	0.357	0.304	0.260	0.225	0.196	0.171	0.150	0.118	0.093	0.075
1.34	0.408	0.342	0.290	0.248	0.214	0.185	0.162	0.142	0.110	0.087	0.069
1.36	0.392	0.329	0.278	0.237	0.204	0.176	0.153	0.134	0.103	0.081	0.064
1.38	0.378	0.316	0.266	0.226	0.194	0.167	0.145	0.127	0.097	0.076	0.060
1.40	0.365	0.304	0.256	0.217	0.185	0.159	0.138	0.120	0.092	0.071	0.056
1.42	0.353	0.293	0.246	0.208	0.177	0.152	0.131	0.114	0.087	0.067	0.052
1.44	0.341	0.282	0.238	0.199	0.169	0.145	0.125	0.108	0.082	0.063	0.049
1.46	0.330	0.273	0.227	0.191	0.162	0.139	0.119	0.103	0.077	0.059	0.046
1.48	0.320	0.263	0.219	0.184	0.156	0.133	0.113	0.098	0.073	0.056	0.043
1.50	0.310	0.255	0.211	0.177	0.149	0.127	0.108	0.093	0.069	0.053	0.040
1.55	0.208	0.235	0.194	0.161	0.135	0.114	0.097	0.083	0.061	0.046	0.035
1.60	0.269	0.218	0.179	0.148	0.123	0.103	0.087	0.074	0.054	0.040	0.030
1.65	0.251	0.203	0.165	0.136	0.113	0.094	0.079	0.067	0.048	0.035	0.026
1.70	0.236	0.189	0.153	0.125	0.103	0.086	0.072	0.060	0.043	0.031	0.023
1.75	0.212	0.177	0.143	0.116	0.095	0.079	0.065	0.054	0.038	0.027	0.020
1.80	0.209	0.166	0.133	0.108	0.088	0.072	0.060	0.049	0.034	0.024	0.017
1.85	0.198	0.156	0.125	0.100	0.082	0.067	0.055	0.045	0.031	0.022	0.015
1.85	0.198	0.156	0.125	0.100	0.082	0.067	0.055	0.045	0.031	0.022	0.015
1.90	0.188	0.147	0.117	0.094	0.076	0.062	0.050	0.041	0.028	0.020	0.014
1.95	0.178	0.139	0.110	0.088	0.070	0.057	0.046	0.038	0.026	0.018	0.012
2.00	0.169	0.132	0.104	0.082	0.066	0.053	0.043	0.035	0.023	0.016	0.011
2.10	0.154	0.119	0.092	0.073	0.058	0.046	0.037	0.030	0.019	0.013	0.009

Table 8.12a Continued

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η	N=2.8	N=3.0	N=3.2	N=3.4	N=3.6	N=3.8	N=4.0	N=4.2	N=4.6	N=5.0	N=5.4
1.85	0.198	0.156	0.125	0.100	0.082	0.067	0.055	0.045	0.031	0.022	0.015
1.90	0.188	0.147	0.117	0.094	0.076	0.062	0.050	0.041	0.028	0.020	0.014
1.95	0.178	0.139	0.110	0.088	0.070	0.057	0.046	0.038	0.026	0.018	0.012
2.00	0.169	0.132	0.104	0.082	0.066	0.053	0.043	0.035	0.023	0.016	0.011
2.10	0.154	0.119	0.092	0.073	0.058	0.046	0.037	0.030	0.019	0.013	0.009
2.2	0.141	0.107	0.083	0.065	0.051	0.040	0.032	0.025	0.016	0.011	0.007
2.3	0.129	0.098	0.075	0.058	0.045	0.035	0.028	0.022	0.014	0.009	0.006
2.4	0.119	0.089	0.068	0.052	0.040	0.031	0.024	0.019	0.012	0.008	0.005
2.5	0.110	0.082	0.062	0.047	0.036	0.028	0.022	0.017	0.010	0.006	0.004
2.6	0.102	0.076	0.057	0.043	0.033	0.025	0.019	0.015	0.009	0.005	0.003
2.7	0.095	0.070	0.052	0.039	0.029	0.022	0.017	0.013	0.008	0.005	0.003
2.8	0.089	0.065	0.048	0.036	0.027	0.020	0.015	0.012	0.007	0.004	0.002
2.9	0.083	0.060	0.044	0.033	0.024	0.018	0.014	0.010	0.006	0.004	0.002
3.0	0.078	0.056	0.041	0.030	0.022	0.017	0.012	0.009	0.005	0.003	0.002
3.5	0.059	0.041	0.029	0.021	0.015	0.011	0.008	0.006	0.003	0.002	0.001
4.0	0.046	0.031	0.022	0.015	0.010	0.007	0.005	0.004	0.002	0.001	0.000
4.5	0.037	0.025	0.017	0.011	0.008	0.005	0.004	0.003	0.001	0.001	0.000
5.0	0.031	0.020	0.013	0.009	0.006	0.004	0.003	0.002	0.001	0.000	0.000
6.0	0.022	0.014	0.009	0.006	0.004	0.002	0.002	0.001	0.000	0.000	0.000
7.0	0.017	0.010	0.006	0.004	0.002	0.002	0.001	0.001			
8.0	0.013	0.008	0.005	0.003	0.002	0.001	0.001	0.000			
9.0	0.011	0.006	0.004	0.002	0.001	0.001	0.000	0.000			
10	0.009	0.005	0.003	0.002	0.001	0.001	0.000	0.000			
20	0.003	0.002	0.001	0.001	0.000	0.000	0.000	0.000			

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Table 8.12b Bakhmeteff function for $\eta<1$

η	N=2.8	N=3.0	N=3.2	N=3.4	N=3.6	N=3.8	N=4.0	N=4.2	N=4.6	N=5.0	N=5.4
0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.02	0.20	0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020
0.04	0.40	0.040	0.040	0.040	0.040	0.040	0.040	0.040	0.040	0.040	0.040
0.06	0.60	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060
0.08	0.80	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
0.10	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
0.12	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120
0.14	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
0.16	0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160
0.18	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
0.20	0.201	0.200	0.200	0.200	0.200	0.200	0.200	0.200	0.200	0.200	0.200
0.22	0.221	0.221	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220
0.24	0.241	0.241	0.241	0.240	0.240	0.240	0.240	0.240	0.240	0.240	0.240
0.26	0.262	0.261	0.261	0.261	0.260	0.260	0.260	0.260	0.260	0.260	0.260
0.28	0.282	0.282	0.281	0.281	0.281	0.280	0.280	0.280	0.280	0.280	0.280
0.30	0.303	0.302	0.302	0.301	0.301	0.301	0.300	0.300	0.300	0.300	0.300
0.32	0.324	0.322	0.322	0.322	0.321	0.321	0.321	0.321	0.320	0.320	0.320
0.34	0.344	0.343	0.343	0.342	0.342	0.341	0.341	0.341	0.340	0.340	0.340
0.36	0.366	0.363	0.363	0.363	0.362	0.362	0.361	0.361	0.361	0.360	0.360
0.38	0.387	0.384	0.384	0.383	0.383	0.382	0.382	0.381	0.381	0.381	0.380
0.40	0.408	0.405	0.405	0.404	0.403	0.403	0.402	0.402	0.401	0.401	0.400
0.42	0.430	0.426	0.428	0.425	0.424	0.423	0.423	0.422	0.421	0.421	0.421
0.44	0.452	0.448	0.448	0.446	0.445	0.444	0.443	0.443	0.442	0.441	0.441
0.46	0.475	0.470	0.470	0.468	0.466	0.465	0.464	0.463	0.462	0.462	0.461
0.48	0.497	0.492	0.492	0.489	0.488	0.486	0.485	0.484	0.483	0.482	0.481
0.50	0.521	0.514	0.514	0.511	0.509	0.508	0.506	0.505	0.504	0.503	0.502
0.52	0.524	0.536	0.538	0.534	0.531	0.529	0.528	0.527	0.525	0.523	0.522
0.54	0.568	0.559	0.559	0.556	0.554	0.551	0.550	0.548	0.546	0.544	0.543
0.56	0.593	0.583	0.583	0.579	0.576	0.574	0.572	0.570	0.567	0.565	0.564
0.58	0.618	0.607	0.607	0.603	0.599	0.596	0.594	0.592	0.589	0.587	0.585
0.60	0.644	0.631	0.631	0.627	0.623	0.620	0.617	0.614	0.611	0.608	0.606
0.61	0.657	0.644	0.644	0.639	0.635	0.631	0.628	0.626	0.622	0.619	0.617
0.62	0.671	0.657	0.657	0.631	0.647	0.643	0.640	0.637	0.633	0.630	0.628
0.63	0.684	0.669	0.669	0.664	0.659	0.655	0.652	0.649	0.644	0.641	0.6238
0.64	0.698	0.683	0.683	0.677	0.672	0.667	0.664	0.661	0.656	0.652	0.649
0.64	0.698	0.690	0.683	0.677	0.672	0.667	0.664	0.661	0.656	0.652	0.649
0.65	0.712	0.703	0.696	0.689	0.684	0.680	0.676	0.673	0.667	0.663	0.660
0.66	0.727	0.717	0.709	0.703	0.697	0.692	0.688	0.685	0.679	0.675	0.672
0.67	0.742	0.731	0.723	0.716	0.710	0.705	0.701	0.697	0.691	0.686	0.683
0.68	0.757	0.746	0.737	0.729	0.723	0.718	0.713	0.709	0.703	0.698	0.694
0.69	0.772	0.761	0.751	0.743	0.737	0.731	0.726	0.722	0.715	0.710	0.706
0.70	0.787	0.776	0.766	0.757	0.750	0.744	0.739	0.735	0.727	0.722	0.717
0.71	0.804	0.791	0.781	0.772	0.764	0.758	0.752	0.748	0.740	0.734	0.729
0.72	0.820	0.807	0.796	0.786	0.779	0.772	0.766	0.761	0.752	0.746	0.741
0.73	0.837	0.823	0.811	0.802	0.793	0.786	0.780	0.774	0.765	0.759	0.753
0.74	0.854	0.840	0.827	0.817	0.808	0.800	0.794	0.788	0.779	0.771	0.766
0.75	0.872	0.857	0.844	0.833	0.823	0.815	0.808	0.802	0.792	0.784	0.778
0.76	0.890	0.874	0.861	0.849	0.839	0.830	0.823	0.817	0.806	0.798	0.791
0.77	0.909	0.892	0.878	0.866	0.855	0.846	0.835	0.831	0.820	0.811	0.804
0.78	0.929	0.911	0.898.	0.883	0.872	0.862	0.854	0.847	0.834	0.825	0.817
η	N=2.8	N=3.0	N=3.2	N=3.4	N=3.6	N=3.8	N=4.0	N=4.2	N=4.6	N=5.0	N=5.4

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0.79	0.949	0.930	0.914	0.901	0.889	0.879	0.870	0.862	0.849	0.839	0.831
0.80	0.970	0.950	0.934	0.919	0.907	0.898	0.887	0.878	0.865	0.854	0.845
0.81	0.992	0.971	0.954	0.938	0.925	0.914	0.904	0.895	0.881	0.869	0.860
0.82	1.015	0.993	0.974	0.958	0.945	0.932	0.922	0.913	0.897	0.885	0.875
0.83	1.039	1.016	0.996	0.979	0.965	0.952	0.940	0.931	0.914	0.901	0.890
0.84	1.064	1.040	1.019	1.001	0.985	0.972	0.960	0.949	0.932	0.918	0.906
0.85	1.091	1.065	1.043	1.024	1.007	0.993	0.980	0.969	0.950	0.935	0.923
0.86	1.119	1.092	1.068	1.048	1.031	1.015	1.002	0.990	0.970	0.954	0.940
0.87	1.149	1.120	1.095	1.074	1.055	1.039	1.025	1.012	0.990	0.973	0.959
0.88	1.181	1.151	1.124	1.101	1.081	1.064	1.049	1.035	1.012	0.994	0.978
0.89	1.210	1.183	1.155	1.131	1.110	1.091	1.075	1.060	1.035	1.015	0.999
0.90	1.253	1.218	1.189	1.163	1.140	1.120	1.103	1.087	1.060	1.039	1.021
0.91	1.294	1.257	1.225	1.197	1.173	1.152	1.133	1.116	1.088	1.064	1.045
0.92	1.340	1.300	1.266	1.236	1.210	1.187	1.166	1.148	1.117	1.092	1.072
0.93	1.391	1.348	1.311	1.279	1.251	1.226	1.204	1.184	1.151	1.123	1.101
0.94	1.449	1.403	1.363	1.328	1.297	1.270	1.246	1.225	1.188	1.158	1.134
0.95	1.518	1.467	1.423	1.385	1.352	1.322	1.296	1.272	1.232	1.199	1.172
0.96	1.601	1.545	1.497	1.454	1.417	1.385	1.355	1.329	1.285	1.248	1.217
0.97	1.707	1.644	1.590	1.543	1.501	1.464	1.431	1.402	1.351	1.310	1.275
0.976	1.773	1.707	1.649	1.598	1.554	1.514	1.479	1.447	1.393	1.348	1.311
0.980	1.855	1.783	1.720	1.666	1.617	1.575	1.536	1.502	1.443	1.395	1.354
0.985	1.959	1.880	1.812	1.752	1.699	1.652	1.610	1.573	1.508	1.454	1.409
0.990	2.106	2.017	1.940	1.873	1.814	1.761	1.714	1.671	1.598	1.537	1.487
0.995	2.355	2.250	2.159	2.079	2.008	1.945	1.889	1.838	1.751	1.678	1.617
0.999	2.931	2.788	2.663	2.554	2.457	2.370	2.293	2.223	2.102	2.002	1.917

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Table 8.12c $\Phi(\eta) = \eta - B(\eta)$ function

η	N=2.8	N=3.0	N=3.2	N=3.4	N=3.6	N=3.8	N=4.0	N=4.2
1.001	1.398	1.183	1.007	0.855	0.724	0.609	0.507	0.416
1.005	0.813	0.644	0.501	0.379	0.274	0.183	0.102	0.031
1.010	0.562	0.409	0.281	0.172	0.079	0.003	0.074	0.137
1.015	0.413	0.271	0.151	0.050	0.037	0.113	0.179	0.237
1.02	0.307	0.171	0.058	0.038	0.120	0.192	0.254	0.309
1.03	0.156	0.030	0.075	0.184	0.240	0.305	0.362	0.412
1.04	0.046	0.073	0.172	0.255	0.326	0.387	0.440	0.486
1.05	0.040	0.154	0.248	0.327	0.394	0.452	0.502	0.546
1.06	0.112	0.222	0.312	0.388	0.452	0.507	0.554	0.596
1.07	0.174	0.280	0.367	0.440	0.501	0.554	0.599	0.639
1.08	0.229	0.331	0.415	0.485	0.545	0.595	0.639	0.677
1.09	0.278	0.377	0.459	0.527	0.584	0.633	0.675	0.711
1.10	0.323	0.419	0.499	0.564	0.620	0.667	0.708	0.743
1.11	0.364	0.458	0.535	0.599	0.653	0.699	0.738	0.772
1.12	0.402	0.494	0.569	0.632	0.684	0.728	0.766	0.799
1.13	0.438	0.528	0.601	0.662	0.713	0.756	0.793	0.825
1.14	0.471	0.559	0.631	0.690	0.740	0.782	0.818	0.849
1.15	0.503	0.589	0.660	0.718	0.766	0.807	0.842	0.872
1.16	0.533	0.618	0.687	0.743	0.791	0.831	0.885	0.894
1.17	0.562	0.645	0.712	0.768	0.814	0.853	0.887	0.915
1.18	0.589	0.671	0.737	0.792	0.837	0.875	0.908	0.936
1.19	0.616	0.696	0.761	0.815	0.859	0.896	0.928	0.955
1.20	0.641	0.720	0.784	0.837	0.880	0.917	0.948	0.974
1.22	0.689	0.766	0.828	0.879	0.921	0.958	0.985	1.011
1.24	0.735	0.809	0.869	0.918	0.959	0.992	1.021	1.045
1.26	0.778	0.850	0.909	0.956	0.995	1.027	1.055	1.078
1.28	0.819	0.889	0.946	0.992	1.030	1.061	1.087	1.110
1.30	0.858	0.927	0.982	1.026	1.063	1.093	1.119	1.140
1.32	0.896	0.963	1.016	1.060	1.095	1.124	1.149	1.170
1.34	0.932	0.998	1.050	1.092	1.126	1.155	1.178	1.198
1.36	0.967	1.031	1.082	1.123	1.158	1.184	1.207	1.226
1.38	1.002	1.064	1.114	1.154	1.186	1.213	1.235	1.253
1.40	1.035	1.096	1.144	1.83	1.215	1.241	1.261	1.280
1.42	1.067	1.127	1.174	1.212	1.243	1.268	1.289	1.306
1.44	1.099	1.158	1.204	1.241	1.271	1.295	1.315	1.332
1.46	1.130	1.187	1.233	1.259	1.298	1.321	1.341	1.357
1.48	1.160	1.217	1.261	1.296	1.324	1.347	1.367	1.382
1.50	1.190	1.245	1.289	1.323	1.351	1.373	1.392	1.407
1.55	1.262	1.315	1.356	1.389	1.415	1.436	1.453	1.467
1.60	1.331	1.382	1.421	1.452	1.477	1.497	1.513	1.526
1.65	1.399	1.447	1.485	1.514	1.537	1.556	1.571	1.583
1.70	1.464	1.511	1.547	1.575	1.597	1.614	1.628	1.640
1.75	1.538	1.573	1.607	1.634	1.655	1.671	1.685	1.696
1.80	1.591	1.634	1.667	1.692	1.712	1.728	1.740	1.751
1.85	1.652	1.694	1.725	1.750	1.768	1.783	1.795	1.805
1.90	1.712	1.753	1.783	1.806	1.824	1.838	1.850	1.859
1.95	1.772	1.811	1.840	1.862	1.880	1.893	1.904	1.912

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Table 8.12c Continued

η	N=2.8	N=3.0	N=3.2	N=3.4	N=3.6	N=3.8	N=4.0	N=4.2
1.95	1.772	1.811	1.840	1.862	1.880	1.893	1.904	1.912
2.00	1.831	1.868	1.896	1.918	1.934	1.947	1.957	1.965
2.1	1.946	1.981	2.008	2.027	2.042	2.034	2.063	2.070
2.2	2.059	2.093	2.117	2.135	2.149	2.160	2.168	2.175
2.3	2.171	2.202	2.225	2.242	2.255	2.265	2.272	2.278
2.4	2.281	2.311	2.332	2.348	2.360	2.369	2.376	2.381
2.5	2.390	2.418	2.438	2.453	2.464	2.472	2.478	2.483
2.6	2.498	2.524	2.543	2.557	2.567	2.575	2.581	2.585
2.7	2.605	2.630	2.648	2.661	2.671	2.678	2.683	2.687
2.8	2.711	2.735	2.752	2.764	2.773	2.780	2.785	2.788
2.9	2.817	2.840	2.856	2.865	2.876	2.882	2.888	2.800
3.0	2.922	2.944	2.959	2.970	2.978	2.983	2.988	2.991
3.5	3.441	3.459	3.471	3.479	3.485	3.489	3.492	3.494
4.0	3.954	3.960	3.978	3.985	3.990	3.993	3.995	3.996
4.5	4.463	4.475	4.483	4.489	4.492	4.495	4.496	4.497
5.0	4.969	4.980	4.987	4.991	4.994	4.998	4.997	4.998
6.0	5.978	5.986	5.991	5.994	5.996	5.998	5.998	5.999
7.0	6.883	6.990	6.994	6.996	6.998	6.998	6.999	6.999
8.0	7.987	7.992	7.995	7.997	7.998	7.999	7.999	
9.0	8.980	8.994	8.996	8.998	8.999	8.999		
10.0	9.991	9.995	9.997	9.998	9.999	9.999		

Table 8.13 Bresse function

η	ψ(η)	η	ψ(η)	η	ψ(η)	η	ψ(η)
0.00	0.00	0.90	1.218	1.002	1.953	1.18	0.509
0.10	0.10	0.91	1.257	1.005	1.649	1.20	0.479
0.20	0.20	0.92	1.300	1.010	1.419	1.25	0.420
0.30	0.302	0.93	1.348	1.02	1.191	1.30	0.373
0.40	0.407	0.94	1.403	1.03	1.060	1.35	0.335
0.50	0.517	0.95	1.467	1.04	0.970	1.40	0.304
0.60	0.637	0.96	1.545	1.05	0.896	1.50	0.257
0.65	0.703	0.97	1.644	1.06	0.838	1.60	0.218
0.70	0.776	0.98	1.783	1.07	0.790	1.70	0.190
0.75	0.857	0.990	2.017	1.08	0.749	1.80	0.166
0.80	0.950	0.995	2.250	1.09	0.712	1.90	0.146
0.82	0.993	0.998	2.690	1.10	0.681	2.00	0.132
0.84	1.040	0.999	2.788	1.12	0.626	2.50	0.082
0.86	1.092	1.000	∞	1.14	0.580	3.00	0.055
0.88	1.151	1.001	2.184	1.16	0.541	more bigger	$1/2\ \eta^2$

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2- Numerical Methods

Direct Step Method

The direct step method is a simple method applicable to prismatic channels. Consider two vertical sections 1 and 2 at a distance Δx apart in Fig. 8.62.

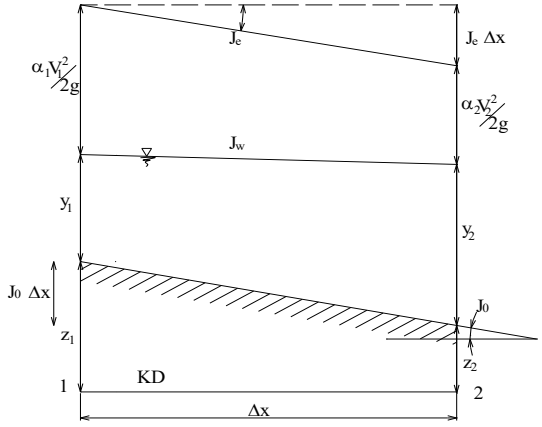


Fig. 8.62 Direct step method

Energy equation;

$$J_0\Delta x + y_1 + \alpha_1 \frac{V_1^2}{2g} = y_2 + \alpha_2 \frac{V_2^2}{2g} + J_e\Delta x \tag{8.148}$$

or,

$$\Delta x = \frac{E_2 - E_1}{J_0 - J_e} \tag{8.149}$$

in a finite difference form as;

$$\frac{E_{i+1} - E_i}{x_{i+1} - x_i} = J_{0m} - J_m \tag{8.150}$$

where,

$$J_{om} = \frac{J_{0i} + J_{0i+1}}{2} \quad ; \quad J_m = \frac{J_i + J_{i+1}}{2} \tag{8.151}$$

J_{0m} and J_m are the slope of channel and energy line respectively. Hence, if y, V, J, E, are known in the section "i" , the location, x_{i+1} , of selected depth, y_{i+1} , may be calculated i.e. Δx

is found. The finite difference equations;

$$x_{i+1} = x_i + \frac{E_{i+1} - E_i}{J_{0m} - J_m} \tag{8.152}$$

y_i, V_i, E_i, J_i are known in initial section x_i ;

$$E_i = y_i + \frac{V_i^2}{2g} \quad ; \qquad J_i = \frac{V_i^2}{C^2R_1} = \frac{V_i^2}{k^2R_i^{4/3}} \quad ; \qquad V_i = \frac{Q}{A_i}$$

The following procedure may be used;

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a) Select y_{i+1} , then compute A_{i+1} and R_{i+1}

b) Q is known, then calculate V_{i+1} , E_{i+1} and J_{i+1} .

c) Using the values in the sections (i) and (i+1), compute specific energies and then x_{i+1} .

Standard Step Method

The standard step method is applicable to non-prismatic channels and therefore to natural rivers. The station positions are predetermined and the objective is to calculate the surface elevations, and hence the depths, at the stations. A trial and error method is employed (see Fig.8.63. The solution is more complex than the direct step method because x_i, x_{i+1}, E_i, J_0 are known, E_{i+1} and J_{i+1} or J_m are not known.

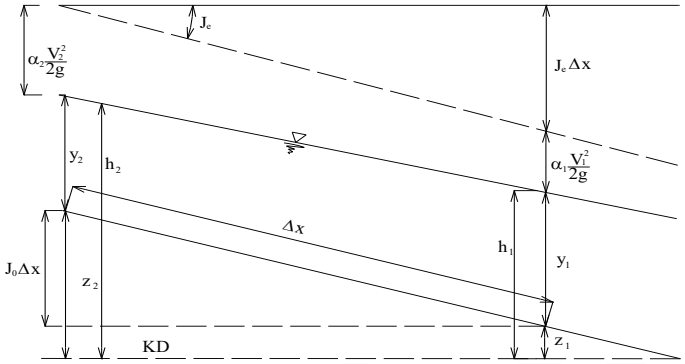


Fig. 8.63 Standard step method

Applying the energy equation between the sections (1) and (2) as shown in Fig.8.63.

$$y_1 + \alpha\frac{V_1^2}{2g} + \Delta h_L = J_0\Delta x + y_2 + \alpha\frac{V_2^2}{2g} \qquad (8.153)$$

assume $h_1 = y_1$, $h_2 = J_0\Delta x + y_2$ and $\alpha_1 = \alpha_2 = \alpha$

$$h_1 + \alpha\frac{V_1^2}{2g} + \Delta h_L = h_2 + \alpha\frac{V_2^2}{2g}$$

where

$$H_1 = h_1 + \alpha\frac{V_1^2}{2g}, \qquad H_2 = h_2 + \alpha\frac{V_2^2}{2g}$$

energy equation may be rearranged,

$$H_1 + \Delta h_L = H_2 \qquad (8.154)$$

The following procedure may be used;

a) H_1 is known. Assume Δx .

b) Assume h_2 , by adding a small amount to h_1

c) Compute $y_2=h_2-z_2$

d) ComputeA_2, P_2, R_2 using y_2

e) Obtain $Je_2 = \frac{n^2V_2^2}{R_2^{4/3}}$ and $J_m = \frac{Je_1 - Je_2}{2}$

f) Calculate $\alpha \frac{V_2^2}{2g}$ and $H_2 = h_2 + \alpha \frac{V_2^2}{2g}$

g) Compute $H_{(2)} = H_1 + J_m \Delta x$

h) Compare H_2 and $H_{(2)}$, if the difference is not within prescribed limits (e.g. 0.001 m), re-estimate h_2 and repeat until agreement is reached;

ı) h_2, y_2 and $H_{(2)}=H_2$ are then recorded and h_2 and H_2 become h_1 and H_1 for succeeding station.

8.3.7.4.2 Non-prismatic Channels

The water surface profiles are additionally to prismatic channels function of channel shape. The calculation will not be discussed in this section.

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6

Dimensional Analysis and Modeling

6.1 Introduction

The method of dimensional analysis is used in every field of engineering especially in such fields as fluid dynamics and thermodynamics where problems with many variables are handled. Engineers are most often interested in knowing the relationship between various physical quantities, which affect a fluid phenomenon. Most physical quantities have dimensions; for example distance has a dimension of length and velocity has dimension of length/time. Some physical quantities have no dimensions; for example, Angular displacement θ is dimensionless. Dimensional analysis is a technique whereby physical quantities may be arranged in dimensionless parameters. Dimensionless parameters play an important role in understanding the flow of fluid in the laboratory- i.e., under experimental conditions.

Hydraulic engineering structures or machines can be designed using;

(i) pure theory

(ii) empirical methods

(iii) semi-empirical methods which are mathematical formulations based on theoretical concepts supported by suitably designed experiments or

(iv) physical models

(v) mathematical models which is also called computational fluid dynamics (CFD).

The purely theoretical approach in hydraulic engineering is limited to a few cases of laminar flow, for example the Hagen Poisseuille equation for the hydraulic gradient in the laminar flow of an incompressible fluid in a circular pipeline.

Empirical methods are based on correlations between observed variables affecting a particular physical system. Such relationships should only be used under similar circumstances to those under which the data were collected. Due to the inability to express the physical interaction of the parameters involved in mathematical terms some such methods are still in use. One well-known example is in the relationship between wave height, fetch, wind speed and duration for the forecasting of ocean wave characteristics.

Semi-empirical methods were obtained from theoretical concepts and experiments designed on the basis of dimensional analysis; it is universally applicable to all Newtonian fluids. A good example of a semi-empirical relationship is the Colebrook-White equation for the friction factors in turbulent flow in pipes.

Dimensional analysis also forms the basis for the design and operation of physical scale models, which are used to predict the behavior of physical scale models that are used to predict the behavior of their full-sized counterparts called “prototypes”. Such models, which are generally geometrically similar to the prototype, are used in the design of aircraft, ships, submarines, pumps, turbines, harbors, breakwaters, river and estuary engineering works, spillways, etc.

While mathematical modeling techniques have progressed rapidly due to the advent of high-speed computers, enabling the equations of motion coupled with semi-empirical relationships to be solved for complex flow situations such as pipe network analysis, pressure transient in pipeline, unsteady flows in rivers and estuaries, etc., there are many cases, particularly where localized flow patterns cannot be mathematically modeled, when physical models are still

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needed.

6.2 Dimensional Analysis

In a laboratory a considerable amount of work and finance would be required to establish a relationship between the dependent and independent variables. The basis of dimensional analysis is to condense the number of separate variables involved in a particular type of physical system into a smaller number of non-dimensional groups of the variables. So, low expense of time and money can be obtained by using dimensionless parameters. Dimensional parameters play an important role in understanding the flow of fluid in laboratory, i.e. the experimental conditions. Using dimensionless parameters, limited experimental results can be applied to physical conditions involving much larger dimensions, and often different fluid properties. Also, the concept of dimensional analysis coupled with the understanding of the mechanism of flow, may be used to generalize the experimental data. This would mean that limited and selective experimental results could be applied to a broad category of larger physical problems.

The arrangement of the variables in the groups is generally chosen so that each group has a physical significance.

All physical parameters can be expressed in terms of a number of basic dimensions; in engineering the basic dimensions, mass (M), length (L) and time (T) are sufficient for this purpose.

Dimensional Analysis is used for:

1. To control of the equations,
2. To control of the units,
3. Derivation of the equations,
4. One of the basic methods for the similarity,
5. Systematic way to perform an experiment.

6.2.1 Physical Significance of Non-Dimensional Groups

The main components of force, which may act on a fluid element, are those due to viscosity, gravity, pressure, surface tension and elasticity. The resultant of these components is called the inertial force and the ratio this force to each of the force components indicates the relative importance of the force types in a particular flow system.

For example the ratio of inertial force to viscous force,

$$\frac{F_i}{F_\mu} = \rho \frac{L^3 L T^{-2}}{\tau L^2}$$

$$\tau = \mu \frac{dV}{dy} = \mu L T^{-1} L^{-1}$$

$$\frac{F_i}{F_\mu} = \rho \frac{L^2 T^{-1}}{\mu} = \rho \frac{L V}{\mu} = \rho \frac{LV}{\mu} = \frac{L V}{\nu}$$

where L is a typical length dimension of the particular system.

The dimensionless term $\rho \frac{L V}{\mu}$ is in the form of the Reynolds number. Low Reynolds numbers indicate a significant dominance of viscous forces in the system which explains why this non-dimensional parameter may be used to identify the type of flow, i.e. whether laminar or turbulent.

Similarly it can be shown that the Froude number is the ratio of inertial force to gravity force in the form

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$Fr = \frac{V^2}{g\,L}$ (usually expressed as $\mathrm{Fr} = \frac{\mathrm{V}}{\sqrt{\mathrm{g\,L}}}$)

The most popular dimensionless numbers in hydraulics;

1) Reynolds Number $= \frac{\text{Inertia Force}}{\text{Viscous Force}} = \frac{\mathrm{V\,L}}{\nu}$ (6.1)

2) Froude Number $= \frac{\text{Inertia Force}}{\text{Gravitational Force}} = \frac{\mathrm{V}}{\sqrt{\mathrm{g\,L}}}$ (6.2)

3) Weber Number $= \frac{\text{Inertia Force}}{\text{Surface Tension Force}} = \rho\frac{\mathrm{V^2\,L}}{\sigma}$ (6.3)

4) Euler Number $= \frac{\text{Pressure Force}}{\text{Inertia Force}} = \frac{\mathrm{p}}{\rho\,\mathrm{V^2}}$ (6.4)

5) Cauchy Number $= \frac{\text{Inertia Force}}{\text{Compressibility Force}} = \frac{\rho\,\mathrm{V^2}}{\mathrm{E}}$ (6.5)

6) Mach Number $= \sqrt{\text{Cauchy Number}}$ (6.6)

7) Strouhal Number $= \frac{\text{Inertia(local)Force}}{\text{Inertia(convective)Force}} = \frac{\mathrm{fL}}{\mathrm{V}}$ (6.7)

Table 6.1 Symbols and dimensions of some basic quantities

Physical Quantity	**Symbol**	**Dimensional Form**
Length	l	L
Time	t	T
Mass	m	M (or $\mathrm{K\,L^{-1}\,T^2}$)
Velocity	V	$\mathrm{L\,T^{-1}}$
Acceleration	a	$\mathrm{L\,T^{-2}}$
Discharge	Q	$\mathrm{L^3\,T^{-1}}$
Force	F, K	$\mathrm{M\,L\,T^{-2}}$ (or K)
Pressure	P	$\mathrm{M\,L^{-1}\,T^{-2}}$ (or $\mathrm{K\,L^{-2}}$)
Power	P, N	$\mathrm{M\,L^2\,T^{-3}}$
Density	ρ	$\mathrm{M\,L^{-3}}$ (or $\mathrm{K\,L^{-4}\,T^2}$)
Dynamic viscosity	μ	$\mathrm{M\,L^{-1}\,T^{-1}}$ (or $\mathrm{K\,L^{-2}\,T}$)
Kinematic viscosity	ν	$\mathrm{L^2\,T^{-1}}$
Surface tension	σ	$\mathrm{M\,T^{-2}}$
Bulk modulus of elasticity	K, E	$\mathrm{M\,L^{-1}\,T^{-2}}$

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6.2.3 The Buckingham π Theorem

Buckingham, in 1915, presented a theorem based on model experiments. A fundamental question we must answer is how many dimensionless products are required to replace the original list of variables? The answer to this question is supplied by the basic theorem of dimensional analysis that states the following:

If an equation involving n variables is dimensionally homogeneous. It can be reduced to a relationship among m=n-r independent dimensionless products, where r is the minimum number of reference dimensions required to describe the variables.

The dimensionless products are frequently referred to as "pi terms" and the theorem is called the Buckingham π theorem. Buckingham used the symbol π to represent a dimensionless product, and this notation is commonly used.

The pi theorem is based on the idea of dimensional homogeneity. Let A1, A2, A3,…., A_n be the n quantities involved, such that

$$f(A_1, A_2, A_3, ..., A_n) = 0 \tag{6.8}$$

Then these quantities can be arranged into (n-r) independent parameters π_1, π_2, π_3,…,π_{n-r} ,such as a way that

$$F(\Pi_1, \Pi_2, \Pi_3, ..., \Pi_{n-m}) = 0 \tag{6.9}$$

In fluid mechanics r=3 and therefore each π term basically contains four of the quantity terms.

Π Parameters ($M^0 L^0 T^0$ or $K^0 L^0 T^0$);

$$\Pi_1 = A_1^{x_1} A_2^{y_1} A_3^{z_1} A_4$$

$$\Pi_2 = A_1^{x_2} A_2^{y_2} A_3^{z_3} A_5$$

$$\Pi_{n-m} = A_1^{x_{n-m}} A_2^{y_{n-m}} A_3^{z_{n-m}} A_n \tag{6.10}$$

one of the π parameters can be written in function of the others,

$$\Pi_1 = \phi(\Pi_2, \Pi_3, \Pi_4, .. ., \Pi_{n-m}) \tag{6.11}$$

A dimensional analysis can be performed using a series of distinct steps as follows:

Step 1- List all the variables that are involved in the problem.

Step 2- Express each of the variables in terms of basic dimensions.

Step 3- Determine the required number of pi terms.

Step 4- Select a number of repeating variables, where the number required is equal to the number of reference dimensions (usually the same as the number of basic dimensions).

Step 5-From a π term by multiplying one of the non-repeating variables by the product of repeating variables each raised to an exponent that will make the combination dimensionless.

Step 6- Repeat Step 5 for each of the remaining repeating variables.

Step 7- Check all the resulting pi terms to make sure they are dimensionless.

Step 8- Express the final form as a relationship among the pi terms and think about what it means.

One of the most important and difficult steps in dimensional analysis is the selection of variables. It is often helpful to classify variables into three groups- geometry, material properties, and external effects.

Typically, in fluid mechanics, the required number of reference dimensions is three, but in some problems only one or two are required.

There is not a unique set of pi terms for a given problem. Once a correct set of pi terms is obtained, any other set can be obtained by manipulation of the original set. It is permissible to exponentiate any π group, e.g. π^{-1}, $\pi^{1/2}$, π^2, etc. to form a new group this will not alter the functional form. It should be emphasized, however, that the required number of pi terms cannot be reduced by this manipulation; only the form of the pi terms is altered.

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Which form for the pi terms is best? Usually our only guideline is to keep the pi terms as simple as possible. The final choice remains an arbitrary one and generally will depend on the background and experience of the investigator.
A useful physical interpretation can often be given to dimensionless groups. Special names along with physical interpretations are given to the most common dimensionless groups.
Dimensional analysis greatly facilitates the efficient handling, interpretation, and correlation of experimental data. If only one pi term is involved in a problem, it must be equal to a constant, π_1=c. For problems involving only two pi terms, results of an experiment can be conveniently presented in a simple graph, $\pi_1=f(\pi_2)$. For problems involving more than two or three pi terms, it is often necessary to use a model to predict specific characteristics, $\pi_1=f(\pi_2, \pi_3)$.

6.4 Physical Modeling

A physical model is a scaled representation of a hydraulic flow situation. A hydraulic model may help the decision-makers to visualize and to picture the flow field, before selecting a "suitable" design. In civil engineering applications, a physical hydraulic model is usually a smaller size representation of the prototype (i.e. the full-scale structure). The physical system for which the predictions are to be made is called the *prototype*.
In a physical model, the flow conditions are said to be similar to those in the prototype if the model displays similarity of form (*geometric similarity*), similarity of motion (*kinematic similarity*) and similarity of forces (*dynamic similarity*).
The ratio of a model variable to the corresponding prototype variable is called the *scale* (r) for that variable.
"Geometric Similarity"; Geometric similarity requires that the ratio of any length in the model L_m to the corresponding length in the prototype L_p should be the same everywhere. This ratio is usually known as scale factor (r_L).

$$r_L = \frac{(L_r)_m}{(L_r)_p} = \frac{(L_2)_m}{(L_2)_p} \tag{6.12}$$

"Kinematic Similarity"; The term kinematic similarity means that at corresponding points in the prototype and in the model, the velocities (V_p ,V_m respectively) and accelerations should have a constant ratio.

$$r_V = \frac{V_m}{V_p} = \frac{L_m}{L_p}\frac{T_p}{T_m} = \frac{r_l}{r_t} \tag{6.13}$$

Where r_v ; velocity scale, r_t ;timescale. For acceleration;

$$r_a = \frac{a_m}{a_p} = \frac{\frac{L_m}{T_m^2}}{\frac{L_p}{T_p^2}} = \frac{L_m}{L_p}\frac{T_p^2}{T_m^2} = \frac{r_l}{r_t^2} \tag{6.14}$$

r_a ;acceleration scale
"Dynamic Similarity"; Dynamic similarity term means that at corresponding points in the model and prototype, the forces should have a constant ratio. A fluid may be subjected to various types of forces. Some possible types of forces are;
1- Pressure force, caused by the differences of pressure between various points in the fluid.

2- Viscous force, caused by the viscosity of the fluid.

3- Elastic force, caused by the compressibility of the fluid.

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4- Surface tension force.

5- Gravity force.

6- Inertia force. From Newton's second law of motion, inertia forces are equal –ma-i.e., a hypothetical force (equal to-ma) is required to bring the acceleration of the fluid particle to zero.

$$r_F = \frac{F_m}{F_p} = \frac{(ma)_m}{(ma)_p} = \frac{\left(\rho L^3 \frac{L}{T^2}\right)_m}{\left(\rho L^3 \frac{L}{T^2}\right)_p} = r_\rho \frac{r_L^4}{r_t^2} = r_\rho r_l^3 r_a \qquad (6.15)$$

where r_F ; force scale

The points corresponding to both in the model and prototype are called *homologue points*.

Models for which one or more similarity requirements are not satisfied are called *distorted models*. A distorted model is a physical model in which the geometric scale is different between each main direction. For example, river models are usually designed with a larger scaling ratio in the horizontal directions than in the vertical directions. The scale distortion does not disturb seriously the flow pattern and it usually gives good results.

With a distorted scale model, it is possible to select small physical models. In addition to the economical and practical benefits, distorted models also have the following advantages compared with non-distorted models;

1-the flow velocities and turbulence in the model are larger.

2-the time scale is reduced.

3-the model Reynolds number is larger, improving the prototype-to-model dynamic similarity, and

4-the larger vertical scale allows a greater accuracy on the flow depth measurements.

Scale effects may be defined as the distortion introduced by effects (e.g. viscosity, surface tension) other than the dominant one (e.g. gravity in the free-surface flows). They take place when one or more dimensionless parameters differ between model and prototype. Scale effects are often small but they are not always negligible altogether.

The procedures for designing models should be made so that the model and prototype will behave in a similar fashion. It will be provided by model design conditions, also called similarity requirements or modeling laws. Model similitude can be established by a number of different methods

1–Similitude by Dynamic Analysis: The dynamic scale ratios are equal in model and prototype.

2–Similitude by Dimensional Analysis: The similarity requirements for a model can be readily obtained with the aid of dimensional analysis. Similarity between a model and a prototype is achieved by equating pi terms.

3–Similitude by Differential Equations: Similarity laws can be directly developed from the equations governing the phenomenon of interest. Governing equations expressed in terms of dimensionless variables lead to the appropriate dimensionless groups.

6.4.1 Dynamic Similarity

The force ratios must be similar for model and prototype;

$$\frac{(F_a)_m}{(F_a)_p} = \frac{(F_g)_m}{(F_g)_p} = \frac{(F_\mu)_m}{(F_\mu)_p} = ...$$

Where; F_a; inertia force, F_g; gravity force, F_μ viscous force. From the above equation it follows that if any two of the force ratios are equal, the third force ratio must also be equal to

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these force ratios. Therefore, if only three forces are involved, the dynamic similarity would be achieved when any two of the forces are in direct proportion. For example, in a flow phenomenon, presume inertia and viscous forces are involved; then for dynamic similarity;

$$\frac{(F_\mu)_m}{(F_a)_m} = \frac{(F_\mu)_p}{(F_\mu)_m}$$

There is no need to consider the third force, because the similarity condition for this force is automatically satisfied.

The acceleration of particles and, hence, the inertia forces play an important role in all fluid flow phenomena, and for this reason it has become customary to include inertia force as one of the two forces considered for dynamic similarity. The other force may be one of the pressure, viscous, elastic, surface tension or gravity forces. For example;

$$\frac{(F_g)_m}{(F_a)_m} = \frac{(F_g)_p}{(F_a)_p}$$

A specific number represents the ratio of inertia force to one of the other forces; for example, the ratio of the inertia force to the gravity force is represented by the Froude number. Therefore, it follows that dynamic similarity can be achieved by making the Mach, Reynolds, Froude or Weber number the same (as the case may be) in both the model and the prototype. The scale ratios are obtained from these numbers.

6.4.1.1 Froude Model

If the inertia forces and gravity forces are dominant forces in a physical system, the Froude number will satisfy dynamic similarity between model and prototype.

$$\frac{(F_g)_m}{(F_a)_m} = \frac{(F_g)_p}{(F_a)_p}$$

or;

$$\frac{(\rho g L^3)_m}{(\rho L^3 L T^{-2})_m} = \frac{(\rho g L^3)_p}{(\rho L^3 L T^{-2})_p}$$

Assuming; $\rho_m=\rho_p$

$$\frac{(L^3)_m}{(L^4 T^{-2})_m} = \frac{(L^3)_p}{(L^4 T^{-2})_p}$$

$$L/T = V$$

$$\frac{V_m^2}{g L_m} = \frac{V_p^2}{g L_p}$$

or

$$Fr_m = Fr_p$$

Therefore, Froude number must be same for the model and the prototype at the corresponding points.

$$r_{Fr} = \frac{Fr_m}{Fr_p} = 1 \qquad (6.16)$$

The scale ratios;

$$r_{Fr} = \frac{V_m^2 L_p}{V_p^2 L_m} = 1$$

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$$r_V = \sqrt{r_l} \tag{6.17}$$

For discharge

$$r_Q = \frac{Q_m}{Q_p} = \frac{A_m V_m}{A_p V_p}$$

$$r_Q = r_V \frac{L_m^2}{L_p^2}$$

$$r_Q = r_V r_l^2 \tag{6.18}$$

or using $\left(r_V = \sqrt{r_l}\right)$;

$$r_Q = r_l^{5/2} \tag{6.19}$$

6.4.1.2 Reynolds Model

If the inertia forces and viscous forces are dominant forces in a physical system, the Reynolds number will satisfy dynamic similarity between model and prototype.

$$\frac{\left(F_\mu\right)_m}{\left(F_a\right)_m} = \frac{\left(F_\mu\right)_p}{\left(F_a\right)_p}$$

or

$$\frac{\left(\mu L^2 L T^{-1} L^{-1}\right)_m}{\left(\rho L^3 L T^{-2}\right)_m} = \frac{\left(\mu L^2 L T^{-1} L^{-1}\right)_p}{\left(\rho L^3 L T^{-2}\right)_p}$$

$$L/T = \nu$$

$$\frac{V_m L_m}{\nu_m} = \frac{V_p L_p}{\nu_p}$$

or

$$Re_m = Re_p$$

Therefore, Reynolds number must be same for the model and the prototype at the corresponding points.

$$r_{Re} = \frac{Re_m}{Re_p} = 1 \tag{6.20}$$

The scale ratios;

$$r_{Re} = \frac{V_m L_m \nu_p}{V_p L_p \nu_m}$$

$$r_V = r_l^{-1} r \tag{6.21}$$

Assuming $r_\nu = 1$

$$r_V = r_l^{-1} \tag{6.22}$$

For discharge

$$r_Q = r_V r_l^2 \tag{6.23}$$

$$r_Q = r_l \tag{6.24}$$

Stating that in order for the results of model testing to be valid for predicting the behavior of the prototype, the following conditions should be met may sum up the above discussion on

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similarity;
1-The model should be geometrically similar to the prototype.
2-Dynamic similarities must be present.
The condition of kinematic similarity is automatically met if the above two conditions are present.
Note: Both Froude and Reynolds similitude are not satisfied at the corresponding points between models and prototypes. Because scale ratios for velocity at the corresponding points in order to satisfy the both numbers;

$\sqrt{r_l} = r_l^{-1}$

$r_l = r_l^{-2}$

$r_l = 1 \quad ; \quad L_m = L_p$

which means that model equals to prototype. Therefore the dominant forces must be considered to find the scale ratios. Table 6.2 summarizes the scaling ratios for the Froude and Reynolds similitude. In general case, the engineer must choose a proper geometric scale. The selection procedure is an iterative process.
Step 1- Select the smallest geometric scale ratio to fit within the constraints of the laboratory.
Step 2- For r_L, and for the similitude criterion (e.g. Froude or Reynolds), check if the discharge can be scaled properly in the model, based upon the maximum model discharge.
Step 3- Check if the flow resistance scaling is achievable in the model.
Step 4- Check the model Reynolds number for the smallest test flow rate.
Step 5-Choose the convenient scale.

Table 6.2 Scale ratios for Reynolds and Froude similarity

			Scales	
Parameter	Symbol	Unit	Froude	Reynolds
Length	L	L	r_l	r_l
Area	A	L^2	r_l^2	r_l^2
Volume	V	L^3	r_l^3	r_l^3
Time	T	T	$r_l^{1/2}$	$r_l^2 r_\gamma^{-1}$
Velocity	V	LT^{-1}	$r_l^{1/2}$	$r_l^{-1} r_\gamma$
Acceleration	A	LT^{-2}	1	$r_l^{-3} r_\gamma^2$
Angular velocity	ω	T^{-1}	$r_l^{-1/2}$	$r_l^{-2} r_\gamma$
Discharge	Q	L^3T^{-1}	$r_l^{5/2}$	$r_l\ r_\gamma$
Unit discharge	Q	L^2T^{-1}	$r_l^{3/2}$	r_γ
Mass	M	M	$r_l^3 r_\rho$	$r_l^3 r_\rho$
Force	P, K	MLT^{-2}	$r_l^3 r_\gamma$	$r_\mu^2 r_\rho^{-1}$
Pressure	P	$ML^{-1}T^{-2}$	$r_l r_\gamma$	$r_l^{-2} r_\mu^2 r_\rho^{-1}$
Impulse and Momentum	I	MLT^{-1}	$r_l^{7/2} r_\rho$	$r_l^2 r_\mu$
Energy and Work	E	ML^2T^{-2}	$r_l^4 r_\gamma$	$r_l r_\mu^2 r_\rho^{-1}$
Power	N	ML^2T^{-3}	$r_l^{7/2} r_\gamma$	$r_l^{-1} r_\mu^3 r_\rho^{-2}$

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